Mpemba relaxation beyond triplet tests

Comprehensive research report

Research report and reproducible mathematical certificate

12 October 2026

A gradual course in thermal relaxation, from fractions to exact proof

Begin at the beginning. The first part develops the needed mathematics through 40 lessons, worked examples, and exercises with answers. The second part retains the complete technical research report and its reproducible certificates. Read them as a connected course, or use the linked contents and reading map to move between intuition and proof.

How to read this unusually detailed report

This report has two jobs. It teaches the mathematics needed to understand a thermal relaxation counterexample, and it records the research arguments precisely enough to check them. The first job requires patience. We begin with fractions and probabilities, not with a matrix full of unfamiliar symbols.

The central question is simple to state. Prepare two copies of the same system, one hotter than the other. Put both in contact with the same colder environment. Can the hotter copy later be closer to equilibrium? In the models studied here, the answer can be yes. A second question is more subtle: can every three-state test miss a reversal in a four-state system? A certified example in this report shows that this too can happen. Understanding why requires learning what the state of the system is, what its dynamics do, what "closer" means, and how a proof differs from a convincing numerical plot.

Starting knowledge. You need ordinary arithmetic and a willingness to read an equation one symbol at a time. High-school algebra helps. Calculus and linear algebra are developed as they become necessary. This is not a replacement for complete courses in those subjects: it is a carefully chosen route through the parts this research uses. No knowledge of Markov chains, spectral theory, statistical mechanics, convex order, interval arithmetic, or proof assistants is assumed at the start.

The learning sequence. Lessons 1–9 build the language: logical claims, fractions, probabilities, weighted averages, temperature, exponentials, Gibbs distributions, derivatives, and covariance. Lessons 10–17 build the dynamics: rates, matrices, equilibrium, eigenmodes, detailed balance, distances, and slow amplitudes. Lessons 18–24 work through the counterexample and the structural explanation. Lessons 25–30 bridge to classification, degeneracy, rigorous computation, finite-time physics, and formal proof. Lessons 31–32 give exercises with solutions and a map into the research text. Lessons 33–40 then slow down the bridge to continuous-time certificates and physical models, deriving random clocks, rigorous tails, target-time bounds and intermediate-state dynamics through worked examples.

How much to read at once. For a first pass, read one lesson, reproduce its small numerical example, and stop when the last paragraph makes sense. For a second pass, derive the displayed identities without looking at the intermediate lines. For a research pass, follow the references into Part II and inspect the associated certificates. Being able to recite a conclusion is different from being able to reconstruct its argument; the worked checks are intended to close that gap.

What the labels mean. An *illustration* is chosen to make an idea easy to see. It need not exhibit a Mpemba reversal. An *identity* follows exactly from its stated assumptions. A *certified result* refers to the particular mathematical claim and verification described in the research text. A *diagnostic calculation* provides evidence or intuition, but does not by itself certify a continuum of parameters. The new teaching examples do not expand the scope of the existing Lean formalization or turn an unfinished proof audit into a completed one.

Keep three questions separate throughout. What has been proved for an abstract finite-state model? What has been verified for the particular parameter choices? What has been demonstrated in a physical experiment? This report supplies mathematical and computational results for models. The physical-model study explains its idealizations explicitly.

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Part I. From basic arithmetic to the research frontier

Lesson 1. What would count as an answer?

Start with a familiar comparison. Suppose two cups of water start at different temperatures. Saying that one "cools faster" is incomplete. It might mean a steeper initial temperature slope, a shorter time to a target temperature, or a smaller distance from the final state after ten minutes. Those statements are different. A curve can start steeply and later flatten. A system can cross one target first and a different target second. We therefore choose a mathematical comparison before trying to prove anything.

Here the system has finitely many states. Its condition is described by a list of probabilities. Temperature is used to prepare that list; after the dynamics begin, the list need not correspond to any single temperature. A bath is the environment with fixed temperature. Both preparations evolve with the same rules toward the same bath equilibrium. We compare distances between their probability lists and that equilibrium.

An implication is a promise with a condition. Write $A \Rightarrow B$ for "if A holds, then B holds." A proposed universal test says that whenever certain three-state tests fail, a four-state reversal cannot occur. To disprove such a statement, we must produce a system in which the condition A is true and the promised conclusion B is false. Producing a reversal alone is not enough: the system must also satisfy every condition of the claim being tested.

There is an important asymmetry. One valid counterexample disproves a universal claim. A million examples consistent with that claim do not prove it. Sampling can miss a thin set, a boundary, or a region outside the sample box. Conversely, failure to find a counterexample is not evidence that a particular proof is correct. A proof must explain why every allowed case is covered.

Read the quantifiers. "There exists" means that at least one object works. "For every" means that none of the allowed objects may fail. In symbols,

$$\exists M \text{ with a reversal} \qquad \text{and} \qquad \forall M \text{ no reversal}$$

make very different claims. The letter M will later denote a matrix of rates. For now it is simply the name of the dynamical rule. A universal bound also needs quantifiers: a bound for four states is not automatically a bound for all numbers of states, and a bound for long-time amplitudes is not automatically a bound for target times.

The proof obligations to remember. Our counterexample must have legitimate probabilities, distinct ordered energies, a finite positive bath temperature, positive transition rates, probability-conserving dynamics, detailed balance, the stated slow-mode property, failure of all the prescribed triplet tests, and a strict reversal. Each adjective has a job. Later lessons explain how each is checked.

Checkpoint. If a three-state test succeeds but a reversal is observed, have we disproved a claim whose premise requires all three-state tests to fail? No. The condition of the implication has not been met. Checking the premise is half of checking a counterexample.

Lesson 2. Fractions, ratios, inequalities, and precision

A fraction a/b, with $b \neq 0$, means the number that becomes a when multiplied by b. Fractions are useful here because they can be exact. The decimal 0.995 equals $995/1000 = 199/200$ exactly. The displayed decimal 0.3333 is usually only an approximation to $1/3$. Knowing which situation applies matters when a proof depends on the sign of a small difference.

A small exact calculation. The bath weights of the primary example are

$$q_1 = 1, \qquad q_2 = \frac{199}{200}, \qquad q_3 = \frac{2}{5}, \qquad q_4 = \frac{1}{400}.$$

The common denominator 400 gives

$$Q = q_1 + q_2 + q_3 + q_4 = \frac{400 + 398 + 160 + 1}{400} = \frac{959}{400}.$$

Dividing each weight by Q therefore gives

$$\left(\frac{400}{959}, \frac{398}{959}, \frac{160}{959}, \frac{1}{959}\right).$$

You can verify normalization before touching a computer: the numerators sum to 959. Notice the distinction between a *weight*, which is merely positive, and a *probability*, whose whole list must sum to one.

Ratios cancel shared scales. If q_i and q_j are both multiplied by the same positive number, q_j/q_i does not change. Similarly, the ratio of two amplitudes is independent of a shared normalization. This is why some intermediate choices are conventions rather than physical assumptions. However, multiplying all energies by a number without compensating the temperature generally changes the physics; not every rescaling is harmless.

Inequalities carry direction. For $x < y$ and $c > 0$, $cx < cy$. For $c < 0$, $cx > cy$. A frequent mistake later is to multiply an inequality by a negative quantity and forget to reverse it. Another is to divide by a number before proving it is nonzero. The research proofs pay particular attention to zero cases for this reason.

For example, $-0.094 < -0.093$, but $|-0.094| > |-0.093|$. Absolute value measures distance from zero and discards the sign. Our slow overlaps in the primary cooling example are negative. A more negative overlap means a larger amplitude, not a smaller one. We will compare absolute values explicitly.

Percentages need a denominator. Reducing a quantity from 100 to 75 is a 25% reduction relative to 100: $(100 - 75)/100 = 1/4$. The reverse increase is $25/75 = 1/3$, not 25%. When we define amplitude advantage as $1 - |b_h|/|b_c|$, the denominator is the colder preparation's amplitude. A different denominator would define a different percentage.

Checkpoint. If $x \in [1.001, 1.002]$ and $y \in [1.005, 1.006]$, then $y - x \geq 0.003 > 0$. This proves an ordering even though neither value is known exactly. If the intervals overlap, the same argument may be inconclusive. Inconclusive does not mean false.

Lesson 3. A state is a list of probabilities

A finite-state model begins by choosing mutually exclusive alternatives. A single system occupies one alternative at a time. The probability p_i is the chance of finding it in state i, or the limiting fraction of a large ensemble found there. An ensemble means many independently prepared copies. The list

$$p = (p_1, \ldots, p_n)^\top, \qquad p_i \geq 0, \qquad \sum_{i=1}^{n} p_i = 1$$

is a probability vector. The superscript $\top$ means "transpose": it turns a horizontal list into a column. The symbol $\sum$ means add the indicated terms. Thus $\sum_{i=1}^{3} p_i = p_1 + p_2 + p_3$.

Why a distribution contains more information than a mean. Take three energy levels $E = (0, 1, 2)$. The probability vectors

$$p = (1/2, 0, 1/2)^\top, \qquad r = (0, 1, 0)^\top$$

both have mean energy 1. Nevertheless they are very different distributions: one has no systems in the middle state, while the other has all systems there. A statement about mean energy alone cannot in general determine a statement about distance between full distributions. This distinction matters when comparing different definitions of the Mpemba effect in the literature.

The geometry is simple before it is high-dimensional. For two states, $p_2 = 1 - p_1$, so one number specifies the distribution. The allowed points form a line segment. For three states, $p_3 = 1 - p_1 - p_2$ with $p_1, p_2 \geq 0$ and $p_1 + p_2 \leq 1$; the allowed points form a filled triangle. For four states they form a solid tetrahedron. In general this set is called a *simplex*. The name is less important than its two rules: nonnegative entries and total mass one.

Differences are not probabilities. If p and r are probability vectors, $z = p - r$ has entries summing to zero. Its entries may be negative. For example,

$$(0.6, 0.4)^\top - (0.75, 0.25)^\top = (-0.15, 0.15)^\top.$$

The negative first entry means a deficit relative to the reference, not a negative physical probability. Eigenvectors describing deviations from equilibrium likewise contain signs. Confusing a signed deviation with a probability vector makes much of spectral theory seem paradoxical.

What the model leaves out. A finite-state model may describe energy levels, conformations, coarse regions of a landscape, or another set of alternatives. Choosing these states is a modeling decision. Mathematical consistency of their probability evolution does not establish that they describe a particular experimental material. That additional identification requires physical evidence.

Checkpoint. Is $(0.4, 0.5, 0.2)$ a probability vector? No: it sums to 1.1. Is $(-0.1, 0.1, 0)$ a possible difference of probability vectors? Yes. Its total is zero, and appropriate reference probabilities make both original vectors nonnegative.

Lesson 4. Averages, spread, and covariance

An observable f assigns a number f_i to state i. Energy is one observable. The mean or expectation of f in distribution p is

$$\langle f\rangle_p = \sum_i p_i f_i.$$

This is a weighted average, not an unweighted average unless all probabilities are equal. For $p = (1/2, 1/3, 1/6)$ and $E = (0, 1, 2)$,

$$\langle E\rangle_p = \tfrac{1}{2}\,0 + \tfrac{1}{3}\,1 + \tfrac{1}{6}\,2 = \tfrac{2}{3}.$$

The brackets $\langle\cdot\rangle$ are notation for an average; they do not represent an extra operation beyond this sum.

Variance measures spread around the average. Define

$$\mathrm{Var}_p(f) = \sum_i p_i (f_i - \langle f\rangle_p)^2 = \langle f^2\rangle_p - \langle f\rangle_p^2.$$

To derive the second form, expand the square. The middle contribution is $-2\langle f\rangle_p \sum_i p_i f_i = -2\langle f\rangle_p^2$, and the last is $\langle f\rangle_p^2 \sum_i p_i = \langle f\rangle_p^2$. Their sum gives the stated identity. Because the first definition is a sum of nonnegative terms, variance cannot be negative.

In our example, $\langle E^2\rangle = 1/3 + 4/6 = 1$, so $\mathrm{Var}(E) = 1 - (2/3)^2 = 5/9$. A zero variance would mean that the observable has one value on every state with positive probability.

Covariance measures joint variation. For observables f and g,

$$\mathrm{Cov}_p(f, g) = \langle fg\rangle_p - \langle f\rangle_p \langle g\rangle_p.$$

Positive covariance means that larger-than-average values of one tend, in the weighted sum, to accompany larger-than-average values of the other. It is not a causal statement. The same observable paired with itself gives variance. A useful identity, obtained by expanding the products, is

$$\mathrm{Cov}_p(f, g) = \frac{1}{2}\sum_{i,j} p_i p_j (f_i - f_j)(g_i - g_j) = \sum_{i<j} p_i p_j (f_j - f_i)(g_j - g_i).$$

For the first equality, the two squared-index terms contribute $2\langle fg\rangle$, and the two cross terms contribute $-2\langle f\rangle\langle g\rangle$. The factor $1/2$ removes that two. The second equality groups the equal (i, j) and (j, i) contributions.

Why this identity will matter. If f and energy both increase with the state index, every pairwise product $(f_j - f_i)(E_j - E_i)$ is nonnegative. Their covariance is then nonnegative. No cancellation can change that sign. If f is not ordered, different pairs may compete. Later the observable f will be a slow eigenfunction. Its shape will determine whether varying the preparation temperature can reverse the usual amplitude order.

Checkpoint. Adding a constant c to every f_i leaves $\mathrm{Var}(f)$ and $\mathrm{Cov}(f, g)$ unchanged. This follows immediately from the difference form: $(f_j + c) - (f_i + c) = f_j - f_i$. Scaling f by c scales its covariance by c and its variance by c^2.

Lesson 5. Energy, temperature, and units

Energy levels are numbers E_i assigned to the states. We label them in increasing order,

$$E_1 < E_2 < \cdots < E_n.$$

The temperature T is positive. Statistical mechanics often uses inverse temperature

$$\beta = \frac{1}{k_B T},$$

where k_B is Boltzmann's constant. The product βE_i is dimensionless. An exponential of an energy measured in joules would be meaningless without such a conversion; the number placed in an exponential must carry no unit.

The inverse reverses the ordering. Higher temperature means lower β. With a colder bath at T_b, two cooling preparations satisfy

$$T_h > T_c > T_b \quad \Longleftrightarrow \quad 0 < \beta_h < \beta_c < \beta_b.$$

The subscripts h and c mean hotter and colder *relative to each other*. Both are hotter than the bath. In a heating experiment they can both be colder than the bath. It is worth writing the inverse-temperature ordering on paper before every sign argument.

Changing the energy zero changes no probabilities. Only energy differences matter for the Gibbs probabilities introduced next. If E_i is replaced by $E_i + C$, every exponential weight is multiplied by $e^{-\beta C}$. That common factor cancels in normalization. We can therefore choose $E_1 = 0$. This choice does not say that the ground state has no physical energy; it selects a reference level.

Choosing bath units. Let $\widetilde{E}_i = \beta_b(E_i - E_1)$ and $\widetilde{\beta} = \beta/\beta_b$. Then $\widetilde{\beta}\widetilde{E}_i = \beta(E_i - E_1)$, so the probabilities are unchanged. In these units the bath inverse temperature is 1. A preparation with $\widetilde{\beta} = 1/10$ is at ten times the bath's absolute temperature. Dropping the tildes simplifies notation, but does not turn inverse temperature into ordinary temperature.

Do not confuse three rescalings. Shifting all energies changes their zero. Scaling energy and inverse temperature reciprocally changes their units. Multiplying every transition rate by a positive factor changes the clock. These are distinct operations. Setting an energy gap to one and also setting the bath inverse temperature to one cannot generally be done as two independent harmless choices; their product contains physical information.

Checkpoint. If the bath has $\beta_b = 1$, then $\beta = 0.01$ is a hot preparation, not a cold one. The limiting value $\beta = 0$ represents infinite temperature. A proof at that endpoint alone is not yet a proof for two finite initial temperatures; a continuity argument or finite choices must complete it.

Lesson 6. Why exponentials and logarithms appear

The exponential e^x is the positive function with $e^0 = 1$ and derivative e^x. It obeys $e^{x+y} = e^x e^y$. One definition that also supports rigorous computation is

$$e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \cdots,$$

where $m! = 1 \cdot 2 \cdots m$, with $0! = 1$. For a real number x, the series converges. For negative x, $e^x = 1/e^{-x}$ is still positive. It never becomes negative just because the exponent is negative.

Repeated fractional change. A quantity that loses a fixed *fraction per unit time* satisfies $y'(t) = -ky(t)$ with $k > 0$. The prime means the instantaneous rate of change; Lesson 8 defines it precisely. Its solution is $y(t) = y(0)e^{-kt}$. Differentiating checks the equation directly. At time $1/k$ the quantity is multiplied by e^{-1}, about 0.368. The number $1/k$ is a characteristic decay time, not the time at which the quantity becomes exactly zero. An exponential with a nonzero initial value reaches zero only in the limit $t \to \infty$.

Logarithms undo exponentials. For $a > 0$, $\log a$ is the real number satisfying $e^{\log a} = a$. We use the natural logarithm throughout. Thus $\log(ab) = \log a + \log b$ and $\log(a/b) = \log a - \log b$. Solving $q = e^{-E}$ gives $E = -\log q$. For $0 < q < 1$, $\log q < 0$, so this energy is positive. Smaller weights correspond to larger energies.

The primary bath weights therefore give

$$E_1 = 0, \quad E_2 = \log(200/199), \quad E_3 = \log(5/2), \quad E_4 = \log 400.$$

Their order follows without decimal approximations because $1 < 200/199 < 5/2 < 400$ and the logarithm is increasing.

Real powers connect preparation to bath. With $q_i = e^{-E_i}$ in bath units,

$$e^{-\beta E_i} = e^{\beta \log q_i} = q_i^{\beta}.$$

For $\beta = 1/2$, this is a square root. For $\beta = 1/1000$, it is a positive thousandth root. A rational bath weight need not produce a rational hot preparation weight. That does not make exact certification impossible: rational intervals can enclose the positive root, with their validity checked by integer powers.

Checkpoint. The equation $e^{-kt} = a$ with $0 < a < 1$ gives $t = -\log(a)/k > 0$. Taking logarithms converts a multiplicative decay target into an additive time formula. This simple fact will later explain why an amplitude percentage and a time-saving percentage are different.

Lesson 7. Constructing a Gibbs distribution by hand

The thermal or Gibbs preparation at inverse temperature β is

$$\pi_i(\beta) = \frac{e^{-\beta E_i}}{Z(\beta)}, \qquad Z(\beta) = \sum_j e^{-\beta E_j}.$$

The symbol Z is called the partition function. At this stage, its job is simply to normalize positive weights. Every numerator is positive and the numerators sum to Z, so the probabilities are positive and sum to one.

Illustration with three states. Use $E = (0, \log 2, \log 4)$ and $\beta = 1$. The unnormalized weights are $(1, 1/2, 1/4)$, because $e^{-\log 2} = 1/2$. Their sum is $7/4$. Dividing gives

$$\pi(1) = (4/7, 2/7, 1/7)^\top.$$

At $\beta = 0$, every weight is 1, so $\pi(0) = (1/3, 1/3, 1/3)^\top$. At $\beta = 2$, the weights are squared, giving $(1, 1/4, 1/16)$ and hence $\pi(2) = (16/21, 4/21, 1/21)^\top$. Increasing inverse temperature concentrates probability in the low-energy states.

Ratios are even simpler than probabilities. Normalization cancels from a probability ratio:

$$\frac{\pi_j(\beta)}{\pi_i(\beta)} = e^{-\beta(E_j - E_i)}.$$

For the example, the ratio of the third probability to the first at $\beta = 1$ is $1/4$. At $\beta = 2$ it is $1/16$. This is how energy gaps control relative occupancy. The value of a single probability depends on every state through Z, while the ratio of two probabilities depends only on their energy gap.

The bath and the preparation have different jobs. We write $\pi_b = \pi(\beta_b)$ for the bath equilibrium. This fixed distribution helps determine the dynamics. We vary β to choose initial distributions $p(0) = \pi(\beta)$. Once the bath is fixed, varying the preparation does not change the matrix of transition rates. Comparing two preparations with two different matrices would be a different experiment.

What happens after time zero? Although $p(0)$ lies on the Gibbs curve $\beta \mapsto \pi(\beta)$, the evolving probability vector usually leaves that curve. One cannot generally assign a single instantaneous temperature to the entire vector. The master equation must evolve all probabilities, rather than just a scalar temperature.

Checkpoint. In the illustration, $\langle E \rangle_{\pi(1)} = (4/7)\log 2$, and $\mathrm{Var}_{\pi(1)}(E) = (26/49)(\log 2)^2$. To check the variance, first compute $\langle E^2 \rangle = (6/7)(\log 2)^2$ and subtract the square of the mean. The three-state illustration is a teaching example, not the four-state counterexample.

Lesson 8. Derivatives, local change, and Taylor expansion

For a function $f(x)$, the derivative is the limiting slope

$$f'(x) = \lim_{h \to 0} \frac{f(x+h) - f(x)}{h}.$$

A positive derivative means increasing values locally; a negative derivative means decreasing values locally. A zero derivative does not alone identify a maximum. The function x^3 has derivative zero at 0 but keeps increasing through it. The function x^2 has a minimum there. One needs signs on either side or additional information about higher derivatives.

The rules we use most often. The product rule is $(fg)' = f'g + fg'$. The quotient rule is $(f/g)' = (f'g - fg')/g^2$ when $g \neq 0$. The chain rule gives $(e^{u(x)})' = u'(x)e^{u(x)}$. These rules make the Gibbs derivative accessible. Let $r_i = e^{-\beta E_i}$. Then $r_i' = -E_i r_i$ and

$$Z' = \sum_i r_i' = -\sum_i E_i r_i = -Z\langle E \rangle_\pi.$$

Therefore

$$\begin{aligned} \pi_i' &= \frac{r_i' Z - r_i Z'}{Z^2} \\ &= \frac{-E_i r_i Z + r_i Z \langle E \rangle_\pi}{Z^2} \\ &= -(E_i - \langle E \rangle_\pi)\pi_i. \end{aligned}$$

Every line follows from substitution and cancellation. A state below the current mean energy gains probability as β increases; a state above it loses probability. Summing the last line gives zero, as required because the total probability remains one.

Taylor expansion is a controlled local approximation. Near a point x, a sufficiently smooth function has

$$f(x+h) = f(x) + hf'(x) + \tfrac{1}{2}h^2 f''(x) + \text{remainder}.$$

An approximation without a remainder estimate is useful intuition, but not a rigorous numerical enclosure. For example, $\log(1+u) = u - u^2/2 + O(u^3)$ as $u \to 0$. Here $O(u^3)$ means that the absolute remainder is bounded by a constant times $|u|^3$ sufficiently near zero. It does not mean the remainder is literally zero.

The notation $r(h) = o(h^2)$ is stronger than merely tending to zero: it means $r(h)/h^2 \to 0$. These symbols will let us separate the leading late-time term from faster-decaying errors. Their constants and domains matter whenever an asymptotic statement is turned into a finite numerical guarantee.

Checkpoint. At the three-state Gibbs distribution from Lesson 7, $\pi_1' = \langle E \rangle \pi_1 = (16/49)\log 2 > 0$. The ground-state population increases as the system is prepared colder. The derivative is with respect to inverse temperature, not time.

Lesson 9. The covariance identity that drives the mechanism

Keep an observable f_i fixed while varying the Gibbs preparation. Using Lesson 8,

$$\begin{aligned}\frac{d}{d\beta}\langle f\rangle_\pi &= \sum_i f_i \pi_i' \\ &= -\sum_i f_i E_i \pi_i + \langle E\rangle_\pi \sum_i f_i \pi_i \\ &= -\langle fE\rangle_\pi + \langle E\rangle_\pi \langle f\rangle_\pi \\ &= -\operatorname{Cov}_\pi(f, E).\end{aligned}$$

This identity translates a temperature derivative into a weighted correlation. It is exact, not a numerical approximation. The word "fixed" is essential: if f itself depended on β, an additional term $\sum_i \pi_i f_i'$ would appear.

First application: mean energy. Choose $f = E$. Then

$$\frac{d}{d\beta}\langle E\rangle_\pi = -\operatorname{Var}_\pi(E) < 0$$

for distinct energies and a finite Gibbs distribution. The strict sign follows because all probabilities are positive and not all energies are equal. Thus mean energy decreases when inverse temperature increases. This is the expected behavior of thermal preparations, derived directly from the formula.

Second application: a shaped observable. If $f_1 \le f_2 \le \cdots \le f_n$, the pairwise covariance formula makes $\operatorname{Cov}(f, E) \ge 0$. Its thermal average cannot increase with β. If the values of f bend up and down instead, pair contributions can have opposite signs. Temperature changes their weights. Their competition can therefore change the sign of the derivative.

This is the basic mathematical opening for an anomalous slow amplitude. The observable that matters is not simply energy; it is an eigenfunction of the dynamics. A hot preparation can have more average energy and nevertheless a smaller projection onto that particular eigenfunction. There is no contradiction because these are different weighted sums.

How to use the identity without overusing it. A positive covariance at one preparation gives a negative derivative there. It does not automatically describe every temperature. A zero covariance marks a stationary point of the average, but the derivative may touch zero without changing sign. To prove a reversal, we need either a sign change with the right amplitude orientation or two strictly ordered amplitude values. The exact counterexample uses the latter route.

Checkpoint. The bath determines the eigenfunction. Holding that bath and the generator fixed is precisely what allows the covariance identity to be used for the slow overlap. Differentiating through a family of changing baths would be a different calculation and would need the missing derivative terms.

Lesson 10. Transition rates and a complete two-state solution

A transition rate is a probability per unit time in the small-time limit. If the system is in state j, a rate a_{ij} means that the probability of jumping to a different state i during a short interval h is $a_{ij}h + o(h)$. The rate itself need not be less than one: it has units of inverse time, whereas the small-time jump probability is dimensionless. A rate of 3 per second is perfectly legitimate.

Destination comes first. Throughout this report a_{ij} means a jump *from j to i*. For ordered energies, a_{12} is a downward transition from state 2 to state 1. This convention is easy to reverse accidentally. Read a matrix column as the possible destinations from its starting state.

A model we can solve without matrix machinery. Let the upward rate from state 1 to 2 be 1 and the downward rate from 2 to 1 be 3. Incoming flow to state 2 is $1p_1$; outgoing flow is $3p_2$. Thus

$$p_2' = p_1 - 3p_2 = (1 - p_2) - 3p_2 = 1 - 4p_2.$$

The equilibrium is found by setting the derivative to zero: $p_{2b} = 1/4$ and $p_{1b} = 3/4$. Define $z = p_2 - 1/4$. Its derivative is $z' = p_2' = 1 - 4(z + 1/4) = -4z$, so Lesson 6 gives

$$p_2(t) = \frac{1}{4} + \left(p_2(0) - \frac{1}{4}\right) e^{-4t}, \qquad p_1(t) = 1 - p_2(t).$$

This is a complete solution for every initial probability vector.

Check the solution rather than trusting it. At $t = 0$ it gives the chosen initial value. Differentiating yields $-4(p_2(0) - 1/4)e^{-4t} = 1 - 4p_2(t)$, so it satisfies the equation. As $t \to \infty$, the exponential goes to zero and $p_2(t) \to 1/4$. For $t \geq 0$, the solution is a convex combination of $p_2(0)$ and $1/4$, meaning a weighted average with nonnegative weights that sum to one, so it remains between zero and one. Conservation, positivity, and relaxation have each been checked separately.

Why this model does not exhibit the desired reversal. For thermal preparations above this bath, the hotter preparation has the larger state-2 population. Its initial deviation from $1/4$ is therefore larger. Both deviations are multiplied by the same positive e^{-4t}. Their ordering cannot reverse. A single scalar decay leaves no room for competition among differently decaying components. Extra states create additional components, although extra states alone do not guarantee a reversal.

Lesson 11. Matrices are bookkeeping for many flows

A matrix is a rectangular array of numbers. Multiplying a matrix by a column vector means taking one row at a time and adding entrywise products. Our two-state model becomes

$$\frac{d}{dt}\begin{pmatrix} p_1 \\ p_2 \end{pmatrix} = \underbrace{\begin{pmatrix} -1 & 3 \\ 1 & -3 \end{pmatrix}}_{M} \begin{pmatrix} p_1 \\ p_2 \end{pmatrix}.$$

The first row says $p_1' = -p_1 + 3p_2$; the second says $p_2' = p_1 - 3p_2$. Nothing new has been assumed by writing the two equations compactly.

The master equation for n states. Set $M_{ij} = a_{ij} \geq 0$ for $i \neq j$, and put the negative total exit rate on the diagonal:

$$M_{jj} = -\sum_{i \neq j} a_{ij}.$$

Then $p' = Mp$ means

$$p_i' = \sum_{j \neq i} a_{ij} p_j - \left(\sum_{j \neq i} a_{ji} \right) p_i.$$

The first sum is incoming probability per time, and the second is outgoing probability per time. Notice the reversed indices in the outgoing rate. A diagonal entry is not a negative jump probability. It records the loss from that state.

Conservation comes from columns. Every column of M sums to zero. Therefore

$$\frac{d}{dt} \sum_i p_i = \sum_i \sum_j M_{ij} p_j = \sum_j \left(\sum_i M_{ij} \right) p_j = 0.$$

If the initial total is one, it stays one. Row sums need not vanish under this column-vector convention. Accidentally imposing row conservation would describe a transposed convention rather than the stated model.

Why nonnegative probabilities remain nonnegative. Choose r at least as large as every exit rate, with $r > 0$, and define $P = I + M/r$. Here I is the identity matrix, with ones on its diagonal and zeros elsewhere. Every entry of P is nonnegative and every column sums to one. Thus P maps probability vectors to probability vectors. The exact solution can be written

$$e^{Mt} = e^{-rt} \sum_{k=0}^{\infty} \frac{(rt)^k}{k!} P^k.$$

The coefficients are nonnegative and sum to one. This expresses the solution as a weighted average of probability-preserving maps, establishing positivity. The matrix exponential is defined by the same power series as the scalar exponential, with matrix powers replacing scalar powers.

Checkpoint. An Euler step $p(t+h) \approx (I + hM)p(t)$ can give negative probabilities if h is too large. That is a failure of the numerical step, not of the exact continuous-time process. The series above also becomes a useful certificate method because its omitted tail can be bounded explicitly.

Lesson 12. Equilibrium, stationarity, and detailed balance

A stationary distribution π_b satisfies $M\pi_b = 0$. Starting there gives $p(t) = \pi_b$ for all time. Stationarity says that total incoming and outgoing flow cancel at each state. A stronger condition, detailed balance, says that each pair balances separately:

$$M_{ij}\pi_{bj} = M_{ji}\pi_{bi} \qquad (i \neq j).$$

The two sides are equilibrium probability currents in opposite directions. Their equality makes the dynamics reversible in the probabilistic sense. It does not mean that a nonequilibrium preparation fails to relax.

Detailed balance implies stationarity. For a fixed state i,

$$\begin{aligned}(M\pi_b)_i &= \sum_{j\neq i} M_{ij}\pi_{bj} + M_{ii}\pi_{bi} \\ &= \sum_{j\neq i} M_{ji}\pi_{bi} - \sum_{j\neq i} M_{ji}\pi_{bi} = 0.\end{aligned}$$

The first equality uses matrix multiplication; the second uses detailed balance and the diagonal definition. Every cancellation is visible.

Constructing reversible rates from weights. Since $\pi_{bi} = q_i/Q$, detailed balance is equivalent to $a_{ij}q_j = a_{ji}q_i$. Choose a downward rate $a_{ij} > 0$ for $i < j$ and set

$$a_{ji} = a_{ij}\frac{q_j}{q_i}.$$

All weights are positive, so the upward rate is positive. If $q_j < q_i$, the upward rate is smaller than its downward partner. Choosing the diagonal last ensures conservation. This construction makes the desired bath stationary automatically.

An explicit check. For the two-state example, take $q = (1, 1/3)$ and downward rate $a_{12} = 3$. Then $a_{21} = 3(1/3) = 1$, as before. Normalization gives $\pi_b = (3/4, 1/4)$. The flow from 2 to 1 is $3(1/4) = 3/4$; the flow from 1 to 2 is $1(3/4) = 3/4$. Individual transitions continue at equilibrium even though the probability distribution does not change.

Connectivity matters. If every off-diagonal rate is positive, any state can reach any other. There is then one stationary distribution and relaxation to it. More sparse models can also have this property when their transition graph is connected appropriately. If the graph splits into disconnected components, there can be several stationary directions. Some proof steps survive that boundary and others require separate treatment; positivity assumptions are therefore more than decorative wording.

Lesson 13. Eigenvectors separate the decay patterns

An eigenvector $v \neq 0$ of M satisfies $Mv = \lambda v$. The matrix changes it by the factor λ while keeping it on the same line. A negative factor reverses its sense along that line. The number λ is an eigenvalue. For the differential equation, a component in that direction evolves as $e^{\lambda t}v$, because

$$\frac{d}{dt}(e^{\lambda t}v) = \lambda e^{\lambda t}v = M(e^{\lambda t}v).$$

An eigenvalue is therefore a decay rate in the dynamics, not a probability.

The two-state eigenvectors can be checked by multiplication. For $M = \left(\begin{smallmatrix} -1 & 3 \\ 1 & -3 \end{smallmatrix}\right)$,

$$M\begin{pmatrix} 3/4 \\ 1/4 \end{pmatrix} = 0, \qquad M\begin{pmatrix} 1 \\ -1 \end{pmatrix} = \begin{pmatrix} -4 \\ 4 \end{pmatrix} = -4\begin{pmatrix} 1 \\ -1 \end{pmatrix}.$$

The stationary mode has eigenvalue zero. The deviation mode has eigenvalue -4. Its entries sum to zero, so adding a small multiple to the equilibrium preserves total probability.

Several modes give several clocks. For a reversible connected four-state model, the eigenvalues can be ordered

$$0 = \lambda_1 > \lambda_2 \geq \lambda_3 \geq \lambda_4.$$

The nonzero values are negative. If $\lambda_2 > \lambda_3$, the slowest nonstationary mode is simple, meaning one-dimensional. Its exponential decays less rapidly than the others: e^{-t} outlasts e^{-10t}. At late times, even a small coefficient of the slower exponential can dominate.

A two-exponential illustration. Consider the positive functions $f_h(t) = e^{-t} + 9e^{-10t}$ and $f_c(t) = 2e^{-t} + e^{-10t}$. At $t = 0$, $f_h = 10 > 3 = f_c$. Their difference is $-e^{-t} + 8e^{-10t}$, which is zero exactly when $e^{9t} = 8$. Thus they cross at $t = (\log 8)/9$, and $f_h < f_c$ thereafter. This is an illustration of competing decay coefficients, not a complete thermal Markov counterexample. Its role is to make the mechanism imaginable: more initial total deviation can coexist with less slow deviation.

Checkpoint. The slow eigenvalue is the negative one closest to zero, not the most negative one. A very negative eigenvalue gives a very fast decay. A simple slow mode permits one scalar amplitude; a repeated slow eigenvalue requires a whole projection, discussed in Lesson 26.

Lesson 14. Left eigenvectors measure the mode amplitudes

A right eigenvector describes a direction of change in the probability vector. A left eigenvector is a row vector satisfying $w^\top M = \lambda w^\top$, equivalently $M^\top w = \lambda w$. Multiplying the state by this row gives a scalar mode coordinate. Indeed,

$$\frac{d}{dt}(w^\top p) = w^\top M p = \lambda w^\top p.$$

Thus $w^\top p(t) = e^{\lambda t} w^\top p(0)$ exactly.

Why the bath disappears from the overlap. For a nonzero eigenvalue,

$$0 = w^\top M \pi_b = \lambda w^\top \pi_b.$$

Dividing by $\lambda \neq 0$ gives $w^\top \pi_b = 0$. Consequently the coefficient can be computed as either $w^\top [p(0) - \pi_b]$ or $w^\top p(0)$. For a Gibbs preparation we call it

$$b(\beta) = w^\top \pi(\beta) = \sum_i w_i \pi_i(\beta).$$

It is a weighted average of the values of the eigenfunction w.

Normalization prevents a hidden factor. Choose a corresponding right eigenvector v such that $w^\top v = 1$. Then $b(\beta)v$ is the actual vector component in that mode. If w is multiplied by $c \neq 0$, b is multiplied by c and v must be divided by c. The physical product does not change. Ratios $|b_h|/|b_c|$ are therefore independent of this common normalization.

Return to the complete two-state solution. Take $w = (1, -3)^\top$. Direct multiplication gives $w^\top M = -4w^\top$. With $v = (1, -1)^\top/4$, $w^\top v = (1+3)/4 = 1$. The initial overlap is $b = p_1 - 3p_2 = 1 - 4p_2$. Therefore

$$p(t) = \pi_b + e^{-4t}(1 - 4p_2(0))\frac{(1, -1)^\top}{4}.$$

Its second component equals the scalar formula of Lesson 10. The abstract left/right distinction has recovered the elementary answer exactly.

Covariance now acquires its physical role. Because w is fixed when the preparation varies,

$$b'(\beta) = -\operatorname{Cov}_{\pi(\beta)}(w, E).$$

This is the formula derived in Lesson 9 with $f = w$. The thermal curve and the dynamical mode meet in a single weighted covariance. The rates choose w; the energies and preparation temperature choose the weights in the covariance. The effect depends on both pieces.

Lesson 15. Reversibility turns a difficult matrix into a symmetric one

A real symmetric matrix equals its transpose. Such matrices have real eigenvalues and an orthonormal basis of eigenvectors: mutually perpendicular unit-length vectors in which every vector has a unique expansion. Our probability generator is generally not symmetric in the ordinary coordinates, because upward and downward rates differ. Detailed balance provides the right change of coordinates.

Let $D = \mathrm{diag}(\pi_b)$ be the diagonal matrix of bath probabilities. Its square root has entries $\sqrt{\pi_{bi}}$, and its inverse square root has entries $1/\sqrt{\pi_{bi}}$. All are defined because the probabilities are positive. Set

$$S = D^{-1/2} M D^{1/2}.$$

Detailed balance says $MD = DM^\top$, and therefore

$$S^\top = D^{1/2} M^\top D^{-1/2} = D^{-1/2} M D^{1/2} = S.$$

This change of coordinates is called a similarity transformation. It leaves eigenvalues unchanged: if $Mv = \lambda v$, then $S(D^{-1/2}v) = \lambda(D^{-1/2}v)$.

Why decay rates cannot be positive. Define equilibrium conductances $c_{ij} = M_{ij}\pi_{bj} = c_{ji}$ for $i \neq j$. Writing the generator on observables as $L = M^\top$, direct expansion gives

$$-f^\top D L f = \sum_{i<j} c_{ij}(f_i - f_j)^2 \geq 0.$$

For an eigenfunction $Lf = \lambda f$, the left side equals $-\lambda \sum_i \pi_{bi} f_i^2$. The weighted squared length is positive when $f \neq 0$, so $\lambda \leq 0$. If all conductances are positive, equality requires all values f_i to coincide. Only the constant observable has zero eigenvalue, corresponding to the unique stationary probability mode.

The right inner product. For observables use $\langle f, g\rangle_D = \sum_i \pi_{bi} f_i g_i$. For probability deviations use $\langle z, y\rangle_{D^{-1}} = \sum_i z_i y_i / \pi_{bi}$. These weighted inner products are the natural ones for detailed balance. The different weights arise because a probability deviation and an observable are dual objects. If $Lw = \lambda w$, then Dw is a right eigenvector of M.

Advanced consequence, in plain terms. The spectral analysis is not relying on a lucky numerical property of one matrix. Reversibility supplies a real diagonalization throughout the stated domain. That removes oscillatory complex modes and nontrivial Jordan blocks from this problem. It does not remove the need to check whether the slow eigenvalue is simple or repeated.

Lesson 16. What does it mean to be closer to equilibrium?

To compare whole probability vectors we need a specified distance or divergence. Different choices emphasize different discrepancies. Three used in this project are

$$\mathrm{TV}(p,\pi_b)=\frac{1}{2}\sum_i |p_i-\pi_{bi}|,$$
$$\chi^2(p\|\pi_b)=\sum_i \frac{(p_i-\pi_{bi})^2}{\pi_{bi}},$$
$$D_{\mathrm{KL}}(p\|\pi_b)=\sum_i p_i\log\frac{p_i}{\pi_{bi}}.$$

A norm assigns a length to a vector: it is positive except at zero, scales by the absolute value of a scalar multiplier, and obeys the triangle inequality $\|x+y\|\le\|x\|+\|y\|$. Euclidean length $\sqrt{\sum_i x_i^2}$ is one example. A metric measures distance between two points and also requires symmetry and the triangle inequality. Total variation (TV) is a metric. Pearson chi-square and relative entropy (KL) are divergences; they need not satisfy every metric axiom. We use the word "distance" informally only when the specified formula is clear. For KL, a zero p_i contributes zero by a limit. The bath entries are positive.

A hand calculation. Take $\pi_b=(3/4,1/4)$ and $p=(1/2,1/2)$. The deviations are $(-1/4,1/4)$, so

$$\mathrm{TV}=\tfrac{1}{2}(\tfrac{1}{4}+\tfrac{1}{4})=\tfrac{1}{4},\qquad \chi^2=\frac{1/16}{3/4}+\frac{1/16}{1/4}=\tfrac{1}{3}.$$

The KL value is $\frac{1}{2}\log(2/3)+\frac{1}{2}\log 2=\frac{1}{2}\log(4/3)$. The numerical values differ because the measurements differ. Comparing a TV number with a KL number as though they used the same scale would be meaningless.

Why KL is approximately quadratic near equilibrium. Write $p_i=\pi_{bi}(1+u_i)$, where $\sum_i \pi_{bi}u_i=0$. For small u,

$$(1+u)\log(1+u)=u+\tfrac{1}{2}u^2+O(u^3).$$

Multiplying by π_{bi} and summing cancels the linear term. Hence

$$D_{\mathrm{KL}}(p\|\pi_b)=\tfrac{1}{2}\sum_i \pi_{bi}u_i^2+O\left(\sum_i \pi_{bi}|u_i|^3\right)=\tfrac{1}{2}\chi^2(p\|\pi_b)+\text{higher-order terms}.$$

This is a local expansion around the bath, not an identity at arbitrary probabilities. Rare bath states have large reciprocal weights in chi-square, which is one reason finite-time comparisons require care.

Initially hotter is initially farther in KL. For a Gibbs preparation,

$$D_{\mathrm{KL}}(\pi(\beta)\|\pi_b)=(\beta_b-\beta)\langle E\rangle_{\pi(\beta)}+\log Z(\beta_b)-\log Z(\beta).$$

Differentiate, use $\langle E\rangle'=-\operatorname{Var}(E)$ and $(\log Z)'=-\langle E\rangle$, and the two mean-energy terms cancel:

$$\frac{d}{d\beta}D_{\mathrm{KL}}=(\beta-\beta_b)\operatorname{Var}(E).$$

On the cooling side this derivative is negative. Thus increasing β toward the bath decreases the initial KL divergence, and the hotter preparation starts farther away. This supplies the initial half of a genuine initial-to-eventual reversal.

Lesson 17. Turning a slow-overlap inequality into a reversal

Assume the slow nonzero eigenvalue is simple, so $\lambda_2 > \lambda_3$. Decompose the initial deviation into eigenmodes. At late times,

$$p(t;\beta) - \pi_b = e^{\lambda_2 t} b(\beta) v_2 + R(t;\beta), \qquad R(t;\beta) = O(e^{\lambda_3 t}).$$

After division by $e^{\lambda_2 t}$, the remainder tends to zero because $e^{(\lambda_3 - \lambda_2)t} \to 0$. This is why the slow coefficient determines the eventual order.

For any fixed norm, continuity and homogeneity give

$$\lim_{t\to\infty} e^{-\lambda_2 t} \|p(t;\beta) - \pi_b\| = |b(\beta)| \|v_2\|.$$

If $|b_h| < |b_c|$, the two limiting values have a strictly positive gap. For sufficiently late finite times, the errors are smaller than a fraction of that gap, so the same strict ordering holds. This is an existence argument for a finite late-time interval, not merely an assertion at $t = \infty$.

For KL the coefficient is squared. Combining the spectral expansion with Lesson 16 gives

$$D_{\mathrm{KL}}(p(t;\beta)\|\pi_b) = \frac{1}{2} e^{2\lambda_2 t} b(\beta)^2 \sum_i \frac{v_{2i}^2}{\pi_{bi}} + o(e^{2\lambda_2 t}).$$

The positive factor involving v_2 is shared by both preparations. A smaller absolute overlap therefore gives smaller eventual KL divergence. Since the hotter Gibbs preparation starts farther in KL, continuity forces at least one crossing between the initial and eventual orderings.

Ordinary and strong effects. An ordinary effect can occur with both overlaps nonzero. A strong effect requires $b(\beta) = 0$ at a preparation different from the bath, eliminating the slow mode altogether. The bath itself always has zero overlap and does not count as a nontrivial strong preparation. The primary counterexample uses two nonzero overlaps; it does not need a strong effect.

Amplitude advantage is not time advantage. Define $A = 1 - |b_h|/|b_c|$. If $A = 1/4$, the slow amplitude ratio is $3/4$, and the asymptotic KL ratio is $9/16$. Neither number is automatically a 25% reduction in target time. In a pure one-mode KL approximation with $\lambda_2 = -r < 0$, the time to a small threshold d is

$$t(\beta;d) \approx \frac{1}{2r} \log \frac{K b(\beta)^2}{d}, \qquad t_c - t_h \approx \frac{1}{r} \log \frac{|b_c|}{|b_h|}.$$

The difference is approximately additive in time. Its fraction of t_c depends on the threshold and tends to zero as the threshold tends to zero with fixed nonzero overlaps. Finite-time claims require their own proof.

Lesson 18. Assemble the exact four-state witness

We now have the ingredients to read actual research data. The primary witness uses the weights from Lesson 2, bath inverse temperature 1, and six downward rates

$$(a_{12}, a_{13}, a_{14}, a_{23}, a_{24}, a_{34}) = \left(1, 28, \frac{249}{10}, \frac{1799}{8}, \frac{4983}{200}, \frac{251}{25}\right).$$

These are inputs, not rounded numerical estimates. The reverse rates are computed from detailed balance, and the diagonal entries from conservation.

Build one column completely. The three exits from state 1 are upward transitions:

$$\begin{aligned} a_{21} &= a_{12}\frac{q_2}{q_1} = \frac{199}{200}, \\ a_{31} &= a_{13}\frac{q_3}{q_1} = 28\frac{2}{5} = \frac{56}{5}, \\ a_{41} &= a_{14}\frac{q_4}{q_1} = \frac{249}{10}\frac{1}{400} = \frac{249}{4000}. \end{aligned}$$

Their sum is $(3980 + 44800 + 249)/4000 = 49029/4000$. The first column is therefore

$$\left(-\frac{49029}{4000}, \frac{199}{200}, \frac{56}{5}, \frac{249}{4000}\right)^{\top}.$$

It sums to zero exactly. Repeating this operation for the other states gives the matrix printed in Part II. You do not have to infer its meaning from a wall of fractions: every positive entry is a jump rate, and every negative diagonal entry is the negative sum of exits.

Check one less obvious reverse rate. For the pair $2, 3$,

$$a_{32} = \frac{1799}{8}\frac{2/5}{199/200} = \frac{1799}{8}\frac{80}{199} = \frac{17990}{199}.$$

It follows that $a_{23}q_3 = a_{32}q_2$ exactly. Such checks expose a reversed ratio immediately; using q_2/q_3 would produce the wrong bath.

The two preparations. Take $\beta_h = 1/1000$ and $\beta_c = 1/100$. Both are positive and smaller than the bath value 1, so both are finite cooling preparations. For each,

$$\pi_i(\beta) = \frac{q_i^{\beta}}{\sum_j q_j^{\beta}}.$$

The certified slow eigenfunction, normalized as in Part II, gives

$$-0.0937454280 < b_h < -0.0937454279, \qquad -0.0938488357 < b_c < -0.0938488356.$$

Therefore

$$|b_c| - |b_h| > 0.0938488356 - 0.0937454280 = 0.0001034076 > 0.$$

A stricter enclosure is recorded in the certificate. The gap is small but strict. A small counterexample still refutes a universal implication.

Two witnesses, two sets of data. The focused paper uses a companion with rational slow eigenvalue and rational eigenfunction. It changes two rates. Its formal proof and its numerical values must not be silently substituted for the primary witness above. The shared mechanism makes them related examples, not identical data.

Lesson 19. Work through a triplet test instead of treating it as a black box

A four-state system has four three-state subsets: 123, 124, 134, and 234. For an ordered triplet $i < j < k$, define

$$A = a_{ik}, \quad B = a_{jk}, \quad C = a_{ij}, \quad t = q_j/q_i, \quad u = q_k/q_j.$$

The letters are local abbreviations for this triplet. They do not introduce new rates or fit effective ones. The original-rate predicate in Part II is

$$\kappa = \frac{A+B}{C(1+t)}, \qquad G = At - B,$$
$$F = A(1+tu/2) - B(1+u/2) - C(1-t),$$
$$P = (\kappa \geq 1 \text{ and } F > 0) \text{ or } (\kappa \leq 1 \text{ and } G > 0).$$

A predicate is simply a statement that is either true or false for the specified data. The strict and non-strict inequality signs are part of the definition. At $\kappa = 1$, both branches must be considered.

Compute triplet 134 by hand. Here $A = 249/10$, $B = 251/25$, $C = 28$, $t = 2/5$, and $u = 1/160$. First,

$$G = \frac{249}{10}\frac{2}{5} - \frac{251}{25} = \frac{249 - 251}{25} = -\frac{2}{25} < 0.$$

Next,

$$\kappa = \frac{249/10 + 251/25}{28(1+2/5)} = \frac{1747/50}{196/5} = \frac{1747}{1960} < 1.$$

The applicable lower branch therefore fails because G is negative. For completeness, use $tu = 1/400$ to calculate

$$\begin{aligned} F &= \frac{249}{10}\frac{801}{800} - \frac{251}{25}\frac{321}{320} - 28\frac{3}{5} \\ &= \frac{199449 - 80571 - 134400}{8000} = -\frac{7761}{4000} < 0. \end{aligned}$$

The other three triplets are checked in the same manner in the exact table. One failed triplet is not enough; the premise requires all four to fail.

Why isolated triplets are not principal submatrices. If a state is removed, its outgoing transitions from the retained states must also be removed. The diagonal entries must be recomputed so the three-state columns sum to zero. Merely deleting one row and column leaves probability leaking to a state no longer in the model. This distinction is separate from the predicate's use of original off-diagonal rates.

Lesson 20. Simplify the branch logic, including the equality case

Complicated-looking logical tests can sometimes be converted into a small set of inequalities. Define

$$H = A + B - C(1 + t), \qquad d = \frac{1 - t}{1 + t}, \qquad h = \frac{2}{1 + t} + \frac{u}{2}.$$

Because energies are distinct and ordered, $0 < t < 1$. Thus $d > 0$ and $h > 0$. Also $C(1 + t) > 0$, so the sign of $\kappa - 1$ is the sign of H. A direct expansion gives

$$F = hG + dH.$$

To verify it, the coefficients of A, B, and C on the right are, respectively,

$$\frac{2t + 1 - t}{1 + t} + \frac{tu}{2} = 1 + \frac{tu}{2}, \quad \frac{-2 + 1 - t}{1 + t} - \frac{u}{2} = -1 - \frac{u}{2}, \quad -(1 - t).$$

These are exactly the coefficients in F.

Now handle the three cases. If $H > 0$, only the upper branch can establish P. Its failure means $F \leq 0$. Since $F = hG + dH$ with $dH > 0$, this forces $G < 0$ as well. If $H < 0$, failure means $G \leq 0$, and then $F = hG + dH < 0$. If $H = 0$, both branches are eligible, but $F = hG$ with $h > 0$; the strict sign tests agree. Their joint failure means $F \leq 0$ and $G \leq 0$. Conversely, if both are nonpositive, neither strict branch test can succeed. Therefore, including equality cases,

$$\neg P \quad \Longleftrightarrow \quad F \leq 0 \text{ and } G \leq 0.$$

The symbol $\neg$ means "not."

The geometric payoff. For fixed weights, F and G are linear functions of the rates. A condition like $F \leq 0$ describes a half-space. Intersecting the eight half-spaces from four triplets gives a polyhedral cone: a set closed under addition and multiplication by nonnegative numbers. Intersect it with strictly positive rates to obtain the physical domain considered here. The word "cone" does not assume a three-dimensional picture; the rate space has six coordinates.

This reformulation helps exact searches and structural proofs. It does not replace the original predicate with a convenient approximation. The equality above proves equivalence. Boundary cases remain part of the mathematical problem even when the displayed witness lies strictly inside the domain.

Lesson 21. Certify a spectral calculation rather than trust its decimal output

A numerical eigensolver is excellent for finding candidate eigenvalues and vectors. A certificate must additionally establish what those approximations represent. Is the selected eigenvalue truly the slow one? Is it simple? Is the overlap gap larger than all numerical uncertainty? The primary witness answers these questions with exact algebra and rational enclosures.

The characteristic polynomial. The determinant $\det(M - xI)$ is a polynomial in x. Its zeros are the eigenvalues. A determinant tests whether the columns are linearly dependent; if $\det(M - xI) = 0$, there is a nonzero vector v with $(M - xI)v = 0$. Because M is rational, its characteristic polynomial has rational coefficients. The stationary eigenvalue supplies a factor x, leaving a cubic. After clearing denominators, the primary witness's cubic is

$$1273600000000x^3 + 530473008000000x^2 + 35732953114072400x + 510539126345027563.$$

Its three roots lie separately in $(-20, -19)$, $(-60, -59)$, and $(-337, -336)$. Exact sign checks at the endpoints establish a root in each interval. A cubic has only three roots counting multiplicity, so these three distinct intervals account for all roots and establish simplicity. The root in $(-20, -19)$ is the slow nonzero one.

An eigenvector from cofactors. For any square matrix A, its adjugate satisfies $A\,\mathrm{adj}(A) = \det(A)I$. At a simple eigenvalue x, a nonzero column of $\mathrm{adj}(M^\top - xI)$ therefore lies in the nullspace of $M^\top - xI$. Its entries are polynomials in x. The certificate verifies that the chosen column is nonzero and normalizes it without dividing by an interval containing zero. This avoids treating a rounded eigenvector as exact.

Thermal roots can be enclosed by integer powers. Suppose $r = q^{1/m} > 0$, where q is positive rational and m is a positive integer. If positive rational numbers $l < u$ satisfy $l^m < q < u^m$, then $l < r < u$ because positive integer powers are strictly increasing. All comparisons can be reduced to integer arithmetic by clearing denominators. This is how the small preparation exponents can be handled rigorously.

Finally propagate the enclosures. Enclose the eigenvalue, evaluate the cofactor polynomials with interval arithmetic, enclose each Gibbs weight, normalize by a strictly positive partition-function interval, and sum the products. The resulting overlap intervals must be disjoint in the required absolute-value order. If they are not, increase precision or use a sharper method. A printed decimal gap by itself is not the certificate; the outward bounds and their checked construction are.

Lesson 22. Why three-state checks can miss four-state dynamics

A triplet test examines a restricted collection of rates. The slow eigenfunction of the full four-state generator solves a simultaneous system involving every state. Removing one state changes both the equations and the probability-conserving diagonals. There is no general reason for the full-system eigenfunction to be a collection of independent triplet eigenfunctions pasted together.

A linear-algebra analogy. Knowing several smaller blocks of a matrix does not allow one to assign the whole matrix's eigenvectors independently within those blocks. The cross-block entries couple the coordinates. Similarly, a four-state mode can distribute positive and negative weights among all four states so that thermal reweighting produces a collective cancellation absent from any isolated triplet. This explains why a local-to-global implication needs a proof; it is not automatically supplied by the small subsystems.

What the counterexample does and does not say. It demonstrates failure of the stated universal finite-temperature inference. It does not invalidate a distinct theorem with a zero-temperature bath or stronger hypotheses. It also does not say that triplet information is useless. The rest of the report derives substantial restrictions that survive all those failed triplet tests. Ordinary cooling can survive while strong effects and certain heating reversals remain excluded.

Temperature changes a weighted cancellation. The slow overlap is

$$b(\beta) = \frac{w_1 + w_2 e^{-\beta E_2} + w_3 e^{-\beta E_3} + w_4 e^{-\beta E_4}}{Z(\beta)}$$

when $E_1 = 0$. The entries of w need not all have the same sign, because its bath mean is zero. Varying β changes both the signed numerator and its normalization. An intuitive story that only watches the highest energy population misses the redistribution among the other three states. The derivative is governed by the full covariance from Lesson 9.

The controlled-family idea. A single exact witness proves existence. A parameterized family is more informative about mechanism: vary one parameter while retaining admissible rates and triplet failures, then locate where the covariance criterion changes sign. Part II supplies such a family with an exact onset threshold. The family is a way to explain how the effect turns on, not a license to assume that every nearby matrix has the same properties without checking.

Checkpoint. The logical conclusion is narrow and strong: there exists a valid four-state example satisfying the full premise and violating the claimed conclusion. Universal claims fail that way. Claims about prevalence, experimental ease, or maximum possible benefit require additional work.

Lesson 23. Prefix contrasts, cones, and ordered dynamics

To understand what the triplet restrictions still control, it helps to change basis in the space of observables. In this lesson let $q = \pi_b$ be normalized; only ratios matter in the formulas. Define $Q_k = q_1 + \cdots + q_k$ and a prefix contrast

$$h_i^{(k)} = \begin{cases} 1, & i \le k, \\ -Q_k/q_{k+1}, & i = k+1, \\ 0, & i > k+1. \end{cases}$$

It contrasts the first k states against the next state. Its weighted bath mean is $Q_k - q_{k+1}(Q_k/q_{k+1}) = 0$. For example, with four states,

$$h^{(1)} = (1, -q_1/q_2, 0, 0)^\top, \quad h^{(2)} = (1, 1, -Q_2/q_3, 0)^\top.$$

These are signed observables, not probability vectors.

Why these particular vectors are useful. If $j < k$, the vector $h^{(k)}$ equals one wherever $h^{(j)}$ is nonzero. Thus their weighted inner product is the bath mean of $h^{(j)}$, which is zero. They form an orthogonal basis of the bath-mean-zero observables. In this basis, certain rate inequalities become entrywise sign conditions on a smaller matrix. A matrix with nonnegative off-diagonal entries is called a Metzler matrix. Adding a sufficiently large scalar multiple of the identity turns it into a nonnegative matrix without changing its eigenvectors. The Perron argument then supplies a nonnegative eigenvector for its largest eigenvalue. Part II checks the needed hypotheses and boundary cases instead of assuming them from the name of the theorem.

From the triplet inequality to conductance order. Define reduced conductances $k_{ij} = M_{ij}/q_i = M_{ji}/q_j$. Then

$$G_{ijk} = a_{ik}\frac{q_j}{q_i} - a_{jk} = q_j(k_{ik} - k_{jk}).$$

Because $q_j > 0$, the condition $G_{ijk} \le 0$ says exactly $k_{ik} \le k_{jk}$. The contrast representation exploits this ordering. For a simple slow mode it allows an orientation $w = \sum_k x_k h^{(k)}$ with $x_k \ge 0$, not all zero.

Why this excludes a strong cooling zero. Consider one prefix contrast at a hotter preparation $\beta < \beta_b$. For each $i \le k$,

$$\frac{\pi_i(\beta)}{\pi_{k+1}(\beta)} = e^{\beta(E_{k+1}-E_i)} < e^{\beta_b(E_{k+1}-E_i)} = \frac{q_i}{q_{k+1}}.$$

Summing gives $\langle h^{(k)} \rangle_{\pi(\beta)} < 0$. Every nonnegative combination above therefore has strictly negative overlap on the cooling side. It cannot cross zero there. Yet its negative value may first move farther from zero and later return toward the bath: a fixed sign does not imply monotonic magnitude. This is the space left for ordinary cooling reversals.

A distinct order concept. Stochastic monotonicity asks whether dynamics preserve order of increasing observables or cumulative probabilities. It is not the same statement as nonnegativity of prefix-contrast coordinates. Part II derives the appropriate adjacent-difference criterion for that stronger order property and states exactly which observables and divergences it controls.

Lesson 24. Derive the sharp one-quarter amplitude bound

Under the full four-state triplet premise and a simple slow mode, Part II proves the structural inequalities

$$w_1 \geq w_2, \qquad w_1 + w_2 \geq 2w_3, \qquad w_4 \leq 0$$

in the orientation of Lesson 23. The difficult rate-to-eigenfunction argument uses a reset subtraction, a nonnegative contrast matrix, and a Schur complement. We now derive the amplitude bound from these inequalities step by step. This makes clear which part is structural input and which part is elementary thermal averaging.

First condition on the lower three states. Let $m(\beta)$ be the Gibbs average of w conditional on states $1, 2, 3$. Write their three entries as a constant plus

$$u(1, -1, 0) + v(1, 1, -2), \qquad u = (w_1 - w_2)/2 \geq 0, \quad v = (w_1 + w_2 - 2w_3)/6 \geq 0.$$

This decomposition is checked by subtracting the first two coordinates and by adding the first two minus twice the third. Those operations give $2u$ and $6v$, respectively.

Set $E_1 = 0$, $a = E_2$, $b = E_3$, $t = e^{-\beta a}$ and $y = e^{-\beta b}$. The conditional probabilities are $(p_1, p_2, p_3) = (1, t, y)/(1 + t + y)$. Quotient differentiation gives

$$(p_1 - p_2)' = \frac{2at + by - (b - a)ty}{(1 + t + y)^2} > 0.$$

Indeed $b - (b - a)t \geq a > 0$ for $0 < t \leq 1$. Also $(p_1 + p_2 - 2p_3)' = -3p_3' > 0$, since state 3 has the highest energy in the conditional distribution. Nonnegative coefficients u, v now imply $m'(\beta) \geq 0$.

Restore the fourth state. Let $P(\beta) = \pi_1 + \pi_2 + \pi_3$. It increases with β, and for finite positive β it lies strictly between $3/4$ and 1. The lower bound follows because each of the first three Gibbs weights exceeds the fourth weight. The full overlap is

$$b(\beta) = P(\beta)m(\beta) + (1 - P(\beta))w_4.$$

For $0 < \beta_h < \beta_c < \beta_b$, put $r = P_c/P_h$. Then $1 < r < 4/3$. Subtracting rb_h from b_c cancels the common lower-state weight:

$$\begin{aligned} b_c - rb_h &= P_c(m_c - m_h) + (1 - r)w_4 \\ &\geq 0. \end{aligned}$$

Both terms are nonnegative: $m_c \geq m_h$, while both $1 - r$ and w_4 are nonpositive. Lesson 23 gives $b_h, b_c < 0$. Replacing them by minus their absolute values reverses the intuitive direction, yielding

$$|b_c| \leq r|b_h| < \frac{4}{3}|b_h|.$$

Divide by the positive $|b_c|$ and rearrange:

$$\frac{|b_h|}{|b_c|} > \frac{3}{4}, \qquad A = 1 - \frac{|b_h|}{|b_c|} < \frac{1}{4}.$$

That is the one-quarter bound. It applies to the stated amplitude measure under the stated full premise, not to every four-state thermal model.

Why "sharp supremum" does not mean an attained maximum. A supremum is the least upper bound. The numbers $1/4 - 1/m$ approach $1/4$ without equaling it at any finite m. In the same way, the constructed admissible families approach 25% while every finite member stays below. A limiting expression with a zero weight or infinite energy need not itself be admissible; the proof must verify a sequence of finite admissible members. The report supplies that verification, and distinguishes its uniformly gapped, strict-inequality family from the different family used in Lean. The corresponding asymptotic KL reduction approaches $1 - (3/4)^2 = 7/16$, not $1/4$.

One level deeper: what the Schur complement actually does. The structural proof mentions eliminating a coordinate. Here is the algebra behind that phrase. Partition a three-by-three eigenvalue equation into two coordinates $y = (x_1, x_2)^\top$ and a third coordinate $z = x_3$:

$$\begin{pmatrix} A & c \\ d^\top & e \end{pmatrix} \begin{pmatrix} y \\ z \end{pmatrix} = \lambda \begin{pmatrix} y \\ z \end{pmatrix}.$$

Here A is a two-by-two matrix, c, d are two-component columns, and e is a scalar. The bottom row says $d^\top y + ez = \lambda z$. If $\delta = \lambda - e > 0$, solve it as $z = d^\top y/\delta$. Substituting into the top rows gives

$$\left(A + \frac{cd^\top}{\delta} \right) y = \lambda y.$$

Thus the effective two-by-two matrix includes the indirect coupling through the eliminated coordinate. The operation is exact; it is not dropping the third coordinate or assuming that it is small. If c, d have nonnegative entries and $\delta > 0$, the additional coupling has nonnegative entries too.

Why not immediately divide? Because $\delta > 0$ must first be established. For the symmetric version of a matrix, the largest eigenvalue is at least any diagonal entry: evaluate its quadratic form on the corresponding unit coordinate vector and compare with the largest eigenvalue. Equality is a special case. In the report's proof, equality would make that coordinate vector a top eigenvector. Simplicity would then contradict a required nonzero first coordinate. This supplies strictness, permitting division. The zero-first-coordinate case is treated separately before this step.

Finally, for an effective matrix with entries b_{ij}, let $r = x_2/x_1$ when $x_1 > 0$. Its two eigenvalue equations are $b_{11} + b_{12}r = \lambda$ and $b_{21} + b_{22}r = \lambda r$. Eliminate λ to obtain

$$b_{21} + r(b_{22} - b_{11}) - b_{12}r^2 = 0.$$

The structural proof evaluates this polynomial at a proposed lower bound for r. Rate inequalities control its sign, and the positive-root structure controls which side of the root the proposed bound lies on. This is how a statement about rates becomes a statement about the shape of the slow eigenfunction. The report also treats vanishing off-diagonal entries, where a positive-root shortcut alone would be insufficient.

Lesson 25. From plotting a curve to classifying every preparation

A plot of $b(\beta)$ suggests where extrema occur, but a finite collection of samples cannot rule out an extra turn between sample points. A global classification needs a bound on how complicated the curve can be and a finite list of decisive tests.

Remove a redundant coordinate. For four distinct energies with $E_1 = 0$, the ground-state probability has $\pi_1' > 0$ at every finite $\beta \geq 0$. Write

$$g = \pi_1', \qquad u = \frac{\pi_2'}{g}, \qquad v = \frac{\pi_3'}{g}.$$

Since the derivative probabilities sum to zero, $\pi_4' = -g(1 + u + v)$. Consequently

$$b' = g\{A + Bu + Cv\}, \qquad A = w_1 - w_4,\ B = w_2 - w_4,\ C = w_3 - w_4.$$

Because $g > 0$, the sign of b' is exactly the sign of $\Phi = A + Bu + Cv$. Instead of looking at four derivatives separately, we study a planar curve (u, v) and a line $A + Bu + Cv = 0$.

What the determinant calculation establishes. A Wronskian is a determinant whose columns are functions and whose rows are successive derivatives. For two functions it is $W(f, g) = fg' - f'g$. For more functions the determinant records higher-order independence. In this problem the exact exponential structure of the Gibbs probabilities makes relevant Wronskians factor into quantities of known sign. Part II derives the factorization. Its consequences here are $u' < 0$ and strict convexity of v as a function of u.

Strict convexity means that the curve bends consistently in one direction. A line can meet such a curve at at most two points in this setting, with multiplicity counted appropriately. More specifically, if $s = dv/du$, then

$$\Phi' = u'(B + Cs).$$

The report gives an explicit monotone formula for s. Thus Φ has at most one interior extremum unless it is a special constant case. To decide whether b' takes both signs on a bath side, it is enough to test Φ at that side's endpoints, its endpoint limits, and the possible single interior extremum. Equality cases are retained, rather than assigned a sign from a plotting tolerance.

Tangency versus reversal. If Φ merely touches zero and returns to the same sign, b' does not change sign. A stationary point of b has occurred without a reversal of monotonicity. If b itself crosses zero away from the bath, that concerns a strong preparation instead. These are two different root problems. The classification separates them and uses monotone intervals to decide where nonbath zeros of b exist.

The advanced generalization. In higher dimension, exponential functions form a system with controlled zero counts. The report derives a bound of $n-2$ on stationary points of a nonconstant thermal overlap, with multiplicities treated in the relevant interval, and constructs the matching number of strong preparations. The four-state planar picture is a low-dimensional window into that Chebyshev-system argument. The terminology comes after the concrete curve, not in place of understanding it.

Lesson 26. Repeated slow eigenvalues and why one overlap is no longer enough

Suppose the slow eigenvalue $\lambda < 0$ has two independent eigenvectors. Every combination of them decays with the same exponential. Choosing one of those vectors is a coordinate choice, not a preferred physical mode. A preparation may have zero coefficient in one chosen direction but a nonzero coefficient in the other. Calling that a strong effect would be incorrect.

Use the entire projection. In symmetric coordinates from Lesson 15, let P_λ be the orthogonal projection onto the whole slow eigenspace. For a preparation, form

$$y(\beta) = D^{-1/2}(\pi(\beta) - \pi_b), \qquad A_\lambda(\beta) = \|P_\lambda y(\beta)\|_2^2.$$

The subscript 2 denotes ordinary Euclidean length. The leading Pearson chi-square divergence is $e^{2\lambda t}A_\lambda(\beta)$. Leading KL is one half of it. A strong preparation must have $P_\lambda y = 0$, not merely one zero coordinate. For other norms, the shape of the projected vector matters, so a single quadratic coefficient need not decide every distance comparison.

A reset process makes the issue visible. Let

$$M = r(\pi_b \mathbf{1}^\top - I), \qquad r > 0,$$

where $\mathbf{1}$ is the all-ones column. Since $\mathbf{1}^\top p = 1$, $Mp = r(\pi_b - p)$, so

$$p(t) = \pi_b + e^{-rt}(p(0) - \pi_b).$$

Every zero-total deviation is an eigenvector with eigenvalue $-r$. There are $n - 1$ independent slow directions. All deviations shrink by the same factor. There is no need to choose a distinguished vector inside this eigenspace, and doing so cannot reveal a physical effect absent from the full projection.

Where the harder classification begins. Part II studies repeated slow spaces and, under the full triplet premise, classifies the four-state boundary as a reset plus mixing on one adjacent pair. It then proves the appropriate all-time convex-divergence ordering. This is a separate structural result, not something that follows merely from the word "degenerate." Degeneracy means repeated eigenvalues; it does not mean singular probabilities or an invalid generator.

Checkpoint. For a vector with two coordinates $(3, 4)$, one coordinate can be made zero by rotating axes, while its length remains 5. The same principle explains why a basis-dependent zero overlap is inadequate in a repeated eigenspace. The invariant object is the whole projected deviation.

Lesson 27. Numerical evidence, exact arithmetic, and interval proof

Computers do several distinct jobs in this project. A search proposes parameters. A numerical solver estimates their behavior. Exact algebra checks identities. Interval arithmetic encloses uncertain real quantities. A proof assistant checks a formal argument. These jobs reinforce one another, but one does not automatically substitute for another.

Floating-point arithmetic. Ordinary numerical software stores finitely many binary digits. Most real numbers, and many simple decimal fractions, cannot be represented exactly. Each operation can introduce rounding. This is usually harmless for exploration, but the sign of a tiny difference between two nearly equal numbers can be sensitive. Subtracting two quantities around 1 to obtain a result around 10^{-12} demands a much more careful error analysis than merely reporting twelve displayed digits.

Exact rational arithmetic. Integers and ratios of integers can be stored and manipulated exactly. For instance, $249/25 - 251/25 = -2/25$ is an exact negative result. Exact arithmetic is ideal for the rate construction, polynomial coefficients, and cleared inequalities. It does not imply that every answer must be rational: an irrational polynomial root can be specified exactly by its polynomial and a rational isolating interval.

Interval arithmetic. Replace an unknown real number by a closed interval guaranteed to contain it. For $x \in [a, b]$ and $y \in [c, d]$,

$$x + y \in [a + c, b + d], \qquad x - y \in [a - d, b - c].$$

A product is enclosed by the minimum and maximum of the four endpoint products. Division requires that the denominator interval not contain zero. With machine arithmetic, endpoints must be rounded outward so the true answer cannot escape. Functions such as exponential and logarithm likewise require validated enclosures, not ordinary rounded evaluations casually wrapped in brackets.

Dependency can make a valid enclosure too wide. If $x \in [0, 1]$, a naive interval evaluation of $x - x$ gives $[-1, 1]$, although the exact expression is zero. The enclosure is valid but weak because the same variable appeared twice. Algebraic simplification, smaller boxes, or sharper bounds can restore useful precision. A failed interval inequality may mean only that the enclosure is too broad.

Why random perturbations do not certify a box. Testing ten thousand parameter points leaves infinitely many others untested. To prove an inequality on an entire box, bound it on that box; if the bound is inconclusive, split the box and repeat. A complete branch-and-bound certificate must account for every subbox, including ones near difficult boundaries. A pending queue, a timeout, or an enclosure error must remain visible. Success on most of the domain is not a proof on all of it.

Negative controls are part of credibility. A checker should reject intentionally corrupted data and deliberately false claims. Otherwise a script that always reports success could look impressive. The project includes such controls, alongside independent formulations and source-bound receipts. They improve confidence in the implementation without becoming external human peer review.

Lesson 28. From late-time asymptotics to useful finite-time questions

A strict slow-amplitude reversal proves an eventual comparison. It does not tell us whether the crossing occurs before the deviations become experimentally negligible. This motivates a separate question: can a specified model show a substantial advantage over a stated time interval and at stated, nonzero distance thresholds?

Specify the observable and the clock. Let $D_x(t) = D_{\mathrm{KL}}(p_x(t)\|\pi_b)$ for preparation x. A distance advantage of at least 10% means $D_h(t) \le 0.9D_c(t)$ on the specified interval. A floor $D_c(t) \ge 10^{-3}$ prevents the claim from being supported solely by arbitrarily tiny distances. Neither statement is a target-time claim. Define a threshold time instead by

$$t_x(\delta) = \inf\{t \ge 0 : D_x(t) \le \delta\}.$$

A time saving of at least 5% means $t_h(\delta) \le 0.95t_c(\delta)$. Here δ is a distance threshold, not a temperature.

Monotonicity makes threshold roots usable. For the irreducible reversible dynamics studied in the physical section, KL decreases strictly along a nonstationary trajectory. A threshold between its initial value and zero has a unique crossing time. If a candidate time s satisfies $D_h(s) \le \delta$, then $t_h(\delta) \le s$. Thus, taking $s = 0.95t_c(\delta)$, a sufficient condition for the desired time saving is

$$D_h(0.95t_c(\delta)) \le D_c(t_c(\delta)) = \delta.$$

The equation explains the logic; the certificate must still handle the unknown cold root and every allowed parameter vector.

Normalize time carefully. The physical study writes $u = t/\tau$, where $\tau = -1/\lambda_2$ is the relaxation time of the chosen generator. For a fixed parameter vector, both preparations share that same τ, so a fractional target-time comparison is unchanged by normalization. Across a parameter box, however, τ itself changes. A dimensionless window does not automatically specify one shared number of seconds for all models. An attempt frequency is needed to turn the clock into seconds.

The certified physical-model benchmarks. The retained study establishes $D_h \le 0.9D_c$ and $D_c \ge 10^{-3}$ for $0.4 \le u \le 0.8$, and $t_h(\delta) \le 0.95t_c(\delta)$ for every $\delta \in [10^{-3}, 10^{-2}]$, throughout its specified simultaneous $\pm 5\%$ parameter box. The perturbed coordinates are three energy gaps, three shared barrier excesses, and two temperature excesses $T_x/T_b - 1$. The last phrase matters: perturbing a temperature excess is not identical to perturbing the full temperature ratio. The eight continuous obligations and their reductions are recorded in Part II.

Lesson 29. What the physical model adds, and what remains idealized

The exact counterexample was designed to settle a mathematical inference. The physical-model study asks a different question using a constrained thermal network. Its rate law has Arrhenius form

$$a_{ij} = \nu_0 \exp[-\beta_b(B_{ij} - E_j)], \qquad B_{ij} = B_{ji}.$$

Here ν_0 is an attempt frequency and B_{ij} is a shared saddle energy. The exponent uses the departure energy E_j, consistent with the destination-first index convention. A higher barrier above the departure state suppresses that transition.

Detailed balance follows from the symmetric saddle. Multiply the rate by the unnormalized bath weight $q_j = e^{-\beta_b E_j}$:

$$a_{ij}q_j = \nu_0 e^{-\beta_b(B_{ij}-E_j)}e^{-\beta_b E_j} = \nu_0 e^{-\beta_b B_{ij}} = a_{ji}q_i.$$

The departure energy cancels. This is a transparent physical construction of reversible rates, rather than a separate balance constraint fitted after the fact.

Shared parameters reduce arbitrary freedom. The study uses four ordered energies with three gaps and two interleaved basins, $\{1, 4\}$ and $\{2, 3\}$. The saddle is $B_{ij} = \max(E_i, E_j) + h_{ij}$, with one shared excess for each intrabasin edge type and one for all interbasin edges. The six edges therefore do not have six independently fitted barrier excesses. This geometry makes the model more constrained and supplies a kinetic interpretation of the collective relaxation.

Do not merge the two research claims. Two triplets in this physical example satisfy the original triplet test. It is therefore not another witness to the all-triplet-failure conjecture. The abstract counterexample and the robust physical example answer different questions. Keeping them separate prevents a compelling finite-time plot from being used as evidence for a premise it does not satisfy.

A realization is not an experiment. Part II gives an energy-conserving stochastic collision realization of the rates and examines an explicit activated-intermediate construction. These make assumptions more inspectable. They do not identify a particular material, prove that a laboratory contact process obeys the idealization, or include all costs of preparing a hotter initial state. A demonstrated reduction in the model's relaxation time is not automatically a net energy or engineering advantage for a device. That latter question requires an account of preparation, coupling, measurement, and operating constraints.

A productive next layer of work. To connect the model to an experiment, specify what each state represents, how rates are measured, how the bath approximation is maintained, and which observable supplies the stopping criterion. Then compare model predictions with measured uncertainty. The mathematical certificate is valuable because it makes one link in that chain explicit; it does not claim the other links have already been established.

Lesson 30. What a formal proof checks, and how to read its scope

A proof assistant checks a mathematical argument written in a formal language against explicit definitions and inference rules. The central question becomes: what exact proposition was checked? A successful build of one theorem does not imply that every sentence in a surrounding report has been formalized.

Start with the definitions. For this project, a formal counterexample must define the actual generator, its bath weights, the preparation distributions, the triplet predicate, and the chosen distance. If any of these definitions differs from the informal claim, a formally correct theorem could still answer the wrong question. Source-to-statement correspondence is therefore a separate review obligation. The same applies to exact computational certificates.

The covered results. The Lean companion verifies the rational-mode counterexample, an initial-to-eventual Pearson chi-square reversal under its master equation, and the sharp four-state amplitude supremum. It uses a particular sharpness family. The report's stronger all-strict, uniformly gapped family has a separate analytic and computational justification; it is not made formal merely because a related family is formalized. The physical interval certificates are also a distinct verification route.

What reproducibility should expose. A useful reproduction record identifies the inputs, source version, command, output, and interpretation. A cryptographic hash is a compact identifier of file contents. Matching hashes tie an old result to the same bytes; they do not prove those bytes implement the intended mathematics. A changed source invalidates that simple correspondence until the change has been reviewed. This is why the accompanying guide tracks source-bound proof reviews separately from passing replay logs.

A practical reading order for a theorem. First read its hypotheses, including positivity, simplicity, temperature range, and chosen distance. Then read the conclusion literally: existence, all-time ordering, eventual ordering, or a quantitative bound. Next identify which definitions give those words mathematical content. Finally inspect the proof and its dependencies. Running the command comes after knowing what a successful command is meant to establish.

How much confidence is justified? Independent exact arithmetic, alternate spectral arguments, negative controls, interval bounds, and formal checking each address different failure modes. Their agreement is useful evidence. None should be represented as external peer review or experimental confirmation. The report preserves those distinctions so that another researcher can decide what to reuse and what to check independently.

Lesson 31. Exercises with complete worked answers

These problems revisit the dependency chain. Try each calculation before reading its answer. Their purpose is to reveal whether a symbol has acquired a concrete meaning, not to test speed.

Exercise 1: normalize and compare. Given weights $(1, 1/2, 1/8)$, find the probabilities and the ratio of the third to the first. *Answer:* the sum is $13/8$, so the probabilities are $(8/13, 4/13, 1/13)$. Their ratio is $(1/13)/(8/13) = 1/8$, equal to the ratio of the original weights. Normalization changes entries but cancels from pair ratios.

Exercise 2: translate a temperature ordering. If $T_h = 4T_b$ and $T_c = 2T_b$, find the inverse temperatures in bath units. *Answer:* $\beta_h = 1/4$, $\beta_c = 1/2$, and $\beta_b = 1$. Both preparations lie on the cooling side, with $0 < 1/4 < 1/2 < 1$. The preparation called "cold" is still hotter than the bath.

Exercise 3: construct both directions. Let $q = (1, 1/4)$ and choose downward rate $a_{12} = 2$. Find the upward rate, generator, equilibrium, and nonzero eigenvalue. *Answer:* $a_{21} = 2(1/4) = 1/2$, so

$$M = \begin{pmatrix} -1/2 & 2 \\ 1/2 & -2 \end{pmatrix}, \qquad \pi_b = (4/5, 1/5)^\top.$$

The columns sum to zero, and the equilibrium flows are $2(1/5) = (1/2)(4/5) = 2/5$. One eigenvalue is zero; their sum is the trace $-1/2 - 2 = -5/2$, so the other is $-5/2$. Direct multiplication by $(1, -1)^\top$ checks the nonzero mode independently.

Exercise 4: verify a covariance sign. Use $p = (1/2, 1/2)$, $E = (0, 2)$, and $w = (1, -1)$. *Answer:* $\langle E \rangle = 1$, $\langle w \rangle = 0$, and $\langle wE \rangle = -1$, so $\mathrm{Cov}(w, E) = -1$. For a fixed observable in a Gibbs family passing through these probabilities, $b' = 1$ there. The pairwise form gives $(1/2)(1/2)(-2)(2) = -1$ as a second check.

Exercise 5: distinguish sign and magnitude. Suppose certified intervals give $b_h \in [-0.21, -0.20]$ and $b_c \in [-0.24, -0.23]$. Prove a strict magnitude advantage and bound its fraction conservatively. *Answer:* $|b_h| \le 0.21 < 0.23 \le |b_c|$. Thus $1 - |b_h|/|b_c| \ge 1 - 0.21/0.23 = 2/23$, about 8.7%. Using interval midpoints would give an estimate, not this guaranteed bound.

Exercise 6: calculate the two-exponential crossing. For $f_h = e^{-t} + 9e^{-10t}$ and $f_c = 2e^{-t} + e^{-10t}$, determine their ordering. *Answer:* $f_h - f_c = e^{-10t}(8 - e^{9t})$. The prefactor is positive, so the difference is positive before $t = \log 8/9$, zero there, and negative after it. This is a complete sign argument, with no plotting or root-finding tolerance.

Exercise 7: separate amplitude, divergence, and time. If a one-mode amplitude ratio is 0.8, what is the late-time KL ratio? *Answer:* $0.8^2 = 0.64$, giving a 36% asymptotic KL reduction. With slow decay rate r, the leading target-time advance is $\log(1/0.8)/r$. Its percentage depends on the reference target time. No universal percentage time saving follows from the amplitude ratio alone.

Exercise 8: find the missing logical check. A program finds a four-state reversal, but checks only three of the four triplets. Does it certify the counterexample? *Answer:* no. The antecedent requires all four predicates to fail. A single omitted triplet could satisfy the test and invalidate the proposed contradiction even if the reversal itself is real.

Exercise 9: understand a strict supremum. Every allowed model has $A < 1/4$, while a sequence of allowed models has $A_m \to 1/4$. Can one claim an allowed model attains 25%? *Answer:* no. The two facts establish the supremum exactly, but the strict bound rules out attainment. A limit outside the allowed domain does not supply a missing maximizer.

Exercise 10: identify insufficient robustness evidence. A numerical study samples a parameter box, finds no violations, and reports a 5% guarantee everywhere. What is missing? *Answer:* a proof covering every point between samples and numerical error in the evaluated quantities. A complete interval cover can provide such a proof if every subbox is successfully certified and the implementation bounds are valid. Random sample count alone cannot provide it.

Exercise 11: verify a prefix contrast. Use normalized weights $q = (4/10, 3/10, 2/10, 1/10)$. Find $h^{(2)}$ and its squared weighted length. *Answer:* $Q_2 = 7/10$ and $Q_2/q_3 = 7/2$, so $h^{(2)} = (1, 1, -7/2, 0)$. Its weighted mean is $4/10+3/10-(2/10)(7/2) = 0$. Its squared length is $4/10+3/10+(2/10)(49/4) = 63/20$. A signed contrast can have length larger than one; it is not a probability vector.

Exercise 12: test a Schur elimination by substitution. Suppose an eigenvalue equation has $e = -5$, $\lambda = -2$, $c = (1, 2)^\top$, and $d = (3, 4)^\top$. Find the correction to A. *Answer:* $\delta = 3$ and

$$\frac{cd^\top}{\delta} = \frac{1}{3}\begin{pmatrix} 3 & 4 \\ 6 & 8 \end{pmatrix}.$$

The bottom equation gives $z = (3x_1+4x_2)/3$. Substituting cz into the top equation yields exactly this correction. This example checks the elimination algebra; no claim is made that an arbitrary choice of A gives a Markov generator or the proposed eigenvalue.

Exercise 13: detect an illegal division. An interval algorithm encloses a denominator by $[-0.01, 0.02]$ and then reports the quotient using the reciprocals of the endpoints. *Answer:* the step is invalid because the interval includes zero. The reciprocal is unbounded near zero. One must establish a narrower nonzero enclosure, split and analyze the domain with the singularity handled, or reformulate the comparison without that division. A small interval is not enough; its position relative to zero matters.

Exercise 14: turn a distance bound into a time bound. Assume a strictly decreasing $D_h(t)$, and let $D_c(t_c) = \delta$. If $D_h(0.95t_c) \le \delta$, show the time saving. *Answer:* by time $0.95t_c$ the hot trajectory has already reached the threshold. Its first crossing therefore satisfies $t_h \le 0.95t_c$. Subtracting from t_c gives $t_c - t_h \ge 0.05t_c$, a saving of at least 5% relative to t_c. A pointwise comparison $D_h(t_c) < D_c(t_c)$ alone would establish only $t_h < t_c$, without the quantitative 5% margin.

Lesson 32. Map the foundations into the advanced report

The next part preserves the technical research exposition. It uses the language now developed, but it is intentionally denser. Use the following routes to move between explanation and proof rather than trying to absorb every theorem in one uninterrupted reading.

Question	Route through the report
What exactly was contradicted?	Read the source predicate and Section 2.3; match each hypothesis to its witness before reading the eigenvalue calculation. Revisit Lessons 1, 19, and 20.
Why does the overlap imply relaxation reversal?	Read Section 2.4 with Lessons 14, 16, and 17. Track the remainder and the simple-slow-mode assumption.
Can I check the original witness independently?	Read Theorem 1, the cofactor appendix, and Section 20.2. Reconstruct the rates as in Lesson 18 and understand the enclosures as in Lesson 21.
What creates the effect?	Read Section 4.2, then the contrast argument in Section 5. Revisit the covariance identity and prefix contrasts.
Why is the amplitude bound exactly one quarter?	Read Section 14 after Lesson 24. Separate the structural lemma, the conditional-average bound, and the construction proving sharpness. They are three different proof obligations.
What changes without the triplet premise?	Read Section 9 for the four-state classification and Section 10 for repeated slow spaces. Lessons 25–26 explain the geometry and the invariant projection.
How do exact algorithms decide a case?	Read Section 10.1 and the later rational-input classifier. Watch assumptions on energies, exact root isolation, equality handling, and failure reporting.
What survives at finite time?	Read Section 11 for a specified pair and the finite-time order sections for their precise hypotheses. Eventual order alone is not their proof.
What is physically demonstrated by the model?	Read Sections 22.4, 22.5, and 22.6. Keep the two experiments of Lessons 18 and 29 separate.
What has been formalized?	Read Section 20.3 with Lesson 30. Match named theorems to the stated definitions and scope.

A compact symbol dictionary.

Symbol	Meaning and common trap
i, j, k, n	State indices and number of states; an index is a label, not a time.
E_i	Energy of state i; only energy differences affect Gibbs probabilities.
T, β	Temperature and inverse temperature; their orderings are opposite.
β_b, π_b	Fixed bath inverse temperature and its Gibbs distribution.
q_i, Q	Positive bath weights and their sum; some advanced sections explicitly use normalized $q = \pi_b$ instead. Read the local definition.
$\pi(\beta), Z(\beta)$	Gibbs preparation and its partition function.
$p(t)$	Evolving probability vector; generally not itself a Gibbs preparation.

Symbol	Meaning and common trap
a_{ij}, M_{ij}	Off-diagonal rate from j to i; columns, not rows, conserve probability in this convention.
M, L	Probability generator and observable generator $L = M^\top$.
D, S	Diagonal bath-probability matrix and symmetric transform $S = D^{-1/2} M D^{1/2}$.
λ_2, v_2, w_2	Slow nonzero eigenvalue, right eigenvector, and left eigenfunction; simplicity must be checked.
$b(\beta)$	Slow overlap $w_2^\top \pi(\beta)$, with a fixed normalization.
$\mathrm{Cov}, \mathrm{Var}$	Covariance and variance under the indicated probability distribution.
F, G, κ, P_{ijk}	Exact triplet expressions and predicate; P_{ijk} is a logical statement, not a transition matrix.
τ, u	Relaxation time and normalized time $u = t/\tau$ in the physical study; the letter u has different explicitly defined local uses elsewhere.
$\delta, t_x(\delta)$	A distance threshold and its crossing time.
$O(\cdot), o(\cdot)$	Controlled order of a remainder and a relatively vanishing remainder, respectively; neither is a blanket finite-error certificate.

The habit to carry forward. For every advanced statement, ask what varies and what stays fixed; which quantity is ordered; whether the comparison is finite-time or asymptotic; where strictness enters; and how equality or degeneracy is handled. Those questions are often more useful than memorizing a long formula. The technical part that follows supplies the proofs, exact data, source correspondence, implementation details, and explicitly stated open problems.

An extended bridge: from random events to a physical certificate

The previous lessons supplied the vocabulary. These eight additional lessons slow down the transition from that vocabulary to the physical computations. They answer a practical question: how can a finite calculation establish a statement about an entire interval of times? They then examine what it means for a mathematical rate to have a physical interpretation.

Lesson 33. A rate is not a probability: build the clock first

Begin with units. A probability is a number between zero and one. A rate has units of inverse time. A rate of three events per second is perfectly meaningful; a probability of three is not. If an event has constant rate r, its probability of occurring in a very short interval of length h is $rh + o(h)$. The notation $o(h)$ means that the error divided by h tends to zero as h tends to zero. It does not mean that the approximation is exact for a one-second interval.

Derive the waiting law. Let $S(t)$ be the probability that no event has happened by time t. Assume that, conditional on no event so far, the chance in the next short interval is still $rh + o(h)$. Then

$$S(t+h) = S(t)(1 - rh + o(h)), \qquad S'(t) = -rS(t), \qquad S(0) = 1.$$

The function with this derivative and initial value is $S(t) = e^{-rt}$. Thus the probability of at least one event by time t is $1 - e^{-rt}$. For $r = 2$ per second and $t = 0.1$ seconds, this is $1 - e^{-0.2}$, approximately 0.1813. The short-time estimate $rt = 0.2$ is close but different. For $t = 1$ second the estimate $rt = 2$ is not a probability at all.

Why the mean wait is $1/r$. The density of the first event time is re^{-rt}. Its mean is

$$\int_0^\infty tre^{-rt}\,dt = [-te^{-rt}]_0^\infty + \int_0^\infty e^{-rt}\,dt = \frac{1}{r}.$$

The integration by parts has merely moved the factor t onto a derivative. There is also a geometric interpretation: the mean waiting time is the area under the survival curve $S(t)$.

Several possible exits. From state j, let the exit rates to other states be a_{ij}. Independent exponential clocks compete. The probability that none has rung by t is the product $\prod_{i\neq j} e^{-a_{ij}t}$, hence the total exit rate is $r_j = \sum_{i\neq j} a_{ij}$. Conditional on an exit, its destination is i with probability a_{ij}/r_j. This produces the diagonal entry $M_{jj} = -r_j$ and explains why a column sums to zero. The negative diagonal does not describe a negative probability; it describes probability leaving.

Try it. Suppose the rates from one state to two others are 2 and 3 per second. The mean residence time is 1/5 second, and the next destination probabilities are 2/5 and 3/5. Multiplying all rates by ten divides every waiting time by ten. It does not change these destination probabilities or the equilibrium distribution. Consequently, a percentage time advantage alone supplies no absolute number of seconds without a physical rate calibration.

Lesson 34. Replace unequal clocks by one clock and harmless pauses

A deliberately simple two-state example. Use a column probability vector and let

$$M = \begin{pmatrix} -2 & 1 \\ 2 & -1 \end{pmatrix}, \qquad \rho = 2, \qquad P = I + M/\rho = \begin{pmatrix} 0 & 1/2 \\ 1 & 1/2 \end{pmatrix}.$$

The first state exits at rate 2; the second exits at rate 1. Instead ring a clock of rate 2 in either state. At each ring, choose the next state using the current column of P. In state two, half the rings leave the state unchanged. Its actual transition rate is therefore $2 \times (1/2) = 1$, exactly as required. The pauses are real elapsed time.

Generalize only after the example. Choose ρ at least as large as every exit rate. Then $P = I + M/\rho$ has nonnegative entries and unit column sums. Such a matrix is called stochastic: it sends probability vectors to probability vectors. The identity $M = \rho(P - I)$ gives

$$e^{Mt} = e^{\rho t(P-I)} = e^{-\rho t} \sum_{k=0}^{\infty} \frac{(\rho t)^k}{k!} P^k.$$

The factorization is allowed because P commutes with I. It would not be legitimate to split an arbitrary matrix exponential into two factors without checking commutation.

Read every factor as a probability. Write $\mu = \rho t$. A Poisson random variable N has probabilities $\Pr(N = k) = e^{-\mu}\mu^k/k!$. Here N counts clock rings by time t. Conditional on $N = k$, the distribution is $P^k p(0)$. Averaging these conditional distributions gives the displayed formula. This procedure is called uniformization. It is an exact continuous-time construction, not a small-time-step approximation.

At $\mu = 1$, the probabilities of zero, one and two rings are respectively e^{-1}, e^{-1} and $e^{-1}/2$. Their sum is about 0.9197. The missing 0.0803 is the probability of three or more rings; it cannot simply be discarded when a rigorous error bound is required.

Check the two-state equilibrium. Solving $M\pi = 0$ and $\pi_1 + \pi_2 = 1$ gives $\pi = (1/3, 2/3)^\top$. Starting from $(1, 0)^\top$, the exact first component is $p_1(t) = 1/3 + (2/3)e^{-3t}$ and the second is its complement. Differentiating verifies $p_1' = 1 - 3p_1$, which is also the first row of Mp. This small solvable model is a useful check of the matrix convention, the clock interpretation and any implementation of uniformization.

Lesson 35. Bound the part of an infinite sum that you did not compute

First learn the geometric comparison. For $0 \le q < 1$, multiplying $1 + q + q^2 + \cdots$ by $1 - q$ cancels every term except the first, giving $1/(1 - q)$. If a nonnegative series starts with a and every later term is at most q times its predecessor, its whole sum is at most $a/(1 - q)$. Equality is unnecessary; an upper bound suffices.

For Poisson weights $w_k = e^{-\mu}\mu^k/k!$,

$$\frac{w_{k+1}}{w_k} = \frac{\mu}{k+1}.$$

After keeping terms 0 through K, the first omitted weight is w_{K+1}. Every subsequent ratio is at most $q = \mu/(K + 2)$. Provided $q < 1$,

$$R_K = \sum_{k=K+1}^{\infty} w_k \le \frac{w_{K+1}}{1 - \mu/(K+2)}.$$

The $K + 2$ in the denominator is intentional: it bounds the ratio from the first omitted term to the next omitted term. Confusing it with the first omitted term's index is an easy off-by-one mistake.

Work a numerical example. Take $\mu = 1$ and keep $K = 2$. Then $w_3 = e^{-1}/6$ and $q = 1/4$, so $R_2 \le 2e^{-1}/9$, approximately 0.08175. The actual missing mass is approximately 0.08030, below the bound. The bound is a little wider than necessary, but its validity is transparent.

Turn a scalar tail into a vector guarantee. Every component of $P^k p(0)$ lies in $[0, 1]$. If s_i is the computed partial sum for component i, then $p_i(t) \in [s_i, s_i + R_K]$. These independent intervals need not sum to exactly one; they are an enclosure of a probability vector, not themselves a new probability vector. Renormalizing their endpoints without a proof could invalidate the enclosure.

Include arithmetic error as well as truncation error. An exact tail formula does not excuse rounding errors in the partial sum. The nominal checker computes interval bounds for inputs, exponentials, matrix products and logarithms. If the entire probability interval is positive, it can enclose each term $p_i \log(p_i/\pi_i)$ and hence KL. An interval crossing zero requires a valid alternative treatment or a failure to certify; a logarithm of an invalid endpoint is not evidence.

In the replayed nominal calculation, at most 281 terms were needed and the largest reported tail upper bound was below 10^{-45}. The small tail is one part of the certificate. The validity of P, the input correspondence, the outward arithmetic and the later comparison inequalities remain separate necessary parts. More printed digits alone establish none of them.

Lesson 36. Prove a whole time interval using its endpoints

A descending staircase picture. Suppose two distances $D_c(t)$ and $D_h(t)$ are nonincreasing in time. On a cell $[a, b]$, the cold distance is at least its value at the right endpoint, while the hot distance is at most its value at the left endpoint:

$$D_c(t) \ge D_c(b), \qquad D_h(t) \le D_h(a).$$

Therefore a single strict inequality

$$D_h(a) < 0.9 D_c(b)$$

proves $D_h(t) < 0.9 D_c(t)$ for every time in that cell. The endpoints are deliberately mismatched to use the worst case for each curve. Merely checking $D_h(a) < 0.9 D_c(a)$ and the analogous inequality at b would not control a crossing between them.

Why KL decreases here. For a stochastic transition matrix K and positive equilibrium π, let $r_i = p_i/\pi_i$. When $K\pi = \pi$, the output likelihood ratio is

$$\frac{(Kp)_j}{\pi_j} = \sum_i \frac{K_{ji}\pi_i}{\pi_j} r_i.$$

The coefficients in this sum are nonnegative and sum to one. Convexity of $f(r) = r \log r$ says that f of this weighted average is no greater than the weighted average of f. Multiply by π_j, sum over j, and use $\sum_j K_{ji} = 1$ to obtain

$$D(Kp\|\pi) \le D(p\|\pi).$$

Apply this to $K = e^{M(b-a)}$. This argument proves the monotonicity used above. It does not say that $D_h(t)/D_c(t)$ is monotone.

A hand-checkable cell. Suppose outward bounds give $D_h(a) \le 0.0080$ and $D_c(b) \ge 0.0090$. Then $0.9D_c(b) - D_h(a) \ge 0.0001 > 0$. Every interior time passes the ten-percent comparison. If the computed margin interval were $[-0.0001, 0.0002]$, no conclusion would follow; a narrower cell might resolve it.

Join the cells and check the floor. A finite list $t_0 < t_1 < \cdots < t_N$ covers the entire closed interval $[t_0, t_N]$ without gaps. Proving the endpoint inequality on every cell proves the full window. Since cold KL decreases, a lower bound $D_c(t_N) \ge 10^{-3}$ also proves that floor throughout the window. The nominal replay used 80 cells and a minimum certified comparison margin greater than 0.0004408. Its fixed endpoints were approximately 58.89269 and 117.78539 in attempt-time units. The script separately encloses the slow time to justify the window-length requirement; a decimal time chosen from a numerical eigenvalue does not become exact merely by being described as a multiple of that numerical estimate.

Lesson 37. Prove a whole band of target distances

Reverse the graph in your mind. For a positive target δ below the initial distance, define $t_x(\delta)$ as the first time preparation x reaches that target. A smaller target means getting closer to equilibrium, so it takes longer. Thus if $\delta_a < \delta_b$, then $t_x(\delta_a) \ge t_x(\delta_b)$. For the irreducible reversible model and a nonequilibrium preparation, KL strictly decreases until its limiting equilibrium value; positive targets below the initial value have unique crossings.

A root estimate proposes; a sign bracket verifies. A numerical solver might propose $t = 10.1234$. To verify a bracket $[L, U]$, enclose $D(L) - \delta$ strictly above zero and $D(U) - \delta$ strictly below zero. Continuity gives a crossing between them; strict decrease gives uniqueness. An approximate residual close to zero is weaker than these two certified signs. We also need $0 < L < U$ for the ratio bound below.

Choose the unfavorable corners. For every $\delta \in [\delta_a, \delta_b]$,

$$t_h(\delta) \le t_h(\delta_a) \le U_h(\delta_a), \qquad t_c(\delta) \ge t_c(\delta_b) \ge L_c(\delta_b) > 0.$$

Division by a positive denominator preserves the intended order, giving

$$1 - \frac{t_h(\delta)}{t_c(\delta)} \ge 1 - \frac{U_h(\delta_a)}{L_c(\delta_b)}.$$

If the right side exceeds 0.05, every target in that cell has at least five-percent time saving. Notice that the numerator and denominator use opposite ends of the target cell. This conservative mismatch is exactly what covers its interior.

Try the ratio with ordinary numbers. Suppose the hottest trajectory's slowest permitted arrival is at most 90 seconds and the cold trajectory's fastest permitted arrival is at least 100 seconds. The saving is at least $1 - 90/100 = 10\%$. If the denominator were instead only known to exceed 92, the bound would fall to about 2.17%, even if all sampled same-target comparisons looked better. The proof must use the bounds it has actually established.

The nominal certificate covers $[10^{-3}, 10^{-2}]$ with 60 target cells. Its conservative lower bound on time saving is approximately 8.12245%. This is a guaranteed lower bound for that nominal model, not the exact minimum. The later eight-obligation certificate handles the full uncertain four-state parameter box. It is a distinct proof, not a conclusion obtained by silently extrapolating the nominal calculation.

Lesson 38. Derive a transition rate from an energy exchange

Begin with one uphill move. Let two system energies be $E_j = 1$ and $E_i = 3$, with barrier energy $B = 5$. Draw a nonnegative reservoir energy Y with density $\beta e^{-\beta Y}$. The move from j is allowed when $1 + Y \geq 5$, namely when $Y \geq 4$. At acceptance, set $Y' = Y + 1 - 3 = Y - 2$. It is nonnegative, and $E_i + Y' = 3 + Y - 2 = 1 + Y = E_j + Y$: combined energy is conserved. The reverse contact sees the same total energy and can undo the move.

Integrate the acceptance probability. For any $c \geq 0$,

$$\Pr(Y \geq c) = \int_c^\infty \beta e^{-\beta y}\, dy = e^{-\beta c}.$$

If contacts for this pair arrive at rate ν_0, the accepted rate from j to i is therefore

$$a_{ij} = \nu_0 e^{-\beta(B - E_j)}.$$

The reverse rate is $a_{ji} = \nu_0 e^{-\beta(B-E_i)}$. Their ratio is $e^{-\beta(E_i - E_j)}$. Multiplying by Gibbs weights gives the stronger flux identity

$$a_{ij} e^{-\beta E_j} = a_{ji} e^{-\beta E_i} = \nu_0 e^{-\beta B},$$

which is detailed balance before the common normalization is inserted.

Separate an always-required gate from the uphill cost. Write $h = B - \max(E_i, E_j)$. Since $B - E_j = h + \max(E_i - E_j, 0)$, the rate becomes

$$a_{ij} = \nu_0 e^{-\beta h} \min\{1, e^{-\beta(E_i - E_j)}\}.$$

In the example at $\beta = 1$, the common gate contributes e^{-2}, the uphill acceptance contributes another e^{-2}, and their product is e^{-4}. The reverse move has only the common e^{-2} gate.

Why simulation may omit attempts but not their time. One can retain only contacts with $Y \geq h$. Their clock has rate $\nu_0 e^{-\beta h}$. Conditional on passing the gate,

$$\Pr(Y - h \geq z \mid Y \geq h) = \frac{e^{-\beta(h+z)}}{e^{-\beta h}} = e^{-\beta z}.$$

Thus a retained mark is h plus a fresh exponential energy. This proves the thinning rule used in the event simulation. Counting every attempted contact as an accepted transition would alter the process; ignoring rejected contacts without adjusting the clock would alter physical time.

What this construction establishes. Under independent fresh thermal marks, independent Poisson pair clocks, constant reservoir density of energy states, and instantaneous updates, the rate identity holds exactly for every allowed barrier and energy. It supplies a stochastic bath interpretation of the same four-state model. It does not identify a material or derive an autonomous microscopic bath. The replayed simulation of 500,000 trajectories per preparation agrees with direct evolution at five times to within 0.00118 per component. That is a diagnostic check of the simulation; the rate identity above is the reason the mathematical generators coincide.

Lesson 39. An intermediate state introduces a second clock

Insert a waiting room between two wells. Suppose the desired well-to-well rates are u from j to i and v from i to j. Insert state s. Use entrance rates $2u$ from j and $2v$ from i, and exit rate b from s to each well. On entering s, the next well is selected with probabilities $1/2$ and $1/2$. Half the visits return to the well just left.

If the clock is stopped during every visit to s, the successful transition rate from j to i is $2u\times(1/2) = u$. This is the trace clock: only time actually spent in the wells is retained. With ordinary physical time, each visit adds a mean wait $1/(2b)$. Those waits cannot be removed when claiming a physical completion time.

Derive the block identity one entry at a time. Order the wells as (j, i). Their direct block, the entrance block, the exit block and the intermediate block are

$$Q_{ww} = \begin{pmatrix} -2u & 0 \\ 0 & -2v \end{pmatrix}, \quad Q_{sw} = \begin{pmatrix} 2u & 2v \end{pmatrix}, \quad Q_{ws} = \begin{pmatrix} b \\ b \end{pmatrix}, \quad Q_{ss} = -2b.$$

The subscripts give destination first, as everywhere else. Multiplication now gives

$$Q_{ww} - Q_{ws}Q_{ss}^{-1}Q_{sw} = \begin{pmatrix} -2u & 0 \\ 0 & -2v \end{pmatrix} + \begin{pmatrix} u & v \\ u & v \end{pmatrix} = \begin{pmatrix} -u & v \\ u & -v \end{pmatrix}.$$

This is exactly the intended two-well generator. Adding the six pair contributions gives the four-well identity in the technical report. This exact algebra identifies the trace generator; it does not assert that the physical-time well probabilities solve that same four-state equation.

Equilibrium and preparation also change. In the expanded model, each intermediate has energy B and phase-space weight $\eta > 0$. Its equilibrium weight is ηe^{-B}, compared with e^{-E_j} for a well. Taking $b = 2/\eta$ in attempt-time units gives mean intermediate residence time $\eta/4$ and matched equilibrium fluxes. At preparation temperature T, all ten states must be prepared with weights $g_a e^{-E_a/T}$. Preparing only the wells would be a different experiment. The phase-space weight η need not be an integer count.

At $\eta = 0.05$, the nominal cold and hot preparations place about 1.90% and 4.07% of their mass in intermediates. Full ten-state calculations retain those populations and every residence time. Their finite-time advantage persists at the tested points. The expanded system also has higher energies and faster rates outside the original all-state caps. It is a separate test of a modeling assumption, not a new certificate inside the original constrained four-state class.

Lesson 40. Read a computational result at the right strength

Four questions before accepting a result. What object was calculated? What inputs were used? Which errors were bounded? Which variables were covered continuously? An answer such as "70 digits" only addresses numerical precision, and even then does not itself establish a bound on every implementation or modeling error.

The independent ten-state calculation directly constructs and exponentiates the full generator at 70-digit working precision. It agrees with the fast spectral implementation within 6.55×10^{-13} in KL at five tested times. Different formulas agreeing is useful evidence against a numerical implementation mistake. Neither implementation uses outward error bounds for that comparison, so it is a high-precision diagnostic, not an interval certificate.

Interpret a large sample precisely. For each of three fixed η values, the study tests 256 corners and 500 seeded interior parameter points. The total is $3(256 + 500) = 2268$. Each point is further tested on finite target and time grids. No sampled case fails the declared benchmarks. This says exactly what happened on those grids. It does not exclude a narrow failure between points, between time samples, between targets, or at an untested value of η. The continuous four-state certificate cannot be transferred to this ten-state extension merely because their trace generators agree.

A percentage needs a denominator and a clock. At $\eta = 0.05$ and target 10^{-3}, the independently computed arrival times are approximately 210.67249 for cold and 190.05131 for hot. Using the cold time as denominator gives

$$\frac{210.67249 - 190.05131}{210.67249} \approx 0.0978826,$$

or about 9.78826%. Dividing by the hot time would answer a different percentage question. These are attempt-time units, not calibrated seconds. The calculation includes post-quench intermediate residence times but does not include the cost or duration of preparing the hotter initial state.

Know what a program's success message establishes. A zero exit code means that the process completed without a reported execution failure. Some exploratory programs record failures in JSON without returning a nonzero code. Read the recorded failure counts, input hashes, unresolved cells, parameter scope and numerical residuals. A certificate runner must also reject incomplete coverage. A numerical search that found nothing has a different meaning from a complete proof that nothing exists.

The final handoff to advanced mathematics. You can now read the physical section by following a chain of explicit claims: construct valid rates; establish equilibrium; evolve preparations; bound numerical error; prove continuous comparisons; and interpret the clock and physical assumptions. The advanced spectral and interval methods make that chain efficient over an uncertain parameter box. They do not replace any link. When a later formula looks forbidding, identify which link it serves before trying to manipulate its symbols.

Part II. Technical research report and exact certificates

The abstract below summarizes the research results in their technical language. All original numbered sections, theorems, equations, and proof text are retained after this teaching part; the foundational examples above use unnumbered displays so that those references keep their original numbering.

Abstract

We disprove the four-state finite-temperature triplet conjecture of Avitan, Factor and Gelbwaser-Klimovsky with a strictly positive reversible generator and an exact certificate. Under the conjecture's original triplet premise, the ordinary cooling amplitude advantage has sharp supremum 25%. A uniformly gapped family approaches this limit with all triplet inequalities strict. Cooling occurs exactly when an endpoint covariance is positive; an equivalent rate test uses one cubic root. Every repeated-slow four-state generator under the premise is a reset plus mixing on one adjacent pair and has normal convex-divergence ordering at every time. In arbitrary dimension, the triplet premise excludes strong effects and asymptotic heating, yet permits ordinary cooling advantages approaching $1 - 3/n$. Thus no dimension-independent amplitude bound below one survives. A weighted prefix-contrast representation also yields finite-time heating exclusion in weighted Euclidean distance, a uniform finite-time relative-entropy ordering window on both bath sides, and all-temperature heating order for every convex divergence in a structured generator family. For arbitrary simple four-state slow modes, an explicit Gibbs Wronskian identity gives a finite asymptotic classification; degenerate modes have a separate distance-specific test. Exact executable classifications cover commensurate rational systems and, under the triplet premise, rational rates without commensurate energies. We distinguish these proved results from unresolved general finite-time questions and the higher-dimensional sharp bound. Lean verifies a rational-mode counterexample, its eventual Pearson chi-square relaxation reversal, and the sharp four-state amplitude supremum; the formalization has explicitly delimited coverage and reproducible build instructions. Classical order, spectral, total-positivity, and transcendence tools are explicitly attributed.

1 Outcome

The proposed triplet test can miss a genuine four-state relaxation reversal. In the example, the hotter preparation begins farther from equilibrium, yet later becomes slightly closer. The reversal is mathematically rigorous and physically small. Every input rate and bath weight is rational. The energies are logarithms of those weights. A cubic with integer coefficients specifies the slow eigenvalue, and cofactor polynomials specify its left eigenvector exactly. Thermal probabilities at the two selected temperatures are algebraic numbers bounded by rational numbers whose correctness is checked by integer powers.

A companion example has an exactly rational slow eigenvalue and left eigenvector. It gives a second, simpler spectral certificate, at the expense of two less compact rates. Both examples are strictly inside the required positive-rate and simple-slow-mode domain. Section 4.2 extends the companion into a parameter family that crosses a proved cooling threshold while all triplet tests continue to fail. No assertion about a strong Mpemba effect is needed: the two displayed slow-mode overlaps are nonzero.

Sections 14 and 15.1 establish the sharp 25% amplitude supremum and a single-endpoint cooling criterion under the original triplet premise. The new exclusion theorem in Section 5 shows that the counterexample has a restricted mechanism: it reverses ordinary cooling while retaining a nonzero slow overlap. The same triplet premise rules out strong effects in every dimension and rules out asymptotic heating. Section 9 supplies a full finite asymptotic classification for arbitrary simple four-state slow modes; Section 10 states the distance-specific extension to other slow multiplicities.

Theorem 15 completes the full-premise degenerate boundary: reset plus one adjacent pair is necessary, and excludes all-time convex-divergence reversals. The same classification holds in every dimension $n \geq 4$ when there is just one faster mode. Corollary 7 implements the simple-mode cooling decision without an eigenvector. These conclusions have mathematical proofs rather than being inferred from the discovery searches.

The Lean companion verifies the rational-mode example in Appendix B, an initial-to-eventual reversal in Pearson chi-square distance under its master equation, and the sharp four-state amplitude supremum. Section 20.3 maps these claims to named theorems and gives reproduction commands. Its sharpness proof uses a simpler family with two triplet equalities; the report's stronger uniform-gap and all-strict perturbation claims are outside that formal coverage.

2 Problem and literature status

The target is the inference that failure of Eq. (3) on every energy-ordered triplet of a four-state reversible thermalizing generator precludes the Mpemba effect. The triplets retain the original off-diagonal rates. This is the conjecture in Avitan, Factor and Gelbwaser-Klimovsky, arXiv:2603.04567v1, submitted 4 March 2026. Its main text and supporting information, particularly S2–S14, were checked on 11 October 2026. The version history listed only v1. Its finite-temperature claim is numerical; the analytic theorem assumes zero bath temperature and stronger triplet inequalities.

The later Gelbwaser-Klimovsky, arXiv:2609.09988v1, submitted 9 September 2026, repeats the inference without providing a general proof. Its examples without an exact three-state mechanism concern a different, stronger predicate. Searches by title, identifier, conjecture, correction and counterexample located no other primary-source resolution. This establishes the scope of the literature check; it does not exclude unindexed or unpublished work. The source-to-counterexample correspondence is made explicit in Section 2.3. Publication priority or exhaustive novelty is not claimed.

For supporting no-Mpemba results we additionally checked Ohga, Hayakawa and Ito, arXiv:2410.06623v2, version-history date 19 October 2024, including Result 2 and SI B–D, on 11 October 2026. Their theorem is a prior application of monotone coupling to absence of mean-energy crossings. Section 4.5 compares the assumptions and observables precisely. The broader assessment in Section 4.6 also checks Fill–Kahn (v3, 6 September 2013), Baake–Sumner (v3, 25 September 2022), and Vu–Hayakawa (v2, 6 January 2025), including their relevant full-text arguments. Supporting applications carry no independent novelty claim.

For definitions we use Lu and Raz, PNAS 114, 5083–5088 (2017), and distinguish the strong effect of Klich et al., Phys. Rev. X 9, 021060 (2019). The asymptotic conclusion is supplemented below by certified finite-time crossing brackets.

2.1 Model and reductions

With column probabilities, $\dot{p} = Mp$, $M_{ij} = a_{ij}$ for $i \neq j$, and $M_{jj} = -\sum_{i \neq j} a_{ij}$. The rate a_{ij} goes from j to i. Write

$$q_i = e^{-\beta_b E_i}, \qquad Q = \sum_i q_i, \qquad \pi_b = q/Q.$$

Detailed balance is $M_{ij} q_j = M_{ji} q_i$. An energy shift sets $E_1 = 0$; measuring energies in bath units sets $\beta_b = 1$. A common positive multiplier of M changes only the time unit. We use $a_{12} = 1$. We neither assume equally spaced energies nor normalize the total energy span. These would impose additional restrictions.

If $D = \mathrm{diag}(\pi_b)$, detailed balance makes $S = D^{-1/2} M D^{1/2}$ symmetric: $MD = DM^{\top}$. Hence the

spectrum is real and the generator is diagonalizable. Its Dirichlet form is

$$-f^\top DM^\top f = \sum_{i<j} c_{ij}(f_i - f_j)^2, \qquad c_{ij} = M_{ij}\pi_{bj} = c_{ji}.$$

In the positive-rate domain the stationary eigenvalue is simple. The counterexample and scalar-overlap classification additionally assume a simple slowest nonzero eigenvalue. Section 10 treats the other multiplicities for specified asymptotic distances, while Theorem 4 does not require simplicity.

2.2 Exact predicate and its negation

For $i < j < k$, put $A = a_{ik}$, $B = a_{jk}$, $C = a_{ij}$, $t = q_j/q_i$, $u = q_k/q_j$. The source's $(-, 0, +)$ correspond to (i, j, k). Its exact test is

$$\kappa = \frac{A + B}{C(1 + t)}, \qquad G = At - B,$$
$$F = A(1 + tu/2) - B(1 + u/2) - C(1 - t),$$
$$P_{ijk} = (\kappa \geq 1 \ \wedge \ F > 0) \ \vee \ (\kappa \leq 1 \ \wedge \ G > 0).$$

Both strict comparisons and both inclusive branch comparisons match Eq. (3) and S9. This is a direct transcription of the published test, using destination-first rate indices.

Lemma 1 (The branch logic reduces to a convex cone)**.** *For positive C and $0 < t < 1$, $\neg P_{ijk}$ is equivalent to $F \leq 0$ and $G \leq 0$.*

Proof. Let $H = A + B - C(1 + t)$, $d = (1 - t)/(1 + t) > 0$, and $h = 2/(1 + t) + u/2 > 0$. Direct expansion gives $F = hG + dH$. If $H > 0$, failure is $F \leq 0$, which forces $G < 0$. If $H < 0$, failure is $G \leq 0$, which forces $F < 0$. At $H = 0$ both branches apply, but $F = hG$, so their strict tests agree. These cases prove both directions. □

For fixed energies the eight linear inequalities define one closed polyhedral cone. The admissible rates are its intersection with the strictly positive orthant. This conclusion does not discard the original branch logic; it proves an equivalent formulation including $\kappa = 1$.

The supporting-information fast-vector formulas divide by $-uG$. We do not use that division. When $G = 0$, direct triplet diagonalization gives nonzero eigenvalues $-A(1 + t + tu)$ and $-[C(1 + t) + Atu]$. If also $H = 0$, they coincide. The predicate remains well-defined whenever $C > 0$. Isolated triplet generators require their diagonals to be recomputed, not inherited as principal submatrices.

2.3 Precise published claim and the contradiction

The target is the finite-temperature conjecture in Avitan, Factor and Gelbwaser-Klimovsky, rather than an extrapolation supplied by this report. Their N-level protocol states: "if all triplets violate the conditions, the N-level system will not exhibit the Mpemba effect." The non-zero-temperature subsection explicitly identifies this as a conjecture. The protocol and SI Eqs. (S10)–(S14) specify the original rates.

Let $\mathsf{ME}_{\mathrm{cool}}(M, E, \beta_b)$ mean that there exist finite $0 < \beta_h < \beta_c < \beta_b$ with $|b_2(\beta_h)| < |b_2(\beta_c)|$, where the bath and generator are fixed and the slow nonzero eigenvalue is simple. Section 2.4 shows that this implies an initial-to-eventual KL-distance reversal. A necessary consequence of the published conjecture, restricted to four states and this regular domain, is

$$\boxed{\bigwedge_{i<j<k} \neg P_{ijk} \quad \Longrightarrow \quad \neg\mathsf{ME}_{\mathrm{cool}}(M, E, \beta_b).} \tag{$\mathcal{C}_4$}$$

This is a universally quantified implication over distinct ordered energies, positive reversible rates and $0 < \beta_b < \infty$. Requiring positive rates and a simple slow mode restricts the domain; it does not enlarge the authors' claim. Restricting the conclusion to cooling makes it weaker. A counterexample to this restricted consequence therefore suffices to refute the unrestricted finite-temperature conjecture, independently of any heating or spectral-boundary convention.

Requirement in $(\mathcal{C}_4)$	Witness in Theorem 1
Four ordered distinct energies	$E = (0, \log(200/199), \log(5/2), \log 400)$.
Finite positive bath	$\beta_b = 1$.
Conservative reversible thermal dynamics	All twelve off-diagonal rates are positive; $\mathbf{1}^\top M = 0$ and $M\,\mathrm{diag}(q)$ is symmetric.
Simple slow nonzero eigenvalue	One root in each of $(-20, -19)$, $(-60, -59)$ and $(-337, -336)$.
Original-rate triplet failures	Exact evaluations below give $F < 0$ and $G < 0$ on all four triplets, with both κ branches represented.
Conclusion contradicted	$\beta_h = 1/1000 < \beta_c = 1/100 < 1$ and $\lvert b_2(\beta_c)\rvert - \lvert b_2(\beta_h)\rvert > 0.0001034076$.

The table records the proof obligations, which the exact construction below discharges. No effective rate, principal-submatrix diagonal, infinite initial temperature or equality case is used. Shifting all energies by the same positive constant also makes $E_1 > 0$, if desired, without changing any rate test or Gibbs distribution.

Three distinct statements. The March paper's zero-temperature theorem assumes $\beta_b \to \infty$, a nondegenerate spectrum and $C > A + B$ for every triplet (SI Eq. S20). Our bath is finite and the latter inequality fails already for 123: $1 < 28 + 1799/8$. Thus the counterexample does not challenge that theorem or the three-state necessity of Eq. (3).

The September paper, Gelbwaser-Klimovsky, Section II, distinguishes Eq. (3) from its exact three-state mechanisms. Write $\mathcal{E}_{ijk}$ for satisfaction of one of those mechanisms. In their nondegenerate three-state setting, necessity gives $\mathcal{E}_{ijk} \Rightarrow P_{ijk}$, but not conversely. Consequently

$$\bigwedge \neg P_{ijk} \implies \bigwedge \neg \mathcal{E}_{ijk}.$$

Necessity supplies no converse to this implication. Its reported four-state effects with no triplet mechanism therefore do not, by that fact alone, establish failure of $(\mathcal{C}_4)$. Section II separately reiterates the original-rate Eq. (3) absence inference. Our witness verifies the stronger premise $\bigwedge \neg P_{ijk}$ explicitly.

General conjecture versus sampled parameter ranges. The March Fig. 2 uses rates from 0.05 to 1 and $\beta_b E_i$ from 0 to 1; the September Fig. 5 uses rates from 0.001 to 1 and $\beta_b E_i$ from 0 to 5. These are sampling ranges, not hypotheses attached to the general inference. Our invariant energy span is $\beta_b(E_4 - E_1) = \log 400 > 5$, outside both boxes. Even the downward-rate ratio $a_{12}/a_{23} = 8/1799 < 0.05$ prevents a common time rescaling from placing all downward rates in the March box. We therefore refute the unrestricted conjecture; we neither reproduce nor contradict the finite sample counts, and make no claim of a counterexample within either sampled box.

2.4 What the cooling witness proves

Choose $Mv_2 = \lambda_2 v_2$, $w_2^\top M = \lambda_2 w_2^\top$ and $w_2^\top v_2 = 1$. Since $\lambda_2 \neq 0$, $w_2^\top \pi_b = 0$. Thus

$$b_2(\beta) = w_2^\top[\pi(\beta) - \pi_b] = w_2^\top \pi(\beta).$$

All initial temperatures here are finite and positive. Cooling has $0 < \beta_h < \beta_c < \beta_b$, with anomalous ordering $|b_2(\beta_h)| < |b_2(\beta_c)|$. Heating instead has $\beta_b < \beta_h < \beta_c$ and requires $|b_2(\beta_c)| < |b_2(\beta_h)|$.

With a simple slow mode,

$$p(t;\beta) - \pi_b = e^{\lambda_2 t} b_2(\beta) v_2 + O(e^{\lambda_3 t}).$$

Consequently a strict overlap comparison gives the same eventual comparison in every fixed norm. For relative entropy,

$$D_{\mathrm{KL}}(p(t;\beta)\|\pi_b) = \tfrac{1}{2} e^{2\lambda_2 t} b_2(\beta)^2 \sum_i v_{2i}^2/\pi_{bi} + o(e^{2\lambda_2 t}).$$

The hotter initial Gibbs state is initially farther in this distance because

$$\frac{d}{d\beta} D_{\mathrm{KL}}(\pi(\beta)\|\pi_b) = (\beta - \beta_b)\,\mathrm{Var}_{\pi(\beta)}(E).$$

Thus the example gives a genuine initial-to-eventual distance reversal. It does not claim reversal at every time or for arbitrary distance functionals without appropriate hypotheses. A strong effect would require a nonbath preparation with zero slow overlap; that is not the witness used here.

Holding the bath, generator and eigenvectors fixed gives

$$b_2'(\beta) = -\,\mathrm{Cov}_{\pi(\beta)}(w_2, E) = -\sum_{i<j} \pi_i(\beta)\pi_j(\beta)(w_{2j} - w_{2i})(E_j - E_i).$$

A stationary point alone does not prove a maximum. We instead certify two strict overlap values on the same bath side. Since $b_2(1) = 0$, their increasing absolute ordering also forces an interior maximum somewhere between β_h and the bath.

3 Main result and exact counterexample

Theorem 1 (Counterexample to the published finite-temperature inference)**.** *There exist four distinct ordered energies, a finite positive bath temperature and a strictly positive reversible generator with a simple slowest nonzero eigenvalue such that all four original-rate predicates P_{ijk} fail, yet two finite cooling preparations obey $|b_2(\beta_h)| < |b_2(\beta_c)|$. Thus the antecedent of $(\mathcal{C}_4)$ holds and its conclusion is false.*

Proof. Set

$$\beta_b = 1, \quad q = (1, 199/200, 2/5, 1/400), \quad E = (0, \log(200/199), \log(5/2), \log 400).$$

These energies are distinct and strictly ordered. The downward rates in the order $(12, 13, 14, 23, 24, 34)$ are

$$(1, 28, 249/10, 1799/8, 4983/200, 251/25).$$

Set upward rates by $a_{ji} = a_{ij} q_j / q_i$ for $i < j$. The full generator is

$$M = \begin{pmatrix} -49029/4000 & 1 & 28 & 249/10 \\ 199/200 & -7280583/79600 & 1799/8 & 4983/200 \\ 56/5 & 17990/199 & -1011751/4000 & 251/25 \\ 249/4000 & 4983/79600 & 251/4000 & -11971/200 \end{pmatrix}.$$

Every off-diagonal entry is positive. Its columns sum to zero and $M\,\mathrm{diag}(q)$ is symmetric. Since $Q = 959/400$, the equilibrium is $\pi_b = (400, 398, 160, 1)/959$.

Exact original-rate triplet evaluations are:

Triplet	κ	F	G
123	7225/57	−1176493/4975	−39403/200
124	3321/133	−32119/1592000	−279/2000
134	1747/1960	−7761/4000	−2/25
234	1391209/12548025	−190400119/1592000	−119/4975

All four predicates fail strictly, with two upper branches and two lower branches. No boundary convention is needed.

The characteristic polynomial is $xC(x)/1273600000000$, where

$$C(x) = 1273600000000x^3 + 530473008000000x^2 + 35732953114072400x + 510539126345027563.$$

Exact opposite endpoint signs show one root in each of $(-337,-336)$, $(-60,-59)$ and $(-20,-19)$. These three disjoint sign-changing intervals exhaust a cubic's roots; all are simple. The largest, denoted r, is λ_2. Approximately,

$$(\lambda_2,\lambda_3,\lambda_4) = (-19.86836952145, -59.91747997894, -336.72876105237).$$

In particular the slow-mode spectral gap exceeds 39 in these time units.

Let $P_i(x)$ be the explicit cofactor polynomials listed in Appendix A. Their common factor corresponds to the first column of $\mathrm{adj}(M^\top - xI)$. Put $w_i = P_i(r)/P_1(r)$. The nonvanishing denominator is certified by the rational isolating interval below. The adjugate identity proves $M^\top w = rw$. It also follows that $q^\top w = 0$. To impose the required biorthogonal normalization set

$$N = \sum_i q_i w_i^2 > 0, \qquad v_i = q_i w_i / N.$$

Detailed balance gives $Mv = rv$ and $w^\top v = 1$. Thus $w = w_2$, $v = v_2$ is a valid consistently normalized pair. Approximately,

$$w = (1, -0.75893729814493, -0.61211930644096, -0.0038663077658171).$$

The relevant overlap is exactly

$$b_2(\beta) = \frac{\sum_i w_i q_i^\beta}{\sum_i q_i^\beta}.$$

For $\beta_h = 1/1000$ and $\beta_c = 1/100$, rational interval certification gives the following deliberately widened enclosures:

$$-0.0937454280 < b_2(\beta_h) < -0.0937454279,$$
$$-0.0938488357 < b_2(\beta_c) < -0.0938488356.$$

Every displayed terminating decimal denotes an exact rational number. In particular

$$|b_2(\beta_c)| - |b_2(\beta_h)| > 0.0001034076 > 0.$$

Both preparations have finite temperatures, $T_h = 1000$ and $T_c = 100$ in units with $k_B = 1$, versus $T_b = 1$. The strict overlap comparison and spectral expansion establish the ordinary cooling Mpemba effect. The verification construction below proves the enclosures without relying on floating-point acceptance tests. This completes the counterexample. □

4 Mathematical explanation and supporting applications

4.1 Why the triplet inference fails

The highest-energy state's slow-mode value is almost zero, while the intermediate states have strongly negative values. At very high initial temperature, occupation of state 4 therefore dilutes the negative slow overlap. The slightly cooler preparation depopulates state 4 enough to increase the magnitude of that overlap. At still lower temperatures, transfer toward the positive ground-state component returns the overlap to zero at the bath. This explanation concerns projection onto a relaxation mode, not merely mean energy.

An isolated triplet has a different eigenvector and a different equilibrium normalization from the full four-state system. Pairwise detailed balance and the inequalities $F, G < 0$ do not control the orientation of the full slow eigenvector relative to the entire Gibbs curve. In particular, nonmonotone slow eigenvectors are permitted. This explains the missing implication in a triplet-only argument.

All rates, F and G are strict; the slow eigenvalue is separated; the overlap gap is strict. Continuous dependence of a simple eigenprojection, Gibbs probabilities and the predicates on rates and energies therefore yields an open neighborhood of counterexamples, including varying energies. This is not an equality-surface artifact. No numerical radius for that neighborhood is claimed.

There is also an explicit one-parameter family. Let $R_0 = \pi_b \mathbf{1}^\top - I$ and $M_\rho = M + \rho R_0$ for $\rho \geq 0$. Its off-diagonal reset rates are $\rho\pi_{bi}$, for which $F = G = 0$ on every triplet. Thus F, G remain unchanged. On the zero-sum subspace $R_0 = -I$, so every nonzero eigenvalue shifts by $-\rho$ and every mode and overlap is preserved. Positivity and the counterexample persist for all $\rho \geq 0$. A subsequent common positive time rescaling also preserves the result. More explicitly, for every $\alpha > 24$, the same q and the downward rates

$$(\alpha - 24, \alpha + 3, \alpha - 1/10, 199\alpha/200 + 200, 199\alpha/200 + 1/25, 2\alpha/5 + 1/25)$$

form a counterexample. Their generator equals $M + (\alpha - 25)(q\mathbf{1}^\top - QI)$. All nonzero eigenvalues shift by $-(\alpha - 25)Q$, while their ordering, modes, overlaps and all F, G remain unchanged. Positivity follows directly from $\alpha > 24$.

4.2 A predictive mechanism and an exact onset threshold

The following construction controls the eigenvector rather than interpreting it after diagonalization. It extends the rational-mode companion in Appendix B, not merely the reset direction: two rates and the slow eigenvector change with a parameter. Throughout the entire parameter interval, the four triplet predicates fail. Nevertheless, the family crosses a sharp boundary between ordinary and anomalous asymptotic cooling.

Lemma 2 (A single-turn criterion for a nearly neutral top state)**.** *Let $E_1 = 0 < E_2 < E_3 < E_4$, $\beta_b = 1$, and suppose the simple slow left mode, normalized with $w_1 = 1$, is*

$$w = (1, -p, -r, z), \qquad p > r > -z > 0, \qquad \sum_i w_i e^{-E_i} = 0.$$

Assume also

$$E_4 > E_2 + E_3, \qquad (1 + r)E_3 > (p - r)(E_3 - E_2).$$

Write $E_\Sigma = E_2 + E_3 + E_4$ and

$$D_0 = (1 - p - r + z)E_\Sigma + 4(pE_2 + rE_3 - zE_4).$$

There is an ordinary asymptotic cooling Mpemba effect at finite positive initial temperatures if and only if $D_0 < 0$. In that case $|b_2(\beta)|$ has exactly one maximum on $0 < \beta < 1$. If $D_0 \geq 0$, it strictly decreases throughout that interval. There is no asymptotic heating reversal for $\beta > 1$ under these hypotheses.

Proof. Let $N(\beta) = \sum_i w_i e^{-\beta E_i}$ and $Z(\beta) = \sum_i e^{-\beta E_i}$, so $b_2 = N/Z$. All three excited-state coefficients are negative, hence $N'(\beta) > 0$. Since $N(1) = 0$, $b_2 < 0$ for $0 \le \beta < 1$ and $b_2 > 0$ for $\beta > 1$.

Define $D(\beta) = Z(\beta)^2 b_2'(\beta)$. Its pair expansion has three positive and three negative terms:

$$\begin{aligned} D(\beta) = c_{12}e^{-\beta E_2} + c_{13}e^{-\beta E_3} + c_{14}e^{-\beta E_4} \\ - c_{23}e^{-\beta(E_2+E_3)} - c_{24}e^{-\beta(E_2+E_4)} - c_{34}e^{-\beta(E_3+E_4)}, \end{aligned}$$

where $c_{1j} = (w_1 - w_j)E_j > 0$ and $c_{ij} = (w_j - w_i)(E_j - E_i) > 0$ for $2 \le i < j \le 4$. In particular $c_{13} > c_{23}$. Multiplying by a positive exponential gives

$$\begin{aligned} J(\beta) := e^{\beta E_4} D(\beta) = c_{12}e^{\beta(E_4-E_2)} + e^{\beta(E_4-E_2-E_3)}(c_{13}e^{\beta E_2} - c_{23}) \\ + c_{14} - c_{24}e^{-\beta E_2} - c_{34}e^{-\beta E_3}. \end{aligned}$$

Every nonconstant summand is strictly increasing for $\beta \ge 0$; the second is a product of positive increasing functions. Thus $J' > 0$. Also $J(1) > 0$ because $N'(1) > 0$ and $N(1) = 0$. Direct evaluation gives $J(0) = D_0$. If $D_0 < 0$, there is exactly one zero $\beta_m \in (0, 1)$, with $b_2' < 0$ before it and $b_2' > 0$ afterwards. As $b_2 < 0$, this is exactly a maximum of its absolute value. Two finite $0 < \beta_h < \beta_c < \beta_m$ give a strict cooling witness. If $D_0 \ge 0$, then $J(\beta) > 0$ for every $\beta > 0$, excluding any such reversal. Finally $J > 0$ and $b_2 > 0$ for $\beta > 1$, giving ordinary heating ordering. □

The condition uses $\beta = 0$ only as an analytic endpoint. Infinite initial temperature is not required for any witness. The single-turn proof is essential: the sign of an endpoint derivative alone would not generally exclude a maximum elsewhere.

Rates that force this geometry while every triplet fails. Keep $q = (1, 199/200, 2/5, 1/400)$ and set

$$p = \frac{759}{1000}, \quad r_0 = \frac{48959}{80000}, \quad \eta = \frac{1}{160}, \quad -\frac{3}{100} \le z \le -\frac{1}{250}.$$

Use $w(z) = (1, -p, -r_0 - \eta z, z)$. The small compensation ηz makes $q^\top w(z) = 0$ exactly. Define the downward rates by

$$a_{12} = 1, \qquad a_{13} = 28, \qquad a_{14} = 249/10, \qquad a_{24} = 4983/200,$$

$$\begin{aligned} a_{23}(z) &= \frac{528238451563 - 2257386450z}{200000(11761 - 500z)}, \\ a_{34}(z) &= \frac{2(1550z^2 - 5989657z + 1197903)}{5(80500z + 48959)}. \end{aligned}$$

As before, $a_{ji} = a_{ij}q_j/q_i$ and the diagonal enforces column conservation. These formulas arise by taking the first component of $Lw = \lambda w$ to define

$$\lambda(z) = -\frac{1550z + 3973343}{200000},$$

then solving its second and fourth components for a_{23} and a_{34}. The third follows from bath orthogonality and stationarity; direct substitution checks all four. This is a rate-level design prescription. At $z = -1/250$ it reproduces Appendix B exactly.

Theorem 2 (A controlled triplet-failing family with a sharp cooling threshold)**.** *Every generator just defined is strictly positive and reversible, fails all four original triplet tests strictly, and has the displayed $w(z)$ as its simple slow mode. Put*

$$U = (1-p-r_0)E_\Sigma + 4(pE_2 + r_0E_3),$$
$$V = (3+\eta)E_4 - (1-\eta)E_2 - (1+3\eta)E_3 > 0, \qquad z_* = U/V.$$

Within the stated closed parameter interval, an ordinary asymptotic cooling effect occurs exactly when

$$\boxed{z > z_*, \qquad -0.017940112 < z_* < -0.017940111.}$$

For $z \le z_*$ there is no asymptotic cooling effect. For $z > z_*$ the cooling amplitude has a unique maximum; the family has no asymptotic heating effect.

Proof. Both rate denominators stay positive. Exact polynomial bounds throughout the interval give $224 < a_{23} < 225$, $10 < a_{34} < 12$, and the following uniform bounds:

Triplet	Upper bound on F	Upper bound on G
123	-230	-196
124	$-1/50$	$-13/100$
134	$-19/10$	$-2/25$
234	-118	$-3/100$

Every comparison is strict; $123, 124$ stay in the upper κ branch and $134, 234$ in the lower branch.

To identify the slow mode uniformly, let M_0 use the four fixed rates above and $a_{23} = 224$, $a_{34} = 10$. An exact $LDL^\top$ congruence of $\mathrm{diag}(q)(M_0^\top + 50I)$ has successive pivots

$$\frac{150971}{4000}, \quad -\frac{494381936093}{12077680000}, \quad \frac{2188881388080131}{19775277443720}, \quad -\frac{432951459178371}{17511051104641048}.$$

There are two negative pivots, so M_0 has exactly two eigenvalues below -50. Increasing conductances increases the reversible Dirichlet form. The min–max principle therefore puts two eigenvalues of every $M(z)$ below -50 as well. The explicit eigenvalue satisfies $-20 < \lambda(z) < -19$; the remaining eigenvalue is the simple stationary zero. Consequently $\lambda(z)$ is the simple slow nonzero eigenvalue, with separation exceeding 30. This rules out realizing the desired geometry in a faster mode by mistake.

The hypotheses of Lemma 2 hold uniformly, and substitution of $r = r_0 + \eta z$ gives $D_0 = U - Vz$. The exact logarithm enclosures described below prove $V > 0$ and the displayed bracket for z_*. Apply the lemma, including its equality case. □

Predictions for finite preparations and for onset. The pair $\beta_h = 1/1000$, $\beta_c = 1/100$ has its own exact threshold. Write

$$u(\beta) = \frac{1 - pq_2^\beta - r_0 q_3^\beta}{Z(\beta)}, \qquad v(\beta) = \frac{q_4^\beta - \eta q_3^\beta}{Z(\beta)}, \qquad b_z = u + zv.$$

Since both overlaps are negative, this pair reverses if and only if

$$z > z_{\mathrm{pair}} := -\frac{u(1/1000) - u(1/100)}{v(1/1000) - v(1/100)} \simeq -0.01480024798314.$$

The denominator is certified positive by integer-power enclosures. For $z_* < z \le z_{\mathrm{pair}}$ other, sufficiently high but finite temperatures witness the effect; this particular pair does not. This distinguishes existence of the effect from success of a chosen temperature grid.

Let $d_* = \partial_\beta D(0, z_*) > 0$ and $\delta = z - z_*$. Analyticity and the implicit function theorem give, as $\delta \downarrow 0$,

$$\beta_m(z) = \frac{V}{d_*}\delta + O(\delta^2),$$
$$|b_z(\beta_m)| - |b_z(0)| = \frac{V^2}{32 d_*}\delta^2 + O(\delta^3).$$

Here $V/d_* \simeq 1.76871$ and $V^2/(32d_*) \simeq 0.943684$ in the fixed normalization $w_1 = 1$. The factors follow from $Z(0) = 4$, $b_z'(0) = -V\delta/16$ and $b_{z_*}''(0) = d_*/16$. The peak is born at the high-temperature endpoint with a linearly opening inverse-temperature window and a quadratically small advantage. Figure 1 illustrates this prediction.

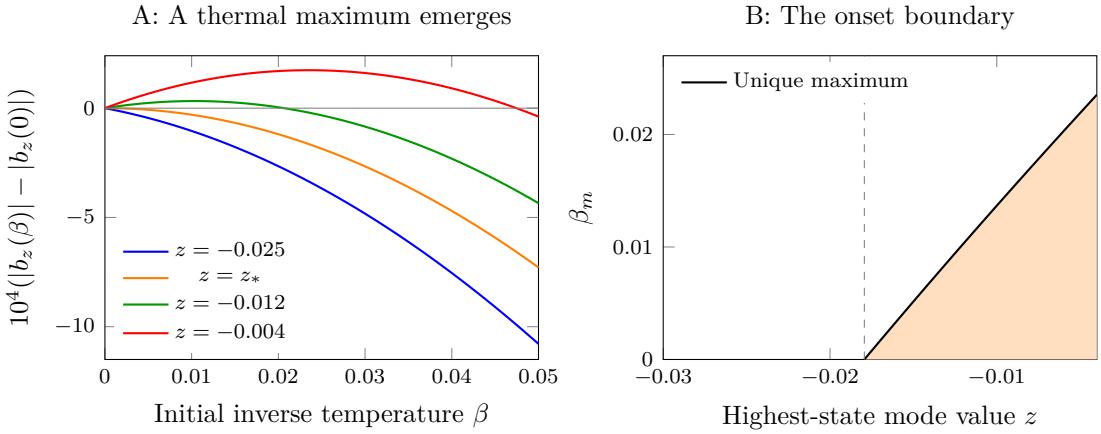


Figure 1: Predictive family of Theorem 2. All four triplet predicates fail throughout. Above z_* the amplitude initially increases with inverse temperature and has exactly one cooling maximum; below z_* it decreases throughout cooling. The shaded region has $\partial_\beta |b_z| > 0$, so any two finite inverse temperatures inside one vertical slice give a witness. The $\beta = 0$ limit is only a reference for the left ordinate. Curves are numerical; the onset and single-turn classification have a continuous-parameter proof.

This makes the neutral-state explanation predictive. Raising z toward zero decreases the negative slow-mode weight carried by state 4, while its small bath weight requires only the weak adjustment ηz in state 3. The two rate formulas enforce that geometry, the spectral comparison makes it dynamically dominant, and $U - Vz < 0$ quantifies when high-temperature occupation of state 4 reverses thermal ordering. The theorem classifies this controlled family and the stated single-turn geometry; it does not classify every four-state counterexample or every finite-time transient crossing. The covariance and spectral tools are established; no exhaustive priority claim is made for this particular construction or lemma.

Continuous-parameter verification. Run `research/mechanism_certificate.py`. It checks the rational eigenvector identities, all original triplet inequalities and the two branches on the entire interval. For each rational inequality, it clears the nonvanishing denominator, substitutes $z = -3/100 + (13/500)t$, and verifies positive Bernstein coefficients on $0 \le t \le 1$. Since the Bernstein basis is nonnegative and sums to one, these are certificates over a continuum, not sampled evidence. The script records every coefficient and the baseline inertia pivots; a second spectral check bounds the remaining quadratic using its value at -50 and the mean of its two real roots.

Logarithms are reduced to arguments in $[1,2]$. With $x = (a-1)/(a+1)$, the positive series $2\sum_{k=0}^{m-1} x^{2k+1}/(2k+1)$ bounds $\log a$ from below, and its remainder is at most $2x^{2m+1}/[(2m+1)(1-x^2)]$. Rational bounds with $m = 45$ certify the onset bracket and coefficient signs. Integer bisection bounds the algebraic Gibbs weights for the finite-pair threshold. Separately, `research/mechanism_plots.py` checks eight family members against symmetric diagonalization and numerical differentiation at 65-digit precision. Those checks and the plotted curves are illustrative cross-checks, not the continuous proof.

4.3 Visible reversal and its size

Figures 2 and 3 separate two facts: the hotter initial state has a smaller slow-mode amplitude, and the full time-dependent distance actually changes ordering. These use the primary generator in Theorem 1, with the bath and generator fixed throughout. The high-temperature amplitude reaches its numerical maximum near $\beta = 0.0237422$.

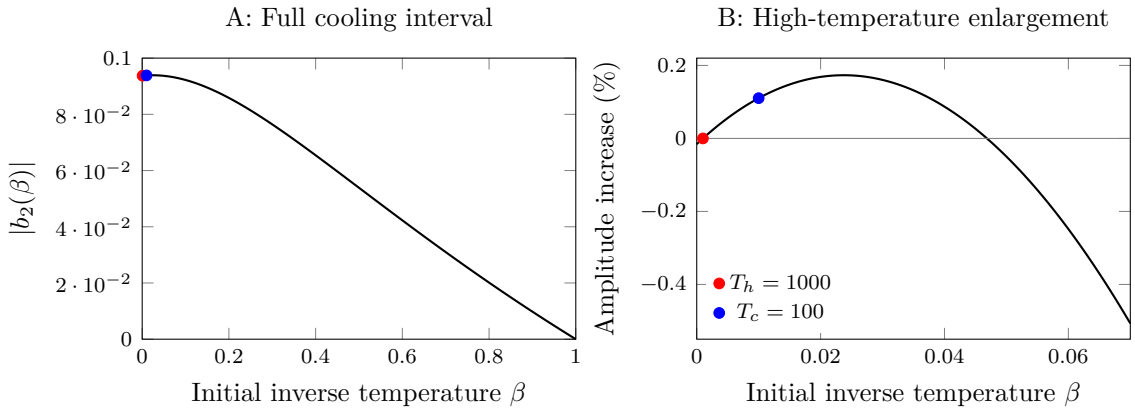


Figure 2: Thermal overlap of the exact primary generator, with $w_1 = 1$ and $T_b = 1$. The broad curve conceals a small high-temperature maximum; the enlargement displays it. The blue preparation has a larger slow amplitude than the hotter red preparation. The enlarged ordinate uses the hot amplitude as its reference; the quoted 0.1102% reduction uses the cold amplitude. Numerical curves use 80-digit arithmetic; the two selected overlaps are independently certified exactly.

Initially $D_h(0) \simeq 1.210137886$ and $D_c(0) \simeq 1.155489557$, so the hotter preparation is farther away in relative entropy. The two distances cross numerically near $t = 0.14967233$. Exact rational bounds prove a crossing in $(0.14, 0.16)$: at the endpoints,

$$D_h(0.14) - D_c(0.14) \in (5.9361291891, 5.9361291893)\,10^{-8},$$
$$D_h(0.16) - D_c(0.16) \in (-1.3100324358, -1.3100324356)\,10^{-8}.$$

Total variation also reverses, with a certified crossing in $(0.10, 0.11)$ and numerical location 0.10436410. The difference between these times is expected: finite-time crossings depend on the distance. Continuity plus the certified signs proves existence; no uniqueness or earliest-crossing theorem is asserted.

The relevant fractional amplitude reduction is

$$\epsilon_b = 1 - \frac{|b_h|}{|b_c|} \simeq 0.0011018531 = 0.11018531\%.$$

For relative entropy the leading term is quadratic, so

$$1 - \lim_{t\to\infty} \frac{D_h(t)}{D_c(t)} = 1 - \frac{b_h^2}{b_c^2} \simeq 0.22024921\%.$$

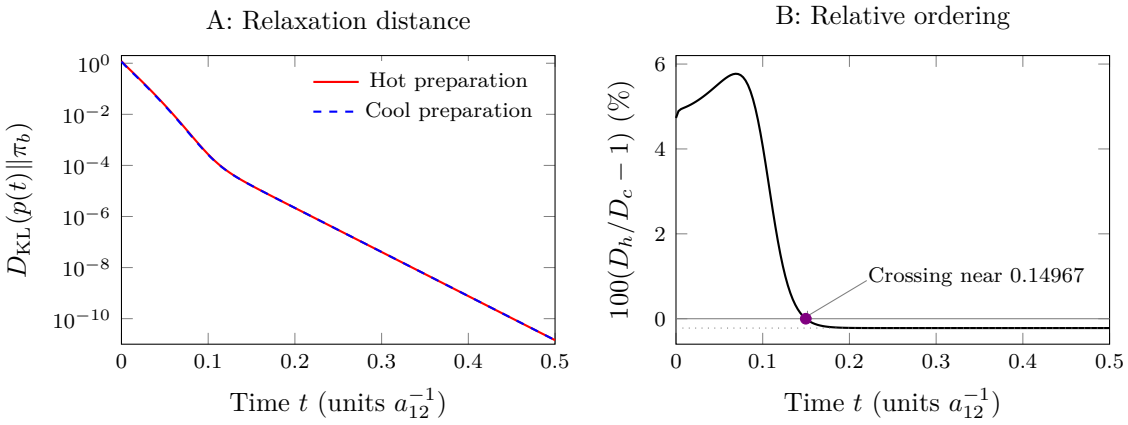


Figure 3: Actual evolution $p(t) = e^{tM}\pi(\beta)$ for the same two preparations. Relative entropy decreases for each, and the hotter state's distance becomes smaller. The ratio panel resolves the crossing that is difficult to see on the logarithmic plot. The horizontal dotted line is the asymptotic ratio difference, -0.2202492%. Rational uniformization and logarithm bounds certify opposite KL orderings at $t = 0.14$ and 0.16; the displayed root location is numerical, with no claim of uniqueness.

Both preparations have the same decay exponent λ_2. This is an ordinary effect, not the exponentially faster strong effect. For an increasingly small common distance threshold, the asymptotic time advance is

$$\Delta t = \frac{\log(|b_c|/|b_h|)}{|\lambda_2|} \simeq 5.5488\,10^{-5}\,a_{12}^{-1}.$$

The slow relaxation time is $1/|\lambda_2| \simeq 0.0503313\,a_{12}^{-1}$, so this advance is only about 0.11025% of one slow relaxation time. It is not a percentage reduction in the full cooling time, which depends on the chosen threshold. The time unit has no assigned experimental value, and $T_h/T_b = 1000$, $T_c/T_b = 100$. These data establish a strict mathematical failure of the triplet inference; they do not establish a practically useful cooling protocol.

4.4 Triplet-rate applications of stochastic monotonicity

Let $L = M^\top$ act on functions, $(Lf)_i = \sum_{j\neq i} a_{ji}(f_j - f_i)$. Define adjacent differences $y_i = f_{i+1} - f_i$, the difference map $Bf = y$, and $(H_0 y)_j = \sum_{\ell<j} y_\ell$. Then $BL = KB$ with $K = BLH_0$, since $L\mathbf{1} = 0$. In the constant-plus-differences basis, L is block triangular with blocks 0 and K, so K has precisely the nonzero generator eigenvalues. For its off-diagonal entries,

$$K_{i\ell} = \begin{cases} \sum_{j\le\ell}(a_{j,i} - a_{j,i+1}), & \ell < i, \\ \sum_{j>\ell}(a_{j,i+1} - a_{j,i}), & \ell > i. \end{cases}$$

Theorem 3 (Application of stochastic monotonicity)**.** *For any finite number of states, strictly ordered energies, a fixed $0 < \beta_b < \infty$, positive reversible rates and a simple slowest nonzero eigenvalue, if every off-diagonal entry of K is nonnegative, there is no asymptotic cooling or heating Mpemba effect at positive initial temperatures.*

Proof. Such a matrix K is called Metzler. After a sufficiently large scalar diagonal shift it is nonnegative. Perron theory gives a nonzero nonnegative eigenvector y at its spectral abscissa, which is λ_2. This does not require irreducibility of K. Choose w with $Bw = y$ and $\pi_b^\top w = 0$. Then $B(Lw - \lambda_2 w) = 0$, so $Lw - \lambda_2 w$ is constant. Its bath expectation is zero, hence it vanishes. Thus w is the slow left eigenfunction, nondecreasing and nonconstant in energy.

The covariance identity now gives $b_2'(\beta) < 0$ for every finite positive β: all its pairwise summands are nonnegative and at least one is strictly positive. Because $b_2(\beta_b) = 0$, its absolute value decreases toward the bath on the cooling side and increases away from the bath on the heating side. These are exactly the normal asymptotic orderings. □

In four states, the six off-diagonal entries are

$$\begin{aligned} K_{12} &= a_{32} + a_{42} - a_{31} - a_{41}, & K_{13} &= a_{42} - a_{41}, \\ K_{21} &= a_{12} - a_{13}, & K_{23} &= a_{43} - a_{42}, \\ K_{31} &= a_{13} - a_{14}, & K_{32} &= a_{13} + a_{23} - a_{14} - a_{24}. \end{aligned}$$

The original failures already ensure the three upper entries are nonnegative, since $G \le 0$ orders upward rates. A corrected, directly checkable sufficient condition is therefore the four original failures together with

$$a_{12} \ge a_{13}, \qquad a_{13} \ge a_{14}, \qquad a_{13} + a_{23} \ge a_{14} + a_{24}.$$

This is a proved sufficient condition, not a proposed necessary characterization.

Corollary 1. *If every triplet has $\kappa \le 1$ and fails the original predicate, then the absence conclusion is true, for any state count under the theorem's assumptions.*

Proof. $H \le 0$ and $G \le 0$ give $A + B \le C(1 + t)$ and $B \ge At$, so $A \le C$. These comparisons make each downward summand in K nonnegative; $G \le 0$ and detailed balance do the same for each upward summand. Apply the theorem. □

A second subfamily consists of downward gap laws $a_{ij} = f(E_j - E_i)$, $i < j$, with positive nonincreasing f. Both $f(x)$ and $e^{-\beta_b x} f(x)$ are nonincreasing, so every required summand is nonnegative. This establishes the same absence conclusion directly, without the false general triplet conjecture. The following comparison locates this argument within established order theory; no general monotonicity novelty is claimed.

4.5 Relation to earlier no-Mpemba theorems

Ohga, Hayakawa and Ito (OHI), Result 2 and SI D already use monotone coupling to exclude Mpemba behavior. Their SI D2 explicitly treats a total order. Our matrix condition is precisely a finite-chain expression of stochastic monotonicity, not a separate physical principle. The relevant distinctions are:

	OHI Result 2	The preceding spectral criterion
State order	Finite partial order; energy respects it	Total order by distinct energies
Dynamics	Irreducible and Gibbs-stationary; reversibility unnecessary	Positive reversible rates
Order condition	Comparable jumps and a monotone coupling	$K = BLH_0$ is Metzler
Bath range	Hypotheses throughout a temperature interval	One fixed finite positive bath
Spectrum	No simplicity requirement	Simple slowest nonzero eigenvalue
Conclusion	No cooling mean-energy crossing at any time	No asymptotic cooling/heating reversal in fixed norms or KL

OHI SI B extends their Result 1, the microscopic characterization, to KL divergence; it does not state Result 2 as an all-time KL theorem. Thus the different observable and time scopes must not be conflated.

Indeed $BL = KB$ gives $Be^{tL} = e^{tK}B$. The Metzler condition is equivalent to e^{tL} preserving increasing functions: sufficiency follows from $e^{tK} \geq 0$, and necessity by differentiation at zero on coordinate vectors of adjacent differences. This also gives a monotone coupling. Choose r large enough that $I + L/r$ is stochastic and $I + K/r$ is entrywise nonnegative. Its transition rows are stochastically ordered. A common Poisson clock of rate r and common-uniform inverse-CDF updates preserve the order of two copies. On a total order, energy compatibility and comparability of jumps are automatic.

For fixed bath, $e^{tL}E$ is then increasing, so

$$\partial_\beta\{\pi(\beta)^\top e^{tL}E\} = -\operatorname{Cov}_{\pi(\beta)}(e^{tL}E, E) \leq 0.$$

This gives all-time energy ordering without reversibility or spectral simplicity. It explains the direct relation to OHI while avoiding an inference from a single observable to all possible distances.

The likelihood-ratio and stochastic-order methods below also have precedent in Fill and Kahn, Ann. Appl. Probab. 23 (2013), 1778–1816, specifically Proposition 3.2, Corollary 3.3 and the density-ratio identity (3.2). They compare different kernels from a common initial law and obtain TV, separation and, with additional assumptions, weighted L^2 comparisons. Here one kernel acts on different Gibbs preparations. Their argument is a methodological precedent, not a verbatim statement of the corollary below. We provide the following consequence for clarity, without a priority claim.

Corollary 2 (Application: all-time convex-divergence ordering)**.** *Let a finite irreducible generator be reversible with positive Gibbs equilibrium, on a total energy order, and let K be Metzler. For any convex Φ finite on positive arguments, with $\Phi(1) = 0$, define*

$$D_\Phi(p\|\pi_b) = \sum_i \pi_{bi}\Phi(p_i/\pi_{bi}).$$

For same-side Gibbs preparations, the farther initial temperature has at least as large a divergence at every $t \geq 0$. This includes KL and total variation, and requires no simple eigenvalue. For KL the inequality is strict at every finite time for distinct initial temperatures.

Proof. Write $p \succeq_{\mathrm{st}} q$ when $\sum_i p_i f_i \geq \sum_i q_i f_i$ for every increasing f. Gibbs laws are stochastically ordered by temperature, by the pairwise covariance identity. A monotone semigroup preserves this order. Thus for cooling $p_h(t) \succeq_{\mathrm{st}} p_c(t) \succeq_{\mathrm{st}} \pi_b$.

Reversibility implies $p_c(t)/\pi_b = e^{tL}[\pi(\beta_c)/\pi_b]$ componentwise. The initial ratio is proportional to $e^{(\beta_b-\beta_c)E_i}$ and is increasing; it stays increasing. Choose a nondecreasing subgradient g of Φ. Convexity and stochastic order give

$$D_\Phi(p_h\|\pi_b) - D_\Phi(p_c\|\pi_b) \geq \sum_i (p_{hi} - p_{ci})g(p_{ci}/\pi_{bi}) \geq 0.$$

For heating, $p_c \preceq_{\mathrm{st}} p_h \preceq_{\mathrm{st}} \pi_b$ and p_h/π_b is decreasing. Apply the same supporting-line inequality at p_h/π_b to obtain $D_\Phi(p_c\|\pi_b) \geq D_\Phi(p_h\|\pi_b)$. KL is strictly convex, and invertibility of e^{tM} keeps two distinct initial distributions distinct at finite t, proving strictness. □

This does not assert all-time ordering for every arbitrary norm. The original spectral proof supplies the every-fixed-norm conclusion asymptotically with a simple slow mode. Our counterexample violates monotonicity explicitly: $K_{21} = a_{12} - a_{13} = -27$. It therefore challenges neither OHI's theorem nor this corollary.

4.6 Broader literature assessment and contribution scope

The starting contribution is the explicit, certified failure of the published original-rate triplet implication. Sections 5–10 add a surviving all-dimension exclusion theorem and a finite four-state asymptotic classification; their precise novelty is assessed separately from the established tools used in their proofs. Section 4.2 also supplies a controlled family and a proved onset classification for a specified eigenvector geometry. This is a problem-specific structural result; its individual publication priority has not been established. The supporting results organize and apply established mathematics. The following comparison was checked on 11 October 2026; it separates mathematical validity from publication priority.

The matrix criterion and reset algebra have direct precedents. Baake and Sumner, *On equal-input and monotone Markov matrices*, v3, 25 September 2022, Fact 3.1 and Proposition 3.11, give the cumulative-coordinate characterization of stochastic monotonicity for matrices and generators. In their row-generator convention, let $T_{ij} = \mathbf{1}_{i \geq j}$. Its coordinates are (f_1, Bf), so

$$T^{-1}LT = \begin{pmatrix} 0 & * \\ 0 & K \end{pmatrix}.$$

The upper row outside the diagonal consists of outgoing-rate tail sums and is automatically nonnegative. Their transformed-generator criterion is therefore exactly our condition that K is Metzler. Our Perron and covariance argument is an application of that order structure, not a new monotonicity criterion.

Their Section 2.1 also treats equal-input generators $C - cI$, with identical rows in C and $C^2 = cC$. Our reset generator is the transpose of this construction, with $C = \mathbf{1}\pi_b^\top$ and $c = 1$. Its scalar action on nonstationary modes is elementary equal-input algebra. The useful calculation here is its compatibility with the particular triplet inequalities, $F(R_0) = G(R_0) = 0$; the general reset construction is not claimed as new.

The divergence corollary also follows from established thermomajorization. Vu and Hayakawa, *Thermomajorization Mpemba Effect*, v2, 6 January 2025, give the Lorenz-curve and all-convex-divergence characterizations of thermomajorization. The connection to our corollary is explicit. In cooling, both p_h/π_b and p_c/π_b are increasing, so both curves use the order $n, n-1, \ldots, 1$. At their common horizontal knots,

$$x_k = \sum_{i=k}^{n} \pi_{bi}, \qquad y_{h,k} = \sum_{i=k}^{n} p_{hi} \geq \sum_{i=k}^{n} p_{ci} = y_{c,k},$$

by stochastic ordering. Linear interpolation orders the entire curves: p_h thermomajorizes p_c. Heating uses the order $1, \ldots, n$ and lower-tail inequalities, so p_c thermomajorizes p_h. This yields the corollary from the known convex-divergence characterization. The supporting-line proof above is an alternative elementary derivation; neither derivation establishes priority for a new general theorem.

This absence argument does not assert a thermomajorization reversal for our counterexample. Vu and Hayakawa's Result 4 distinguishes that stronger notion from the scalar slow-overlap criterion. The counterexample's stated conclusions remain eventual fixed-norm and KL ordering, with separate finite-time KL and TV brackets.

What is claimed, and what is retained only as support.

- **Central result:** the four-state witness satisfying all original-rate triplet failures, its exact certification and the consequent refutation of $(\mathcal{C}_4)$. The checked sources do not supply this witness or settle that precise implication. This bounded finding is not an exhaustive priority guarantee.
- **Established tools:** reversible symmetrization, slow-mode overlaps, Perron theory, stochastic monotonicity, likelihood-ratio evolution, thermomajorization, and equal-input reset algebra. The report claims no methodological novelty for these tools or for rational certification itself.
- **Applications proved here:** the additional rate inequalities, the all-$\kappa \leq 1$ subfamily, the nonincreasing gap-law subfamily, and the all-time convex-divergence corollary. Their role is to explain sufficient conditions that survive the failed conjecture. They are not promoted as independently novel theorems.
- **Problem-specific deductions:** the identity $F = hG + dH$, the feasible-cone formulation, and reset compatibility simplify construction and verification. They are proved directly; their individual publication priority remains unestablished. The open-neighborhood result is a continuity consequence, not a new perturbation principle.

The review covers the exact source conjecture, OHI's coupling result and SI, Fill–Kahn's comparison and density-ratio argument, Baake–Sumner's matrix and generator criteria, and Vu–Hayakawa's thermomajorization formulation. Searches combined Mpemba, triplet, monotonicity, convex divergence, likelihood ratio, nonincreasing rates and equal-input generators. Relevant full-text sections were inspected, rather than inferred from titles. No identical all-lower-branch or gap-law statement was located in this bounded review; that absence is insufficient evidence of novelty. Unindexed work and exhaustive citation tracing remain outside the check. The counterexample remains a contribution independently of supporting priority questions. The added contrast-cone and classification results must be evaluated on their exact hypotheses and conclusions; the bounded search is not a guarantee that these consequences are unprecedented.

Additional sources for the classification and realization arguments. The explicit slow-mode realization is already proved in Klich et al., v3, 18 December 2018, Appendix VIII, particularly Eq. (67). Our rank-one formula is that construction in different normalization, and the rank-r version uses the same projector idea. The Mpemba index and parity criteria are likewise prior results from that work. Teza et al., v3, 16 April 2025, Sections 2.1 and 5.2.3, review those results. Both primary full texts were inspected on 11 October 2026. The university-hosted journal PDF failed to download; the arXiv review version was inspected instead, and differences in the final journal text have not been excluded.

The nonvanishing-Wronskian characterization of extended complete Chebyshev systems is classical; Gasull, Geyer and Mañosas (2020), Lemma 2.3, states the criterion used here. The Gibbs Wronskian formula and the convex-curve test are explicit applications. The weighted Helmert basis, Metzler semigroup positivity and Perron theory are also established. What is supplied here is the rate-to-contrast identity, the resulting asymmetric cooling/heating implications of the triplet premise, and the explicit finite thermal test. Searches combining Mpemba with Helmert, conductance ordering, strong-effect absence, Chebyshev systems and stationary-point counts did not locate matching statements in the inspected primary sources. This is a bounded attribution assessment, not an exhaustive priority claim.

An explicit prior strong-temperature count. A further search located Parker Emmerson, *Organizational Mechanics of Effective Cardinality Transitions*, Zenodo v1, with a stated publication date of 10 July 2026 and repository creation date of 17 July 2026. The full text was inspected on 11

October 2026. Theorem 7.3 applies the classical sign-variation bound to the slow-overlap numerator; Corollary 7.5 subtracts the forced bath zero. Thus the scalar nonbath strong-temperature bound $n-2$ is explicitly present in prior work and is not a novelty claim here. Section 11 gives a four-state sharpness example using a preparation landscape different from the stationary one. Warning 7.13 distinguishes simultaneous mode cancellation from the scalar result. Our rank-r bound $n-r-1$ addresses that simultaneous-cancellation problem; its priority remains subject to further review. The bound on *stationary points* and the derivative-curve classifier concern different objects from zeros of the overlap numerator. No statement about the manuscript's peer-review status is needed for this attribution.

The coexistence of strong cooling and heating in four states is also prior work: Bera, Walker and Vucelja, v1, 8 August 2023, Section IV and Fig. 4, give a four-state ring example and discuss nonoverlap of the strong effects in three states. Those passages were checked in the primary full text on 11 October 2026. Our rational coexistence example is a sharpness and verification construction, not the first observation of that phenomenon. The general derivative test also covers ordinary effects that need not cancel the slow mode.

The sharp full-triplet conclusions. The additional results in Sections 14 and 15.1 concern the precise original-rate premise: a sharp 25% amplitude-advantage supremum and an exact endpoint covariance criterion for ordinary cooling. Their proofs use a restriction on the first three components of the slow eigenfunction, established through a reversible Schur complement and three nonnegative-rate boundary faces. A separate Perron-coupling proof supports the same restriction. These are proposed problem-specific contributions, not new Schur-complement or Perron principles. Searches combining Mpemba with amplitude bounds, triplet constraints, four-state systems and energy-span thresholds located no matching theorem in the inspected primary sources. Expert assessment of priority remains necessary. The spacing-independent exclusion also explains why the March paper's finite-temperature sampling box lies entirely within a region where the original inference is valid.

5 What the triplet premise still guarantees in any dimension

The counterexample invalidates the proposed exclusion of ordinary cooling. A substantial part of the exclusion principle nevertheless survives, under only the G inequalities and for any number of states.

Let $q_i = \pi_i(\beta_b) > 0$ and define the symmetric reduced conductances

$$k_{ij} = \frac{M_{ij}}{q_i} = \frac{M_{ji}}{q_j}, \qquad i \neq j.$$

These are conductances divided by both equilibrium weights. For an ordered triplet $i < j < k$, its original-rate expression is $G_{ijk} = a_{ik}q_j/q_i - a_{jk} = q_j(k_{ik} - k_{jk})$. Thus the condition

$$k_{ik} \leq k_{jk} \quad \text{for all } i < j < k \tag{1}$$

is exactly $G_{ijk} \leq 0$ for every triplet. Failure of every original predicate implies this condition, by the branch-equivalence lemma. No F inequality is used in the following theorem.

Theorem 4 (No strong effect and no anomalous heating)**.** *Consider an irreducible reversible n-state generator, $n \geq 2$, with distinct ordered energies and $0 < \beta_b < \infty$. Suppose (1) holds. Then:*

1. *No nonbath Gibbs preparation at a finite positive temperature has zero projection onto the entire slowest nonstationary eigenspace. Hence there is no strong cooling or strong heating effect, including when that eigenspace is degenerate.*

2. *For any $\beta_f > \beta_n > \beta_b$ and every finite $t \geq 0$,*

$$\sum_i \frac{[p_i(t;\beta_f) - q_i]^2}{q_i} > \sum_i \frac{[p_i(t;\beta_n) - q_i]^2}{q_i}.$$

Thus heating has no finite-time weighted Euclidean reversal. It also has no asymptotic relative-entropy reversal, for any slow multiplicity.

3. *If the slow mode is simple, its thermal overlap can be oriented to be strictly negative throughout cooling, strictly positive throughout heating, and strictly increasing throughout heating. Consequently there is no asymptotic heating effect in any fixed norm.*

These statements do not exclude ordinary cooling, and do not assert finite-time relative-entropy or total-variation ordering.

Proof. Write $L = M^\top$, $Q_k = \sum_{i=1}^k q_i$, and define the weighted prefix contrasts, for $k = 1, \ldots, n-1$, by

$$h_i^{(k)} = \begin{cases} 1, & i \leq k, \\ -Q_k/q_{k+1}, & i = k+1, \\ 0, & i > k+1. \end{cases}$$

They have zero q-mean and are mutually orthogonal in $\langle f, g\rangle_q = \sum_i q_i f_i g_i$. Their squared norms are $\eta_k = Q_k Q_{k+1}/q_{k+1}$. Put $e^{(k)} = h^{(k)}/\sqrt{\eta_k}$. This is a weighted Helmert basis of the entire nonstationary function space.

Detailed balance makes the matrix $A_{kl} = \langle e^{(k)}, Le^{(l)}\rangle_q$ symmetric. For $k < l$, direct multiplication gives

$$\begin{aligned} \sqrt{\eta_k \eta_l} A_{kl} &= Q_{l+1} \sum_{i=1}^{k} q_i (k_{k+1,l+1} - k_{i,l+1}) \\ &\quad + \sum_{j=l+2}^{n} q_j \sum_{i=1}^{k} q_i (k_{k+1,j} - k_{ij}) \geq 0. \end{aligned} \tag{2}$$

To check the identity, note that for every $i \leq k+1 \leq l$, $(Lh^{(l)})_i = -Q_{l+1}k_{i,l+1} - \sum_{j=l+2}^n q_j k_{ij}$. The weighted contrast subtracts its value at $i = k+1$ from the weighted prefix values. Each resulting difference is nonnegative by (1). Thus A is a symmetric Metzler matrix. In particular e^{tA} is entrywise nonnegative for $t \geq 0$, with positive diagonal. This follows by choosing c so that $A + cI$ is entrywise nonnegative and using the exponential power series.

Let $g_k(\beta) = \sum_i e_i^{(k)} \pi_i(\beta)$. Apart from its positive normalization $\sqrt{\eta_k}$, its numerator is

$$\begin{aligned} N_k(\beta) &= \sum_{i=1}^{k} e^{-\beta E_i} - \frac{Q_k}{q_{k+1}} e^{-\beta E_{k+1}} \\ &= e^{-\beta E_{k+1}} \sum_{i=1}^{k} \left[e^{\beta(E_{k+1} - E_i)} - e^{\beta_b (E_{k+1} - E_i)} \right]. \end{aligned}$$

Every gap in this sum is strictly positive. Therefore every coordinate g_k is strictly negative for $\beta < \beta_b$, zero at the bath, and strictly positive for $\beta > \beta_b$.

Each g_k is also strictly increasing on the heating side. Indeed, put

$$T_k = (1 + Q_k/q_{k+1}) e^{-\beta E_{k+1}} + \sum_{j=k+2}^{n} e^{-\beta E_j}, \qquad Z = N_k + T_k.$$

For $\beta > \beta_b$, both N_k, T_k are positive and

$$\frac{N_k'}{N_k} > -E_{k+1}, \qquad \frac{T_k'}{T_k} \le -E_{k+1}.$$

The first inequality follows by differentiating the positive bracketed sum in N_k; the second is its interpretation as the negative weighted average of energies at least E_{k+1}. Consequently $(N_k/Z)' = (N_k' T_k - N_k T_k')/Z^2 > 0$.

The Gibbs likelihood ratio expands as $\pi(\beta)/q = \mathbf{1} + \sum_k g_k(\beta) e^{(k)}$. Under the reversible evolution its nonconstant coordinate vector becomes $e^{tA} g(\beta)$. For two heating preparations farther and nearer the bath, $g(\beta_f) > g(\beta_n) > 0$ entrywise. The nonnegative matrix e^{tA} with positive diagonal preserves both strict inequalities. Taking squared Euclidean norms of the coordinate vectors proves the finite-time assertion.

For the slow-space conclusions let λ_2 be the largest eigenvalue of A and let Π_2 be its orthogonal spectral projector. Symmetry gives

$$\Pi_2 = \lim_{t \to \infty} e^{-\lambda_2 t} e^{tA}.$$

Hence Π_2 is entrywise nonnegative and nonzero. Since $g(\beta)$ has strictly one sign in every coordinate, $\Pi_2 g(\beta) \neq 0$ for every nonbath β. This proves absence of strong effects without assuming simplicity. Moreover $\|\Pi_2 g(\beta)\|^2$ is strictly increasing on the heating side: its derivative is $2(\Pi_2 g)^\top (\Pi_2 g') > 0$. Strictness follows from any positive diagonal entry of the nonzero nonnegative projector, since $g, g' > 0$. The same quantity controls the leading relative-entropy coefficient.

If the slow mode is simple, Perron theory for a symmetric Metzler matrix allows its coordinate eigenvector v to be chosen nonnegative and nonzero. Then $b_2(\beta) = v^\top g(\beta)$ has the stated signs and heating monotonicity. Its scale does not affect any ordering conclusion. □

A larger rate family. The proof in fact requires only that the right-hand side of (2) be nonnegative for each $k < l$; the individual conductance inequalities are sufficient, but not necessary. In four states this larger, explicitly checkable family is given by just three inequalities:

$$\begin{aligned} Q_3(k_{23} - k_{13}) + q_4(k_{24} - k_{14}) &\ge 0, \\ k_{24} - k_{14} &\ge 0, \\ (q_1 + q_2)k_{34} - q_1 k_{14} - q_2 k_{24} &\ge 0. \end{aligned}$$

Every conclusion of Theorem 4 holds under these three aggregate inequalities. More generally there are $(n-1)(n-2)/2$ linear aggregate inequalities. They can be evaluated before any eigendecomposition.

For a simple four-state slow mode in this family, the anomalous cooling region $\{\beta \in (0, \beta_b) : \partial_\beta |b_2| > 0\}$ is either empty or one connected interval. Indeed $b_2 < 0$ on cooling and $b_2' > 0$ throughout heating, while the global derivative bound allows at most two stationary points counting multiplicity. The negative set of b_2' before the bath can therefore have at most one connected component. In this larger family its left endpoint is not determined by the present argument. Under the original full triplet premise, Corollary 5 proves that any such interval starts at $\beta = 0$ and supplies an exact endpoint criterion.

A corrected implication and its mechanism. For every finite n, failure of all the original triplet predicates excludes strong effects and excludes asymptotic heating in relative entropy; with a simple slow mode, heating is excluded in every fixed norm. The finite-time weighted Euclidean statement is stronger for its specified distance. The four-state counterexample therefore occupies a precise remaining possibility: ordinary cooling without eliminating the slow mode. Its small thermal maximum is compatible with a slow overlap that never changes sign on either side of the bath.

The proof uses a different order structure from monotone energy eigenvectors. A positive combination of prefix contrasts can have nonmonotone components, so the counterexample does not conflict with this cone. Nevertheless its weighted prefix sums are nonnegative. More directly, every Gibbs preparation on one side of the bath has coordinates of the same sign in this basis; a nonnegative slow spectral projector cannot cancel all of them. This is the mechanism excluding strong effects. On heating, all those coordinates also grow with the distance from the bath, which explains the additional exclusion. The basis, Metzler positivity, and Perron theory are established tools; the specific rate-to-contrast identity and its Mpemba consequences require a separate priority assessment.

6 A finite-time convex-divergence theorem for contrast chains

A stronger finite-time conclusion is available when cooperation in the contrast basis has a nearest-neighbor structure. This condition concerns the generator in contrast coordinates, not transitions between physical energy levels.

Theorem 5 (Heating order for a tridiagonal contrast generator)**.** *Let an irreducible reversible finite-state generator have strictly ordered energies and positive finite bath temperature. Assume all $G_{ijk} \leq 0$ and that its weighted Helmert contrast matrix is tridiagonal. Then for any two finite heating preparations $\beta_b < \beta_1 < \beta_2$, every convex divergence satisfies*

$$D_\Phi(p_t(\beta_2)\|q) \geq D_\Phi(p_t(\beta_1)\|q) \qquad (t \geq 0), \quad D_\Phi(p\|q) = \sum_i q_i \Phi(p_i/q_i),$$

where Φ is convex and finite on positive arguments, with $\Phi(1) = 0$. For strictly convex Φ the inequality is strict at every finite time. No spectral simplicity or irreducibility in contrast coordinates is required. The proof in fact needs only a Metzler tridiagonal contrast matrix, rather than the original G inequalities themselves.

Proof. Use the unnormalized Helmert matrix H and write the density ratio as $r = \mathbf{1} + Hx$. The contrast generator B is Metzler and tridiagonal, and $x(t, \beta) = P_t x(0, \beta)$, where $P_t = e^{tB}$. On heating both $x(0, \beta)$ and its thermal derivative $y(0, \beta) = \partial_\beta x(0, \beta)$ are strictly positive by the prefix-contrast calculation in Theorem 4.

Put $\delta = \beta - \beta_b > 0$, shift $E_1 = 0$, and write $Z_\delta = \sum_i q_i e^{-\delta E_i}$. Up to a positive coordinate normalization independent of δ,

$$x_k(0, \beta) = \frac{f_k(\delta)}{Z_\delta}, \qquad f_k = \sum_{i \leq k} q_i (e^{-\delta E_i} - e^{-\delta E_{k+1}}) = \delta \int_0^{E_{k+1}} Q(s) e^{-\delta s}\, ds,$$

where $Q(s) = \sum_{E_i \leq s} q_i$. The weighted mean of s in this truncated integral strictly increases with its upper endpoint. Consequently

$$\frac{y_k(0, \beta)}{x_k(0, \beta)} = \frac{1}{\delta} - \frac{Z'_\delta}{Z_\delta} - \frac{\int_0^{E_{k+1}} sQ(s) e^{-\delta s}\, ds}{\int_0^{E_{k+1}} Q(s) e^{-\delta s}\, ds}$$

is decreasing in k, and remains positive.

Every ordered 2×2 minor of P_t is nonnegative. One elementary proof uses the differential equation for the exterior square: its matrix is the second additive compound of B. Tridiagonality ensures that every off-diagonal entry of that compound is nonnegative, since one index can move only to a neighboring unoccupied index without crossing the other. Its exponential is therefore nonnegative, and its entries are precisely the ordered 2×2 minors of P_t. This is the classical total-positivity

implication of Schwarz (1970), Theorem 2; the argument covers zero couplings as well as irreducible contrast chains. For positive vectors x, y, expansion by paired column indices gives

$$(P_t y)_i (P_t x)_j - (P_t y)_j (P_t x)_i = \sum_{k<l} \det(P_t)_{\{i,j\},\{k,l\}} (y_k x_l - y_l x_k) \geq 0 \qquad (i < j).$$

It follows that the evolved ratios $R_k = y_k(t,\beta)/x_k(t,\beta)$ remain nonincreasing and positive. Positivity of the exponential's diagonal guarantees $x, y > 0$, even if some contrast couplings vanish.

First suppose Φ is differentiable, and let $g_k = \sum_i q_i H_{ik} \Phi'(r_i)$ be the coordinate gradient of the divergence. Its weighted prefix sums satisfy

$$S_k := \sum_{j \leq k} x_j g_j = \sum_{i \leq k+1} q_i (r_i - c_k) \Phi'(r_i) \geq 0, \qquad c_k = 1 + \sum_{j>k} x_j.$$

Indeed the higher contrasts are constant on this physical prefix, and $\sum_{i \leq k+1} q_i (r_i - c_k) = 0$. Thus S_k is the prefix mass times the covariance of r with the nondecreasing function $\Phi'(r)$, which is nonnegative by the pairwise covariance identity. Abel summation now yields

$$\partial_\beta D_\Phi = \sum_k R_k x_k g_k = R_{n-1} S_{n-1} + \sum_{k=1}^{n-2} (R_k - R_{k+1}) S_k \geq 0.$$

For strictly convex Φ, the full-prefix covariance S_{n-1} is strictly positive unless r is constant. At finite time the evolved state is not equilibrium, by invertibility of e^{tM}, while $R_{n-1} > 0$. This proves strictness.

For a nondifferentiable convex function, the same argument holds at almost every β, using subgradients at the finitely many density ratios. Subgradients at distinct arguments remain ordered, so the covariance calculation is unchanged, including its strictness for a strictly convex function. On each compact interval of finite heating temperatures, all density ratios are bounded away from zero and infinity; the divergence is therefore locally Lipschitz as a function of β. Integrating its almost-everywhere derivative proves the claimed ordering and strictness. Alternatively, convex smooth approximation proves the nonstrict assertion directly. □

The condition is broader than stochastic monotonicity of physical energy states. For example, the certified three-state generator used in the dimension caveat after Theorem 15 has $G_{123} < 0$ and a two-dimensional, automatically tridiagonal contrast matrix, but its physical adjacent-difference entry is $K_{21} = -14/27 < 0$. Thus this theorem does not merely restate the physical stochastic-order corollary. The argument uses classical total positivity and covariance tools; no novelty claim is made for those underlying principles.

6.1 The contrast-ratio argument does not extend automatically

The decreasing contrast-derivative ratios used in the preceding proof can fail for a physically reachable Gibbs preparation under the full triplet premise. This is an obstruction to that proof route, not a KL counterexample.

Take four bath weights $q_1 > q_2 > q_3 > q_4 > 0$, normalized to sum one, and reduced conductances, in pair order $12, 13, 14, 23, 24, 34$,

$$k^{(0)} = (2, 2, 1, 2, 2, 3).$$

The original tests are

ijk	F_{ijk}	G_{ijk}
123	0	0
124	$-q_1 - q_4/2$	$-q_2$
134	$-q_1 - q_3 - q_4$	$-2q_3$
234	$-q_3 - q_4/2$	$-q_3$

Add any positive multiple $a > 0$ of the constant-downward-one generator: $k_{ij} = k_{ij}^{(0)} + a/q_i$ for $i < j$. Every rate is positive and every F, G is now strict. The two differences

$$d_3 = k_{23} - k_{13}, \qquad d_4 = k_{24} - k_{14}$$

still satisfy $d_4 - d_3 = 1$.

Use orthonormal Helmert coordinates, with contrast generator A, $P_t = e^{tA}$, initial heating coordinates $x(\beta)$, and thermal derivative $y(\beta)$. As $\beta \to \infty$,

$$x_k(\beta) \longrightarrow x_{\infty,k} = 1/\sqrt{\eta_k}, \qquad y(\beta)/y_1(\beta) \longrightarrow (1, 0, 0)^\top.$$

The second limit holds because the first contrast derivative decays on the first energy-gap scale, while subsequent contrast derivatives begin at successively larger gaps. For fixed t the difference of evolved derivative ratios, divided by $y_1(\beta) > 0$, therefore tends to

$$f(t) = \frac{(P_t)_{31}}{(P_t x_\infty)_3} - \frac{(P_t)_{21}}{(P_t x_\infty)_2}.$$

Here $f(0) = 0$, while the exact Helmert off-diagonal identity gives

$$f'(0) = \sqrt{\eta_3} A_{31} - \sqrt{\eta_2} A_{21} = \frac{q_1 Q_3}{\sqrt{\eta_1}}(d_4 - d_3) > 0.$$

Thus $f(t) > 0$ for all sufficiently small positive times. Fix any such time; for a sufficiently large but finite heating inverse temperature,

$$\frac{(P_t y)_3}{(P_t x)_3} > \frac{(P_t y)_2}{(P_t x)_2}.$$

This is an actual thermal evolution satisfying all original triplet failures, but its derivative ratios are not decreasing. A proof of unrestricted finite-time KL heating order under all G must therefore use information beyond that monotone-ratio condition.

7 A uniform finite-time KL window around the bath

The full finite-time KL question under the ordered-conductance condition can also be settled on a quantitative temperature interval, without any additional sparsity or spectral assumptions.

Theorem 6 (A bath-neighborhood KL ordering window)**.** *Assume a finite irreducible reversible generator has distinct energies and every $G_{ijk} \le 0$. Put $c = E_n - E_1 > 0$. On either side of the bath, KL divergence increases strictly with the initial inverse-temperature distance from the bath at every finite time, provided that distance is at most*

$$|\beta - \beta_b|\, c \le \frac{2}{\sqrt{5}}.$$

Thus two same-side preparations within this interval have the usual finite-time KL ordering. Cooling preparations must additionally satisfy $\beta > 0$. The assertion is uniform in time and needs no simple slow mode. The numerical constant is a sufficient bound, not a sharp one.

Proof. Shift $E_1 = 0$ and let $\delta = |\beta - \beta_b| > 0$, $z = \delta c$. Use orthonormal weighted Helmert columns $e^{(k)}$ and let E denote their matrix in this proof. If $\sigma = \operatorname{sign}(\beta - \beta_b)$, the Gibbs density-ratio coordinates

have the form $\sigma x_0(\delta)$, where every component of x_0 is positive. Apart from positive constants, their numerator is, on heating,

$$\sum_{i\le k} q_i(e^{-\delta E_i} - e^{-\delta E_{k+1}}) = \delta \sum_{i\le k} q_i \int_{E_i}^{E_{k+1}} e^{-\delta s}\, ds,$$

and on cooling it is

$$\sum_{i\le k} q_i(e^{\delta E_{k+1}} - e^{\delta E_i}) = \delta \sum_{i\le k} q_i \int_{E_i}^{E_{k+1}} e^{\delta s}\, ds.$$

Divide by the corresponding relative partition function $Z_\sigma = \sum_i q_i e^{-\sigma\delta E_i}$. Both the integral's logarithmic derivative after removal of its leading δ, and that of the partition function, are signed averages of energies in $[0, c]$. Their difference therefore lies in $[-c, c]$. Consequently, coordinatewise,

$$\left(\frac{1}{\delta} - c\right) x_0 \le y_0 := \partial_\delta x_0 \le \left(\frac{1}{\delta} + c\right) x_0.$$

For $z < 1$ both bounds are positive. The contrast semigroup $P_t = e^{tA}$ is nonnegative, with positive diagonal, so the evolved positive vectors $x = P_t x_0$, $y = P_t y_0$ obey the same inequalities. Set $a = 1/\delta - c$, $b = 1/\delta + c$. Their Euclidean angle satisfies

$$\frac{x^\top y}{\|x\|\|y\|} \ge \frac{2\sqrt{ab}}{a+b} = \sqrt{1-z^2}.$$

For completeness, with weights $x_i^2/\|x\|^2$ and ratios $v_i = y_i/x_i \in [a, b]$, the bound follows from $\mathbb{E}[v^2] \le (a+b)\mathbb{E}[v] - ab$ and minimization of $\mathbb{E}[v]^2/\mathbb{E}[v^2]$.

Initially the density ratio lies in a positive interval $[r_-, r_+]$ whose endpoint ratio is exactly $r_+/r_- = e^z$. Its weighted mean is one, so the interval also contains one. The Markov backward semigroup preserves that interval. Thus $r = \mathbf{1} + \sigma E x$ at time t and the whole straight segment from equilibrium to r lie in the same interval. The Hessian of KL in orthonormal contrast coordinates is

$$T_s = E^\top D\, \mathrm{diag}\big((1 + s(r-1))^{-1}\big) E, \qquad mI \preceq T_s \preceq MI, \quad m = 1/r_+, \quad M = 1/r_-.$$

Here $M/m = e^z$. The gradient vanishes at equilibrium. Integrating the Hessian along this segment, put $T = \int_0^1 T_s\, ds$; the sign σ cancels in the outward temperature derivative, giving

$$\partial_\delta D_{\mathrm{KL}} = y^\top T x.$$

The bound $\|T - (M+m)I/2\| \le (M-m)/2$ and the angle estimate yield

$$\partial_\delta D_{\mathrm{KL}} \ge \frac{M+m}{2}\left[\sqrt{1-z^2} - \tanh(z/2)\right]\|x\|\|y\| > 0$$

whenever $0 < z \le 2/\sqrt{5}$: then $\sqrt{1-z^2} \ge z/2 > \tanh(z/2)$. Both vectors are nonzero at finite time. Integration between two same-side preparation temperatures gives the stated strict ordering. The result at the bath itself follows from zero equilibrium divergence and positivity at every distinct finite-time state. □

The argument actually covers the larger explicit range $\sqrt{1-z^2} > \tanh(z/2)$ within $0 < z < 1$; the simpler constant $2/\sqrt{5}$ avoids a transcendental threshold. It does not establish finite-time KL ordering for arbitrarily far preparations under general G-only dynamics. The proof uses exact inequalities rather

than a truncated linear-response expansion, so its time-uniform conclusion remains valid even near spectral degeneracy.

More generally, at any heating temperature define $a = \min_k g'_k/g_k > 0$ and $b = \max_k g'_k/g_k$ from the initial Gibbs contrast coordinates. If the evolved density ratios have condition number $\kappa = \max_i r_i / \min_i r_i$, the same argument gives the finite-time sufficient condition

$$\frac{\kappa - 1}{\kappa + 1} < \frac{2\sqrt{ab}}{a + b} \quad \Longrightarrow \quad \partial_\beta D_{\mathrm{KL}}(p_t(\beta)\|q) > 0.$$

The semigroup preserves the initial coordinate-ratio bounds, while $\kappa \to 1$ under thermalization. This supplies a quantitative route to finite-time certification at an arbitrary fixed heating preparation; it does not settle the derivative's sign before the condition holds.

7.1 A sharp energy-and-bath threshold for the contrast family

The contrast theorem also determines exactly which energy/bath choices exclude ordinary cooling *uniformly over the entire rate family.* Shift $E_1 = 0$, let $a = E_2 > 0$, and let $\mu = n^{-1} \sum_i E_i$ be the arithmetic mean energy, not the bath mean.

Theorem 7 (Uniform cooling threshold)**.** *Fix distinct ordered energies and a positive finite bath. The following are equivalent:*

1. *Every positive reversible generator satisfying $k_{ik} \le k_{jk}$ for $i < j < k$ has no ordinary asymptotic cooling effect in weighted Euclidean distance.*
2. *Either $\mu \le a$, or*
$$\beta_b \le \beta_* := \frac{1}{a} \log \frac{\mu}{\mu - a}. \tag{3}$$

Under condition 2, absence is stronger: it holds at every time in weighted Euclidean distance, for asymptotic relative entropy at any slow multiplicity, and asymptotically in every fixed norm when the slow mode is simple. The same universal equivalence holds for the larger family in which the contrast matrix is Metzler. Failure of condition 2 does not by itself assert existence under the additional original F inequalities.

Proof. For cooling, each contrast amplitude $R_k = -g_k$ is strictly positive. With $\alpha_k = Q_k/q_{k+1} = \sum_{i \le k} e^{\beta_b(E_{k+1} - E_i)}$,

$$R_k = \frac{e^{-\beta E_{k+1}}}{\sqrt{\eta_k Z}} \left(\alpha_k - \sum_{i=1}^{k} e^{\beta(E_{k+1} - E_i)} \right), \qquad \beta < \beta_b.$$

The logarithmic derivative of the bracket is negative and strictly decreasing, while $(\log Z)'' = \mathrm{Var}_{\pi(\beta)} E > 0$. Therefore $(\log R_k)'' < 0$: every cooling contrast amplitude is strictly log-concave. It follows that $g'_k(0) \ge 0$ implies $g'_k(\beta) > 0$ for every $0 < \beta < \beta_b$, including the equality case at 0.

Direct differentiation gives

$$n^2 \sqrt{\eta_k}\, g'_k(0) = n \sum_{i=1}^{k} \left[e^{\beta_b(E_{k+1} - E_i)}(E_{k+1} - \mu) + (\mu - E_i) \right]. \tag{4}$$

For $k = 1$ this has the sign of $\mu - e^{\beta_b a}(\mu - a)$. It is nonnegative precisely under condition 2. We show that this one condition makes every other expression nonnegative. Put $y = E_{k+1}$. If $y \ge \mu$, the sum

equals $\alpha_k(y-\mu)+k\mu-\sum_{i\le k}E_i>0$, because the mean of a proper ordered prefix is below the full arithmetic mean. If $y<\mu$, necessarily $a<\mu$. For every $i\le k$,

$$\frac{\log(\mu-E_i)-\log(\mu-y)}{y-E_i}=\frac{1}{y-E_i}\int_{E_i}^{y}\frac{dt}{\mu-t}\ge\frac{1}{a}\int_0^a\frac{dt}{\mu-t}=\beta_*.$$

The inequality holds because the integrand increases: for $i=1$, the interval starts at zero and extends at least to a; for $i>1$ its entire interval starts at or above a. Thus $\beta_b\le\beta_*$ implies $e^{\beta_b(y-E_i)}\le(\mu-E_i)/(\mu-y)$, making each summand in (4) nonnegative. Consequently all g_k strictly increase throughout cooling. Their negatives are ordered componentwise for two cooling preparations, and the positive contrast semigroup gives the same finite-time norm ordering as in Theorem 4. The spectral projection argument gives the asymptotic conclusions.

Conversely, suppose $a<\mu$ and $\beta_b>\beta_*$. The first contrast has $g_1'(0)<0$, so $|g_1|$ initially increases with inverse temperature and gives finite positive cooling witnesses. Realize $h^{(1)}$ as the simple slow mode with the established rank-one construction

$$M=-I+q\mathbf{1}^\top+\epsilon\frac{Dh^{(1)}(h^{(1)})^\top}{\eta_1}$$

for sufficiently small $0<\epsilon<1$. Positivity and the simple slow eigenvalue follow as before. Its reduced conductances are $k_{ij}=1+\epsilon h_i^{(1)}h_j^{(1)}/\eta_1$. For every $i<j<k$, $h_k^{(1)}=0$, so $k_{ik}=k_{jk}=1$. Thus all G inequalities hold at equality. Its contrast matrix is diagonal, so it also belongs to the larger family. This proves sharpness of the uniform threshold. □

For four equally spaced energies $(0,a,2a,3a)$, the threshold is $\beta_b a\le\log 3$. Throughout that range, failure of all triplet tests really does exclude ordinary cooling as well as the other effects covered above. Outside that range, the larger G-only family permits cooling, but the original F-constrained equal-spacing question is not settled by the sharpness example. For three equally spaced energies, $\mu=a$, so the G-only family has no cooling effect at any finite positive bath temperature.

Corollary 3 (A spacing-independent energy-span exclusion)**.** *For $n\ge3$, write $c=E_n-E_1$. If*

$$\beta_b c\le\frac{n}{n-2},$$

every positive reversible generator with a Metzler contrast matrix has no finite-time cooling reversal in weighted Euclidean distance. The asymptotic conclusions of Theorem 7 hold as well. In particular, these exclusions apply under failure of all original triplet predicates. The constant $n/(n-2)$ is optimal uniformly over energy spacings in the larger G-only family; optimality under the additional F constraints is not asserted.

Proof. Shift $E_1=0$, set $r=a/c$, and put $d=n-2$. The case $\mu\le a$ is already covered. Otherwise $0<r<d/(d+1)$ and $\mu\le c(d+r)/n$. The threshold decreases as μ increases, so

$$c\beta_*\ge\frac{\log(1+r/d)-\log(1-(d+1)r/d)}{r}>\frac{n}{d}.$$

The strict inequality follows by expanding the logarithms: the constant term is n/d, and the coefficient of r^{k-1} for every $k\ge2$ is $[(-1)^{k+1}+(d+1)^k]/(kd^k)>0$. Both series converge on the stated interval. This proves absence, including equality at the advertised cutoff.

For sharpness, let $E_2/c\to0$ and let the remaining $n-2$ distinct energies approach c from below (with $E_n=c$). Then $c\beta_*\to n/(n-2)$. Any fixed larger value of $\beta_b c$ therefore violates the threshold for some strictly ordered finite energy vector. The first-contrast construction in the preceding theorem realizes ordinary cooling with all $G=0$. □

For four states, the spacing-independent exclusion is $\beta_b c \le 2$. This also explains a limitation of the numerical evidence for the original conjecture. The March paper's Fig. 2 samples $\beta_b E_i$ between zero and one, hence $\beta_b c \le 1$. Every triplet-failing sample in that box lies inside the proved exclusion region, for each of the tested state counts $n = 4, \dots, 7$. Its absence of observed counterexamples in that box is thus consistent with a stronger restricted theorem, while it cannot justify the unrestricted inference. The September sampling box with span up to five is not covered by this corollary.

8 A sharp dimension-dependent amplitude bound

The contrast representation also limits the size of ordinary cooling under the G inequalities. Unlike the quarter-limit construction, the following sharp bound concerns the larger family defined by $G \le 0$ alone.

Theorem 8 (Amplitude bound for the contrast cone)**.** *For an n-state system satisfying Theorem 4, with a simple slow mode and $0 < \beta_h < \beta_c < \beta_b$,*

$$\frac{|b_2(\beta_c)|}{|b_2(\beta_h)|} < \frac{n}{2}.$$

Thus any anomalous amplitude advantage is strictly less than $1 - 2/n$. For $n \ge 3$ this bound is sharp as a supremum within the G-only family. The same bound holds whenever the contrast matrix A is Metzler.

Proof. Orient the slow eigenvector so that $b_2 = \sum_k v_k g_k$, with $v_k \ge 0$ and not all zero. On cooling write $R_k = -g_k > 0$ and $m = k + 1$. Apart from a fixed positive factor,

$$R_k(\beta) = \frac{\alpha_k - \sum_{i<m} e^{\beta(E_m - E_i)}}{\sum_i e^{-\beta(E_i - E_m)}}, \qquad \alpha_k = \sum_{i<m} e^{\beta_b(E_m - E_i)}.$$

The positive numerator strictly decreases with β. Put $A_c = \sum_{i \le m} e^{\beta_c(E_m - E_i)} \ge m$. The denominator at β_h is at most $A_c + n - m$, while the denominator at β_c is at least A_c. Therefore

$$\frac{R_k(\beta_c)}{R_k(\beta_h)} < 1 + \frac{n - m}{A_c} \le \frac{n}{m} \le \frac{n}{2}.$$

The amplitude ratio is a weighted average of these ratios with nonnegative weights $v_k R_k(\beta_h)$, proving the bound.

For sharpness fix $n \ge 3$ and let $L \to \infty$. Take $\beta_b = 1$, $E_1 = 0$, $E_2 = 1/L$, and $E_j = (j-2)L$ for $j \ge 3$. Use the first contrast $w = (1, -e^{1/L}, 0, \dots, 0)$ and normalized bath weights q. It is realized as the simple slow mode by the established rank-one construction

$$M = q\mathbf{1}^\top - I + \epsilon \frac{D w w^\top}{w^\top D w}, \qquad \epsilon = \frac{q_1 + q_2}{2}.$$

Every off-diagonal entry is positive: only the 12 pair has a negative rank-one correction, reducing its reset rate by exactly one half. The slow eigenvalue is $-1 + \epsilon$, all other nonzero eigenvalues are -1, and every $G_{ijk} = 0$ because the correction has no component in a state $k \ge 3$. With $\beta_h = L^{-2}$ and $\beta_c = L^{-1/2}$, the thermal numerator is $1 - e^{(1-\beta)/L} \sim -1/L$ for both preparations, while their partition functions tend to n and 2, respectively. Their amplitude ratio tends to $n/2$. These examples do not in general satisfy the F inequalities. □

For the larger G-only four-state family this gives a sharp 50% supremum. The additional original F inequalities improve it to a sharp 25% supremum, proved in Section 14. For any fixed simple mode in the G-only family the corresponding asymptotic time advance is less than $\log(n/2)/|\lambda_2|$. This is a bound in relaxation time units and does not require choosing a distance threshold.

9 A finite classification for every simple four-state slow mode

The triplet test uses local rate information. A replacement can instead use the full generator's slow eigenvector. The result below is necessary and sufficient for ordinary *asymptotic* cooling and heating effects throughout the working domain: four distinct energies, a positive finite bath temperature, a reversible irreducible generator, and a simple slow nonzero eigenvalue. It does not classify all transient crossings or a degenerate slow eigenspace. Computing the full slow mode is part of the test; this is not another local rate-only criterion.

9.1 The geometry of Gibbs derivatives

Shift the energies to $(0, a, b, c)$ with $0 < a < b < c$, without changing their gaps. Write $\pi_i(\beta) = e^{-\beta E_i}/Z(\beta)$, and let a prime always mean a derivative in the initial inverse temperature. Define

$$g = \pi_1' > 0, \qquad u = \frac{\pi_2'}{\pi_1'}, \qquad v = \frac{\pi_3'}{\pi_1'}.$$

For any fixed vector w, including the left slow mode,

$$\frac{d}{d\beta}(w^\top \pi) = g\Phi(\beta), \qquad \Phi = A + Bu + Cv, \quad (A, B, C) = (w_1 - w_4, w_2 - w_4, w_3 - w_4). \tag{5}$$

The identity follows from $\sum_i \pi_i' = 0$.

Lemma 3 (Strictly convex derivative curve)**.** *For finite real β, $u'(\beta) < 0$ and v, considered as a function of u, has strictly positive second derivative. For $\beta \geq 0$, put*

$$c_0 = ab(b-a), \quad c_1 = ac(c-a), \quad c_2 = bc(c-b), \quad \gamma = \frac{c-a}{c-b} > 1, \qquad x = e^{-(c-b)\beta}.$$

Then

$$\frac{dv}{du} = s(x) = \frac{c_2 x^\gamma - c_0}{c_0 + c_1 x}, \qquad s'(x) = \frac{c_0 c_2 \gamma x^{\gamma-1} + c_1 c_2 (\gamma - 1) x^\gamma + c_0 c_1}{(c_0 + c_1 x)^2} > 0. \tag{6}$$

Here $s(0) = -1$, while $s(1) > -1$ need not be positive.

Proof. For functions $h_1, \ldots, h_k$ let $W(h_1, \ldots, h_k)$ denote the determinant with derivative orders $0, \ldots, k-1$ in its rows. The following identity holds for any number n of distinct ordered energies, with $f_i = e^{-\beta E_i}$ and $Z = \sum_i f_i$:

$$W(\pi_1', \ldots, \pi_k') = \frac{(-1)^{k(k-1)/2}}{Z^{k+1}} \left(\prod_{i<j\leq k} (E_j - E_i) \right) \left(\prod_{i=1}^{k} f_i \right) \sum_{j=k+1}^{n} f_j \prod_{i=1}^{k} (E_j - E_i), \quad k < n. \tag{7}$$

Indeed, adjoining a constant column gives $W(\pi_1', \ldots, \pi_k') = W(1, \pi_1, \ldots, \pi_k)$. Multiplying all $k+1$ functions by Z multiplies their Wronskian by Z^{k+1}: the product-rule transformation of the derivative rows is triangular with diagonal Z. Expand the first column of $W(Z, f_1, \ldots, f_k)$. Terms with index at most k vanish by repeated columns; every remaining term is an exponential Vandermonde determinant with the displayed sign.

For $k = 2$ the identity gives $W(g, gu) = g^2 u' < 0$. For $k = 3$ it gives $W(g, gu, gv) = g^3 (u')^3 d^2 v/du^2 < 0$, proving strict convexity. For completeness, a mixed two-column calculation gives

$$W(\pi_1', \pi_3') = \frac{f_1 f_3}{Z^3} \{ab(b-a) f_2 - bc(c-b) f_4\},$$

whereas $W(\pi_1', \pi_2') = -f_1 f_2 \{ab(b-a) f_3 + ac(c-a) f_4\}/Z^3$. Their ratio is dv/du and simplifies to (6). Differentiation gives the strictly positive displayed numerator. □

9.2 The finite test and its equality cases

The endpoint values needed below are explicit:

$$u(0) = 1 - \frac{4a}{a+b+c}, \qquad v(0) = 1 - \frac{4b}{a+b+c},$$
$$u(\infty) = -1, \qquad v(\infty) = 0.$$

At the bath and any other finite temperature, stable formulas are

$$u = \frac{-a + (b-a)e^{-b\beta} + (c-a)e^{-c\beta}}{a + be^{-(b-a)\beta} + ce^{-(c-a)\beta}},$$

$$v = \frac{e^{-(b-a)\beta}\{-b + (a-b)e^{-a\beta} + (c-b)e^{-c\beta}\}}{a + be^{-(b-a)\beta} + ce^{-(c-a)\beta}}.$$

Theorem 9 (Necessary and sufficient finite test)**.** *Let w be a nonzero left eigenvector of the simple slow nonzero mode, with $w^\top \pi(\beta_b) = 0$. Form A, B, C, Φ as in (5). If $C \neq 0$ and $-1 < -B/C < s(1)$, define β_* by the unique solution*

$$s(x_*) = -B/C, \quad 0 < x_* < 1, \qquad \beta_* = -\frac{\log x_*}{c-b}.$$

Otherwise omit β_*. Form the two finite sets of real numbers

$$\mathcal{V}_c = \{\Phi(0), \Phi(\beta_b)\} \cup \{\Phi(\beta_*) : 0 < \beta_* < \beta_b\},$$
$$\mathcal{V}_h = \{\Phi(\beta_b), w_1 - w_2\} \cup \{\Phi(\beta_*) : \beta_b < \beta_* < \infty\}.$$

An ordinary asymptotic cooling effect exists if and only if $\min \mathcal{V}_c < 0 < \max \mathcal{V}_c$*. An ordinary asymptotic heating effect exists if and only if* $\min \mathcal{V}_h < 0 < \max \mathcal{V}_h$*. All inequalities are strict. A zero extremum without opposite signs does not certify an effect. Endpoints $\beta = 0, \infty$ are used only as limits; every effect asserted by the theorem has witnesses at finite positive temperatures.*

Proof. If $C = 0$, then $\Phi = A + Bu$ is monotone (or constant). Otherwise $\Phi' = u'(B + C\, s(x))$. Lemma 3 shows that this has at most one zero for $\beta > 0$, precisely the β_* just specified. If $-B/C$ equals either limiting slope, there is no interior zero. Thus the listed values give the infimum and supremum of Φ on each side of the bath, including its limiting endpoint. Since $g > 0$, they decide whether $b'(\beta)$ takes both positive and negative values on that open side.

It remains to justify the equivalence with a Mpemba ordering; a stationary point alone would not suffice. Write $b = w^\top \pi$, so $b(\beta_b) = 0$. If b' has only one sign on a side, b is monotone there and $|b|$ increases with distance from the bath. There is no anomalous pair. Conversely, suppose b' takes both signs. The strictly convex representation implies at most two zeros of b' in total, counting multiplicities, unless it vanishes identically; the latter would require w constant and is excluded. On the cooling side consider the last sign-changing zero before the bath. On the interval immediately to its right, b has sign opposite to b', because $b(\beta_b) = 0$ and there are no further sign changes. Immediately to its left b keeps that nonzero sign but b' changes sign. Consequently $\partial_\beta |b| > 0$ there, yielding a finite cooling witness. For heating, use the first sign-changing zero after the bath. Immediately to its left b has the same sign as b'; immediately to its right the derivative changes sign while b does not, so $\partial_\beta |b| < 0$. This yields a finite heating witness. Isolated tangencies do not affect this reasoning. If the bath itself is stationary, apply the same argument on the open side. Strict opposite signs at a limiting endpoint persist at nearby finite temperatures by continuity. □

The theorem replaces an unbounded scan of initial temperatures by endpoint evaluations and at most one scalar equation with a unique solution. Its scope is every generator in the stated simple-mode working domain, with no restriction on energy spacing, rate scales, or triplet branches. It is an exact mathematical criterion, not a claim that floating-point arithmetic decides every equality case. With rational commensurate energies in an exponential coordinate and algebraic mode data, polynomial root isolation can certify the test. General transcendental boundary cases require exact arguments or suitable enclosures.

Strong effects can also be classified with finitely many tests. The same construction isolates every zero of b': on either side of the optional β_*, Φ is strictly monotone unless $B = C = 0$. Opposite endpoint signs give one unique root; an exact zero at β_* is a tangency. There are at most two finite positive roots in total. On each side of the bath, insert these stationary temperatures between the bath and the limiting endpoint (0 or ∞). Evaluate b at the resulting ordered knots. Every pair of consecutive nonzero values of opposite sign gives exactly one strong preparation in that open interval, since b is strictly monotone there. Every stationary knot with $b = 0$, other than the bath, is itself a strong preparation, including a tangential zero. A zero at 0 or ∞ is excluded. Intervals adjacent to a zero knot contain no additional zero. This is a complete finite classification of nonbath strong preparations; their actual temperatures, if desired, are the unique roots on those monotone intervals. It preserves the distinction between an ordinary turning point and a strong zero. The floating-point implementation supplies the ordinary-effect test. Section 10.1 also implements strong-root counts exactly for rational rates and commensurate energies; for unrestricted energy ratios the construction here is a mathematical root-isolation prescription.

9.3 A sharp limit on thermal complexity

For four states, Lemma 3 shows that b' has at most two zeros counting multiplicity: a line intersects a strictly convex graph at most twice, with tangency counting twice. More generally, (7) shows that $(\pi_1', \dots, \pi_{n-1}')$ is an extended complete Chebyshev system. This means every nonzero linear combination of its first k functions has at most $k - 1$ zeros counting multiplicity. The implication from nonvanishing prefix Wronskians is the standard repeated weighted Rolle argument (see, for example, Gasull, Geyer and Mañosas, *A Chebyshev criterion with applications*, J. Differential Equations 269 (2020), Lemma 2.3): divide by the first nonvanishing function, differentiate, and repeat; the transformed prefix Wronskians remain nonzero. At each differentiation the number of zeros can decrease by at most one. Thus any nonconstant Gibbs expectation has at most $n - 2$ stationary points. This is a consequence of established Chebyshev-system theory; no new general Wronskian or zero-counting principle is claimed.

Likewise, $Zb = \sum_i w_i e^{-\beta E_i}$ has at most $n - 1$ zeros counting multiplicity by the exponential Vandermonde argument. One is the bath, so there are at most $n - 2$ nonbath strong preparations. These two bounds are sharp even for positive reversible generators with a simple slow mode. Indeed, prescribe $n - 2$ distinct positive nonbath inverse temperatures. Together with the bath they impose $n - 1$ homogeneous interpolation conditions on n coefficients w_i. The exponential Chebyshev property makes the nullspace one-dimensional and its nonzero solution has exactly those simple zeros and no others. Rolle's theorem forces $n - 2$ stationary points between them, attaining the derivative bound as well.

To realize this w, let $D = \operatorname{diag}(\pi_b)$, $N = w^\top D w > 0$, and choose

$$M = -I + \pi_b \mathbf{1}^\top + \epsilon \frac{D w w^\top}{N}, \qquad 0 < \epsilon < \min\left\{1, \frac{N}{\max_i |w_i|^2}\right\}. \tag{8}$$

Its off-diagonal entries are $\pi_{bi}(1 + \epsilon w_i w_j / N) > 0$; its column sums vanish and detailed balance holds.

The eigenvalues are $0, -1+\epsilon, -1, \ldots, -1$, with w the left slow mode. This is the rank-one realization construction of Klich, Raz, Hirschberg and Vucelja, *The Mpemba index and anomalous relaxation*, Appendix VIII, Eq. (67), after time normalization and conversion of conventions. We use their established realization theorem to prove attainability. It need not obey the triplet premise.

In particular, four is the minimum number of distinct states allowing both ordinary cooling and ordinary heating effects at one fixed bath with a simple slow mode: each side needs a sign-changing stationary point, while three states allow at most one globally. Four states attain the bound. Take $E_i = (i-1)\log 2$, $\beta_b = 1$, and

$$w = (-3/32,\ 11/16,\ -3/2,\ 1), \qquad \sum_{i=1}^{4} w_i x^{i-1} = (x-1/4)(x-1/2)(x-3/4).$$

The two nonbath strong preparations are $\beta = \log(4/3)/\log 2 < 1$ and $\beta = 2 > 1$. Choosing $\epsilon = 1/100$ in (8) gives an entirely rational generator, with eigenvalues $0, -99/100, -1, -1$. Both effects follow from the two exact nonbath zeros. This example illustrates attainability, not a second triplet counterexample. Publication priority for these Gibbs-specific consequences requires a separate literature assessment.

10 Degenerate slow spaces and a sharp bound on strong preparations

A scalar overlap does not classify a degenerate slow eigenspace. There is, however, a complete finite test for a specified asymptotic distance, and a simple sharp bound on strong preparations that works for any number of states.

Let $S = D^{-1/2} M D^{1/2}$, and let P be the orthogonal projector onto the entire slowest nonstationary eigenspace of dimension r. Put

$$H = D^{-1/2} P D^{-1/2}, \qquad Q(\beta) = \pi(\beta)^{\top} H \pi(\beta).$$

Since $P\sqrt{\pi_b} = 0$, $H\pi_b = 0$. Spectral expansion gives

$$\sum_i \frac{(p_i(t;\beta) - \pi_{bi})^2}{\pi_{bi}} = e^{2\lambda_2 t} Q(\beta) + o(e^{2\lambda_2 t}).$$

Relative entropy has the same leading term multiplied by $1/2$. Thus strict ordering of Q certifies asymptotic weighted Euclidean and relative-entropy reversals. If two specified preparations have equal Q, subleading terms can decide their eventual ordering; the leading coefficient alone does not decide that pair. The existence test below remains complete over all allowed initial temperatures: if Q' has no sign change on a side, analyticity and the bath zero make Q strictly ordered there. For other norms a degenerate projection can give different orderings; no norm-independent scalar assertion is made. A strong preparation is precisely $Q = 0$ away from the bath, equivalently $PD^{-1/2}\pi(\beta) = 0$.

Theorem 10 (State count and slow-space dimension)**.** *For n distinct energies and a slow eigenspace of dimension r, there are at most $n - r - 1$ distinct nonbath strong Gibbs preparations. For every prescribed set of $n-r-1$ distinct positive nonbath inverse temperatures, and every positive finite bath, this bound is attained by a strictly positive reversible generator with an r-dimensional slow eigenspace.*

Proof. Any $k \le n$ distinct Gibbs vectors are linearly independent. To see this, take their first k coordinates and multiply each column by its positive partition function. The resulting evaluation

matrix of k distinct exponentials has nonzero determinant by the exponential Chebyshev property. Multiplication by $D^{-1/2}$ preserves independence.

The kernel of P has dimension $n-r$ and already contains the transformed bath vector $\sqrt{\pi_b}$. It therefore contains at most $n-r-1$ additional Gibbs vectors. For attainability, let U be the span of the transformed bath and the $n-r-1$ prescribed transformed Gibbs vectors. They are independent, so $\dim U = n-r$. Let P project orthogonally onto $U^\perp$, and define

$$S = -I + ss^\top + \epsilon P, \qquad s = \sqrt{\pi_b}, \qquad M = D^{1/2} S D^{-1/2}.$$

Choose $0 < \epsilon < 1$ sufficiently small that $s_i s_j + \epsilon P_{ij} > 0$ for every $i \neq j$; this is possible because all s_i are positive and there are only finitely many entries. Then M has positive off-diagonal entries, vanishing column sums, detailed balance, and spectrum 0, $-1+\epsilon$ with multiplicity r, and -1 with multiplicity $n-r-1$. Its strong preparations include all the prescribed ones, and the bound just proved excludes any others. □

For four states the maximum number of distinct strong preparations is therefore two, one, or zero when the slow multiplicity is one, two, or three. This bound counts exact absence of the *whole* slow projection; canceling one chosen vector in a degenerate eigenspace is insufficient. It also clarifies why a classification must retain spectral multiplicity. The realization proof extends the rank-one perturbation in Klich et al., Appendix VIII, to an orthogonal projector of rank r. No novelty claim is made for the underlying linear-independence or projector-construction principles; publication priority for the sharp Gibbs-preparation bound remains to be checked. For $r = 1$, the $n-2$ nonbath-zero bound already follows explicitly from Emmerson's Theorem 7.3 and Corollary 7.5, as detailed in Section 4.6; no new scalar strong-temperature bound is claimed. The dimension argument here also covers simultaneous cancellation when $r > 1$.

Finite classification when $r = 2$. Write $f_i = e^{-\beta E_i}$ and $Z = \sum_i f_i$. Then

$$Q' = \frac{R(\beta)}{Z^3}, \qquad R = (f^\top H f)' Z - 2(f^\top H f) Z'.$$

For four states, R contains at most sixteen distinct exponential terms. There are twelve terms $f_i^2 f_k$ with $i \neq k$, with coefficients $2(E_k - E_i)(H_{ii} - H_{ik})$, and four terms with distinct indices i, j, k, with coefficients

$$2\{H_{ij}(2E_k - E_i - E_j) + H_{ik}(2E_j - E_i - E_k) + H_{jk}(2E_i - E_j - E_k)\}.$$

All four pure cubes have zero coefficient; coincident exponent sums can only reduce the count. After combining equal exponents, a nonzero m-term R has at most $m-1 \leq 15$ real zeros counting multiplicity. It cannot vanish identically: that would make Q constantly zero since $Q(\beta_b) = 0$, forcing P to annihilate all Gibbs vectors, which span the full state space.

Isolate these finitely many zeros on $(0, \beta_b)$ and (β_b, ∞), then evaluate the sign of R once on each intervening interval. An ordinary asymptotic cooling effect exists exactly when some cooling interval has $R > 0$; a heating effect exists exactly when some heating interval has $R < 0$. This follows directly by differentiating the nonnegative amplitude Q. Exponential root isolation can be organized by the repeated Rolle procedure: factor out the smallest exponent, differentiate to reduce the number of terms by one, recursively isolate derivative roots, and use monotonicity between those roots. Limiting endpoint signs follow from the leading nonzero term. Exact tangencies must be retained. This is a finite mathematical prescription; it does not assert that floating-point sampling or arbitrary transcendental equality testing is a certified terminating decision algorithm.

For $r = 3$, $S = -\mu(I - ss^\top)$ for some $\mu > 0$, hence $M = \mu(\pi_b \mathbf{1}^\top - I)$. This reset generator is stochastically monotone, so the earlier all-time convex-divergence ordering applies. There is no

asymptotic cooling or heating effect in the distances just specified. Together with Theorem 9, these cases cover all slow multiplicities of irreducible reversible four-state systems for asymptotic relative entropy and weighted Euclidean distance. The arguments also allow zero transition rates when irreducibility and the stated spectral conditions remain valid; disconnected generators require a different target equilibrium and are outside this thermalizing model.

10.1 An exact executable classifier for commensurate energies

The general thermal test involves transcendental quantities. An important subfamily admits a terminating exact implementation, including tangencies, endpoint roots, and algebraic slow eigenvectors. Suppose

$$E_i = m_i\Delta, \qquad m_1 < m_2 < m_3 < m_4 \text{ integers}, \qquad b = e^{-\beta_b\Delta} \in \mathbb{Q} \cap (0,1),$$

and the original transition rates are rational. A shift makes $m_1 = 0$. Put $x = e^{-\beta\Delta}$; the positive-temperature range is $0 < x < 1$, the bath is $x = b$, and $Z(x) = \sum_i x^{m_i}$. Rational rates and bath weights make the generator rational, and its slow eigenvectors algebraic.

For a simple slow mode define $N(x) = \sum_i w_i x^{m_i}$. Then

$$b_2(x) = \frac{N(x)}{Z(x)}, \qquad \operatorname{sgn}\frac{db_2}{dx} = \operatorname{sgn}\{N'Z - NZ'\}.$$

All coefficients belong to the real algebraic field generated by the slow eigenvalue. For a degenerate slow space use the amplitude form of Section 10, so $Q = N/Z^2$ and the derivative numerator is $N'Z - 2NZ'$. The projector is rational in this case: a repeated root of the rational nonzero cubic characteristic polynomial must be rational, and the remaining eigenvalue is rational too. Polynomial spectral projection therefore avoids any square roots of the bath weights.

Factor the derivative numerator into square-free factors, retaining those of odd multiplicity. A Sturm count of their roots in $(b,1)$ and $(0,b)$ decides ordinary cooling and heating, respectively: an effect exists exactly when the corresponding count is nonzero. Indeed, odd-multiplicity roots are precisely derivative sign changes. For the scalar overlap, a fixed derivative sign and the bath zero imply normal amplitude ordering; conversely normal absolute-amplitude ordering forces a fixed sign of the overlap and its derivative. For the nonnegative degenerate amplitude, the derivative has the normal sign near its isolated bath zero, so any reversal requires and is implied by an interior sign change. Even-multiplicity stationary points alone do not count as effects.

A separate square-free Sturm count of N gives the numbers of strong preparations on the two intervals. Factors at $x = 0, b, 1$ are explicitly removed before interval counts: infinite and zero temperature endpoints, and the bath itself, are not finite nonbath preparations.

`exact_commensurate_classifier.py` implements these operations for all three possible slow multiplicities and permits zero rates when the transition graph remains connected. Signs in an algebraic coefficient field are determined by rational isolating intervals for its specified real embedding, refined until interval evaluation excludes zero. Exact field reduction detects zero first, so refinement terminates for every nonzero coefficient. This avoids treating a polynomial representation's leading sign as the sign of its algebraic value. Sturm and square-free arithmetic are classical algorithms; the contribution here is a reproducible implementation of the stated physical classification with all equality cases retained.

Sixteen checks cover irrational and scaled cubic slow eigenvalues, double and triple slow degeneracy, strong roots without sign changes, stationary tangencies without an effect, bath-threshold equality, excluded temperature endpoints, nonuniform commensurate spacing, and connected generators with zero rates. An explicit sign-regression test ensures that the real root α of $\alpha^3 - \alpha - 1 = 0$ satisfies

$\alpha - 2 < 0$. The implementation uses exact SymPy polynomial arithmetic and rational interval refinement, not temperature sampling or a floating-point eigenvector. Its hypotheses on rationality and commensurability are substantive; they are not assumed in the preceding analytic classification.

11 A complete finite-time test for a specified thermal pair

The earlier classification concerns the existence of asymptotic anomalies as the preparation temperature varies. For weighted Euclidean distance there is also a particularly small finite-time test, covering every four-state reversible generator, all slow multiplicities, and any fixed pair of initial temperatures. It is an application of spectral orthogonality and classical exponential zero counting, rather than a new general zero-counting principle.

Write $q = \pi(\beta_b)$ and $D = \mathrm{diag}(q)$. Let P_j be the orthogonal projectors of the symmetric generator onto its distinct nonzero eigenvalues $\mu_1 > \cdots > \mu_m$, where $m \le 3$. For the preparation farther from the bath and the nearer preparation set

$$z_f = D^{-1/2}(\pi(\beta_f) - q), \quad z_n = D^{-1/2}(\pi(\beta_n) - q), \quad A_j = \|P_j z_f\|^2 - \|P_j z_n\|^2.$$

The difference of squared weighted Euclidean distances is exactly

$$\Delta(t) = \sum_{j=1}^{m} A_j e^{2\mu_j t}.$$

Terms with $A_j = 0$ may be discarded, even if the corresponding eigenspace is the slowest one. This handles equal leading amplitudes without inferring the eventual ordering from a coefficient that vanishes.

Theorem 11 (Finite-time weighted-Euclidean classification)**.** *For any fixed pair of distinct positive initial temperatures on the same side of a positive finite bath, a four-state irreducible reversible generator has a weighted-Euclidean Mpemba reversal if and only if the following finite test finds a strictly negative value. After discarding zero coefficients, factor the largest remaining exponential out of Δ. With three terms write*

$$F(t) = A + Be^{-at} + Ce^{-bt}, \qquad 0 < a < b.$$

Evaluate $F(0)$ and its limiting value A. If $BC < 0$ and $-bC/(aB) > 1$, also evaluate

$$t_* = \frac{\log[-bC/(aB)]}{b - a}, \qquad F(t_*).$$

There is a reversal precisely when the minimum of the listed values is negative. For two terms only the two endpoint values are needed; one term cannot change ordering. There are at most two distinct finite crossing times, counting multiplicity. A zero minimum alone does not establish a reversal: it can be a finite tangency or an equality reached only in the limit.

Proof. Orthogonality of the spectral projectors proves the expression for Δ. The derivative of F is $-aBe^{-at} - bCe^{-bt}$; it has the stated single possible positive zero. Thus the listed values contain the global infimum of F on $[0, \infty)$. The exponential factored out is strictly positive. A strictly negative limiting value implies negative values at finite times; a zero limiting value alone does not. For one or two terms the same argument is immediate. The three exponentials form a Chebyshev system, so a nonzero difference has at most two zeros counting multiplicity.

The farther thermal preparation is initially strictly farther in this distance. Indeed, writing $Z(\beta) = \sum_i e^{-\beta E_i}$,

$$1 + \chi^2(\pi(\beta)\|q) = \frac{Z(\beta_b)Z(2\beta - \beta_b)}{Z(\beta)^2},$$

whose logarithmic derivative is $2[\langle E\rangle_\beta - \langle E\rangle_{2\beta-\beta_b}]$. The mean energy strictly decreases with inverse temperature, including when the auxiliary argument $2\beta - \beta_b$ is negative. Hence the initial distance strictly increases away from the bath on either side. In particular $\Delta(0) > 0$, so every negative value found by the test represents an actual crossing from the initial ordering. □

If the eventual coefficient A is positive, a reversal can occur only via an interior negative minimum and two crossings: ordinary ordering returns at sufficiently long times. Such a purely transient reversal in this distance needs at least three distinct nonzero decay rates, and therefore at least four states. Two- and three-state reversible systems cannot have this pattern. This last statement is a spectral necessity, not an existence claim for every four-state generator or for the triplet-constrained family. The result does not transfer automatically to relative entropy or total variation, whose finite-time expansions need not have these three terms.

Four states suffice for a purely transient pair. An exact construction shows that the state-count necessity is sharp. Take $q = (512, 64, 8, 1)/585$, corresponding to energies $(0, 3\log 2, 6\log 2, 9\log 2)$ at bath inverse temperature one. Prepare $p_h = (1, 1/2, 1/4, 1/8)/Z_h$ and $p_c = (1, 1/4, 1/16, 1/64)/Z_c$, so $\beta_h = 1/3$ and $\beta_c = 2/3$. Let $w^{(0)}$ be the coefficient vector of

$$(x - 1/2)(x - 1/4)(x - 1/8),$$

and set $w = w^{(0)} + h/1000$, where $h_i = (i-1) - \sum_j q_j(j-1)$. Choose a nonzero rational vector v satisfying $q^\top v = w^\top D v = p_h^\top v = 0$, and a nonzero rational u orthogonal to $\mathbf{1}, w, v$ in the D inner product. All these vectors are explicitly computed by rational nullspace elimination in `transient_only_chi_certificate.py`. Define

$$P_z = \frac{z z^\top D}{z^\top D z}, \qquad C_0 = (P_v + 2P_u)^\top, \qquad \rho = 1 + \max\left(0, \max_{i\neq j} \frac{(C_0)_{ij}}{q_i}\right), \qquad M = \rho(q\mathbf{1}^\top - I) - C_0.$$

The off-diagonal rates are strictly positive, detailed balance holds, and the eigenvalues are $0, -\rho, -\rho - 1, -\rho - 2$. The squared-distance difference has the form $e^{-2\rho t}(A + Bx + Cx^2)$ with $x = e^{-2t}$, where exact rational projection calculations give

$$A \simeq 1.80809\,10^{-5}, \qquad B \simeq -0.0344769, \qquad C \simeq 3.69493.$$

The certificate proves $A > 0$, $A + B + C > 0$, and $0 < -B/(2C) < 1$ with $A - B^2/(4C) < 0$. An exact Sturm count places both polynomial roots in $(0, 1)$. Thus the pair has normal initial and eventual ordering but a strictly reversed interval between two finite crossings. Their approximate times are 2.36803 and 3.74577; the exact times are $-\frac{1}{2}\log[(-B \pm \sqrt{B^2 - 4AC})/(2C)]$.

This construction is not asserted to satisfy the triplet-failure premise. It proves sharpness of the four-state transient classification, not absence of asymptotic anomalies for other pairs in the same generator. Its common decay rate is large, so the crossing distances are extremely small; it is a mathematical separation of transient and asymptotic ordering, not a practical cooling protocol. The design makes both preparations almost orthogonal to the slow mode, puts their negative distance difference in the middle mode, and their initially positive difference in the fastest mode.

12 Prescribing the slow mode: a two-dimensional feasibility test

The thermal classification concerns an eigenvector of the full generator. There is also a finite inverse test for whether a proposed direction can occur while every original triplet test fails. It reduces the rate-design problem, including spectral ordering, to the intersection of a disk and eight half-planes.

Fix the bath probabilities q and a nonconstant vector w with $q^\top w = 0$. Let $D = \operatorname{diag}(q)$, and choose any full-rank 4×2 matrix U whose columns span

$$\{f : q^\top f = 0,\ (Dw)^\top f = 0\}.$$

For a symmetric 2×2 matrix B, set

$$M_0(B) = -DUBU^\top.$$

This auxiliary matrix need not have nonnegative transition rates; it has zero column sums, detailed balance, and null left vectors $\mathbf{1}, w$.

Theorem 12 (Exact inverse feasibility)**.** *There exists a strictly positive reversible generator for which w is the simple slow left mode and every original triplet predicate fails if and only if some positive definite B satisfies the eight linear inequalities $F_{ijk}(M_0(B)) \le 0$ and $G_{ijk}(M_0(B)) \le 0$. Equivalently, normalize $\operatorname{tr} B = 1$ and write*

$$B = \frac{1}{2}\begin{pmatrix} 1+x & y \\ y & 1-x \end{pmatrix}.$$

The exact feasibility condition is that the resulting eight closed half-planes intersect $x^2 + y^2 < 1$.

Proof. If B is positive definite, $D^{-1/2} M_0 D^{1/2}$ is negative semidefinite, with exactly the two-dimensional kernel spanned by $\sqrt{q}$ and $D^{1/2}w$. Add a reset generator:

$$M = M_0 + \rho(q\mathbf{1}^\top - I).$$

For sufficiently large $\rho > 0$, all off-diagonal entries are strictly positive. The nonstationary eigenvalues are $-\rho, -\rho - \nu_1, -\rho - \nu_2$ with $\nu_1, \nu_2 > 0$, so w is the simple slow mode. The reset contributes zero to every F and G, hence the inequalities persist. Since the physical rates are now positive, the branch-equivalence lemma identifies these with failure of the original predicates.

Conversely, start with any such physical generator, with simple slow eigenvalue $\lambda_2 < 0$. Put $M_0 = M + \lambda_2(q\mathbf{1}^\top - I)$. Its symmetrization is zero on the stationary and slow directions, and strictly negative on their two-dimensional orthogonal complement. Therefore $M_0 = -DUBU^\top$ for a positive definite B. Reset invariance gives the eight inequalities. Positive scaling of B preserves them, so trace normalization loses no solution. A symmetric trace-one 2×2 matrix is positive definite exactly when $x^2 + y^2 < 1$. □

For rational q, w, choose U rational by Gaussian elimination. All eight half-planes then have rational coefficients. Exact feasibility is particularly simple: find the point of their intersection closest to the origin, and compare its squared norm with one. The closest point, when the polyhedron is nonempty, is either the origin, the orthogonal projection of the origin onto one boundary line, or the intersection of two nonparallel boundary lines. Enumerate these finitely many candidates and discard those violating any half-plane. Each remaining squared norm is rational. A minimum below one certifies realizability; a minimum equal to or above one, or no feasible candidate, excludes a simple slow realization. Equality must not be rounded into an admissible positive definite matrix.

This gives a complete checkable inverse characterization at fixed rational bath weights and rational mode direction. It uses established reset and semidefinite geometry, but the half-planes enforce the exact original triplet premise. It is not a claim that eigenvectors depend convexly on rates. It also explains why prescribing a thermal shape can make adversarial searches much more efficient than sampling rate matrices: spectral ordering is enforced by positive definiteness of only a 2×2 matrix.

The necessary slow-mode inequalities are not sufficient. The structural restriction proved in Section 14 does not replace this inverse test. For an exact obstruction, use unnormalized bath weights $\widehat{q} = (5, 4, 2, 1)$ and

$$w = (162, -18, 71, -880).$$

Its bath mean is zero, its three Helmert coefficients are $(80, 2, 80) > 0$, and $w_1 > w_2$, $w_1 + w_2 - 2w_3 = 2 > 0$, $w_4 < 0$. Nevertheless no positive full-triplet generator can have this simple slow direction. To see this, use $\widehat{D} = \operatorname{diag}(\widehat{q})$ and

$$U = \begin{pmatrix} -89/450 & 431/450 \\ -91/360 & -521/360 \\ 1 & 0 \\ 0 & 1 \end{pmatrix}.$$

For the signed core $M_0 = -\widehat{D}UBU^{\top}$, exact expansion gives

$$\frac{482571629}{11676414} F_{123}(M_0) + \frac{496382581}{11676414} F_{124}(M_0) + \frac{16620682809}{77842760} G_{234}(M_0) = \operatorname{tr} B.$$

All coefficients are positive. The triplet constraints would imply $\operatorname{tr} B \le 0$, contradicting $B \succ 0$. Using unnormalized weights only rescales the fast matrix by a positive constant and does not change feasibility. Reset addition leaves these expressions unchanged. This is a Farkas-type separating certificate, checked by `extended_mode_obstruction_certificate.py`; it explains why the new structural inequalities remain necessary rather than a complete realizability characterization.

13 A substantial cooling reversal with a uniform spectral gap

The first counterexample has a small amplitude advantage. This is not a necessary consequence of the triplet inequalities. The following family has an advantage approaching 25%, with a uniformly separated slow mode. The percentage concerns the slow amplitude, not a percentage reduction in the time needed to reach an arbitrary distance threshold.

Theorem 13 (A quarter-amplitude limit)**.** *There are strictly positive four-state reversible generators, with four distinct energies and bath inverse temperature one, for which every original triplet predicate fails and finite preparations $0 < \beta_h < \beta_c < 1$ satisfy*

$$1 - \frac{|b_2(\beta_h)|}{|b_2(\beta_c)|} \longrightarrow \frac{1}{4}.$$

The slow eigenvalue can be fixed at -3, while its separation from the next mode exceeds $1/3$. There is also a perturbed family with all eight F, G inequalities strict, the same amplitude limit, and a slow spectral gap exceeding $1/4$. The independent upper-bound theorem in Section 14 proves that 25% is the sharp supremum under the full triplet premise; it is not attained at finite parameters.

Proof. Take $0 < \varepsilon \le 1/30$ and unnormalized bath weights

$$\widehat{q} = (1, 1 - \varepsilon/3, 3/5, \varepsilon^2), \qquad q = \widehat{q} / \sum_i \widehat{q}_i, \qquad E_i = -\log \widehat{q}_i.$$

They are strictly decreasing. Put $d = 1 - \varepsilon/3$ and

$$r = -\frac{1}{3} - \frac{4\varepsilon}{9} + \frac{5\varepsilon^3}{3}, \qquad w = (1, -4/5, r, -\varepsilon)^{\top}.$$

Direct substitution gives $q^\top w = 0$. Use the fast-space basis

$$U = \begin{pmatrix} -3/5 - (r-1)/3 & \varepsilon^2(5\varepsilon - 4)/9 \\ (r-1)/(3d) & -5\varepsilon^2(\varepsilon+1)/(9d) \\ 1 & 0 \\ 0 & 1 \end{pmatrix}.$$

Its columns have zero q-mean and are q-orthogonal to w. Let $B = \frac{1}{2}\left(\begin{smallmatrix} 1+x & y \\ y & 1-x \end{smallmatrix}\right)$, and determine x, y by the two linear equations

$$F_{124}(-DUBU^\top) = 0, \qquad G_{234}(-DUBU^\top) = 0.$$

Define the physical generator

$$M_\varepsilon = -DU(B/\varepsilon^2)U^\top + 3(q\mathbf{1}^\top - I). \tag{9}$$

These formulas specify the generator using rational arithmetic alone.

The continuous sign verification is supplied by `quarter_limit_certificate.py`. It independently forms the two linear equations from the stated original-rate predicates and proves their coefficient determinant is nonzero. For every rational function whose sign is needed, it removes powers of ε, substitutes $\varepsilon = t/30$, and expands numerator and denominator in the Bernstein basis $\binom{n}{k}t^k(1-t)^{n-k}$. All recorded coefficients have the requisite strict sign. Positivity of the Bernstein basis for $0 < t < 1$ and the strictly positive endpoint coefficients prove the signs on the full interval, not just on a mesh. The 30 certificates establish

- $B_{11} > 0$, $\det B > 0$, and positivity of all twelve transition rates;
- $F_{124} = G_{234} = 0$ identically, with the other six F, G expressions strictly negative;
- writing $J = U^\top DU$, the positive fast eigenvalue increments satisfy

$$\min(\nu_1, \nu_2) > \frac{\det(BJ)/\varepsilon^4}{\operatorname{tr}(BJ)/\varepsilon^2} > \frac{1}{3};$$

- all eight F, G expressions of the auxiliary generator H below are strictly negative.

The first strict inequality follows from $ab/(a+b) < \min(a,b)$ for $a, b > 0$. Detailed balance, column conservation and $M_\varepsilon^\top w = -3w$ are checked symbolically. Thus the spectrum is $0, -3, -3-\nu_1, -3-\nu_2$, as required. The certificate's rational coefficients, together with the short checker, are part of the proof.

To compute the thermal limit set $L = \log(1/\varepsilon)$ and choose $\beta_h = L^{-2}$, $\beta_c = L^{-1/2}$. Both are finite positive cooling preparations and are correctly ordered on the stated interval. At β_h all four unnormalized Gibbs weights tend to one. At β_c the first three tend to one and the fourth tends to zero, since $e^{-\beta_c E_4} = e^{-2\sqrt{L}}$. Meanwhile $w \to (1, -4/5, -1/3, 0)$, whose first three entries sum to $-2/15$. Hence

$$b_h \longrightarrow -\frac{1}{30}, \qquad b_c \longrightarrow -\frac{2}{45}, \qquad \frac{|b_h|}{|b_c|} \longrightarrow \frac{3}{4}.$$

The strict anomalous ordering therefore holds for all sufficiently small positive ε.

For strict triplet inequalities, let H have reduced conductances

$$(k_{12}, k_{13}, k_{14}, k_{23}, k_{24}, k_{34}) = (100, 10, 1, 11, 2, 3)$$

at the same bath weights, with conserving diagonal, and put $\widetilde{M}_\varepsilon = M_\varepsilon + \varepsilon^4 H$. Linearity and the certified signs give all eight strict inequalities. Its symmetrized perturbation has operator norm at most $100\varepsilon^4$: for any function f, its Dirichlet form is bounded by

$$\sum_{i<j} q_i q_j k_{ij} (f_i - f_j)^2 \le 100 \sum_{i<j} q_i q_j (f_i - f_j)^2 = 100 \operatorname{Var}_q(f).$$

Weyl's bound preserves a gap greater than $1/3 - 200\varepsilon^4 > 1/4$. Work in the nonstationary symmetric subspace, where the original slow eigenvector is isolated by a gap greater than $1/3$. Resolving a perturbed unit eigenvector into its original slow component and its orthogonal complement gives an orthogonal component at most $100\varepsilon^4/(1/3 - 100\varepsilon^4)$. Orient the two eigenvectors to have positive inner product and give them the same weighted norm. Since $\|w\|_q \le 1$, their weighted difference is less than $500\varepsilon^4$. Also $q_i \ge \varepsilon^2/3$. Consequently one can normalize the perturbed left eigenvector so that

$$\max_i |\widetilde{w}_i - w_i| < 1000\varepsilon^3.$$

Every thermal overlap changes by at most this amount, uniformly in the initial temperature. The nonzero limiting overlaps and their ratio persist. □

A finite, strict example. Take $\varepsilon = 10^{-50}$, $\beta_h = 1/10000$ and $\beta_c = 1/50$ in (9), and optionally add $\varepsilon^4 H$. The integer-root interval checker `quarter_limit_thermal_certificate.py` proves that the unperturbed amplitude advantage lies between 22.6171597478% and 22.6171597506%. The above perturbation error bound proves an advantage greater than 22% also for the strict family. Every energy and temperature is finite. These large energy and temperature separations are a mathematical construction, not an experimentally validated parameter regime.

The physical mechanism is simple in this limit. Raising inverse temperature slightly empties the highest state, whose slow-mode component is approaching zero, while scarcely changing the other three relative populations. Their normalization changes from four-state to three-state weighting. Their slow overlap therefore grows by a factor approaching $4/3$. Neither a vanishing slow spectral gap nor cancellation of the slow mode is responsible. For relative entropy the asymptotic distance advantage approaches $1 - (3/4)^2 = 7/16$. The asymptotic time advance approaches $\log(4/3)/3$, or $\log(4/3)$ slow relaxation times, rather than a 25% reduction of an unspecified cooling time.

How quickly the quarter limit is approached. Write $a = \log(5/3)$ and $L = \log(1/\varepsilon)$. For all sufficiently large L the unperturbed family has a unique cooling amplitude maximum at

$$\beta_m = \frac{\log(4L/(13a))}{2L} + O\left(\frac{\log L}{L^2}\right).$$

Its advantage over the infinite-temperature limiting amplitude satisfies

$$1 - \frac{|b_2(0)|}{|b_2(\beta_m)|} = \frac{1}{4} - \frac{13a}{16L}\left[\log\frac{4L}{13a} + 1\right] + O\left(\frac{(\log L)^2}{L^2}\right). \tag{10}$$

Here $\beta = 0$ is only a limiting reference; finite sufficiently hot preparations approach the same advantage.

To verify these expansions, put $y = e^{-a\beta}$ and $x = e^{-2L\beta}$. Uniformly on bounded cooling intervals, the positive amplitude is

$$R(\beta) = |b_2(\beta)| = \frac{y/3 - 1/5}{2 + y + x} + O(\varepsilon),$$

with derivative error $O(L\varepsilon)$ near the maximum. Its derivative is initially positive for large L, while $b_2'(1) > 0$ by the contrast theorem. The two-stationary-point bound and these opposite endpoint signs

force exactly one cooling stationary point; a second zero, even a tangency, would violate the bound counting multiplicity. The limiting amplitude at any fixed $0 < \beta < 1$ strictly decreases with β, so this maximum tends to zero. It satisfies $L\beta_m \to \infty$, since the positive normalization derivative would otherwise remain of order L. Differentiating the displayed leading ratio gives the stationarity equation

$$x = \frac{13ay}{5(2L-a)y - 6L}\,[1 + o(1)] = \frac{13a}{4L}\left[1 + O\left(\beta_m + \frac{1}{L}\right)\right].$$

The exponentially small remainder does not change the indicated algebraic orders. More explicitly, the exact numerator of the positive amplitude is $-1 + (4/5)(1-\varepsilon/3)^\beta - ry + \varepsilon x$, and its denominator is $1 + (1-\varepsilon/3)^\beta + y + x$. On $0 \le \beta \le 1$, their differences from the displayed leading numerator and denominator are $O(\varepsilon)$; their derivative differences are $O(L\varepsilon)$. The denominator stays bounded below. For the leading ratio, multiplying its derivative by $15(2+y+x)^2$ gives exactly

$$x\big[(10L - 5a)y - 6L\big] - 13ay.$$

Near the maximum $y \to 1$, so the coefficient of x is asymptotic to $4L$ and the exponentially small derivative error remains negligible. Taking logarithms first gives $2L\beta_m = \log(4L/(13a)) + O(\beta_m + 1/L)$. Absorbing the $O(\beta_m)$ term into the left side proves $\beta_m = O(\log L/L)$; substituting this bound back gives the stated $O(\log L/L^2)$ remainder. This intermediate bound is needed before the sharper remainder can be asserted. Substituting into

$$R(\beta_m) = \frac{2}{45}\left[1 - \frac{13a}{6}\beta_m - \frac{x}{3} + O(\beta_m^2 + \beta_m x + x^2 + \varepsilon)\right], \qquad R(0) = \frac{1}{30} + O(\varepsilon),$$

proves the two expansions. The leading deficit is of order $\log L/L$, equivalently $\log E_4/E_4$. Thus a nonvanishing spectral gap does not imply that a near-quarter amplitude advantage requires only modest energy separation.

The numerical figure uses the finite hot preparation $\beta_h = L^{-2}$, whereas (10) uses the limiting reference $\beta_h = 0$. The finite-hot numerator differs by $O(1/L)$, since its highest-state weight is $e^{-2/L}$ instead of one. Both advantages approach $1/4$, but their subleading corrections are not identical. `research/check_quarter_approach.py` verifies the stationarity and Taylor-coefficient algebra symbolically and independently recomputes five plotted maxima at 100-digit precision. Its finite residual table is a diagnostic illustration of the remainder scales, not a proof of a big-O bound; the analytic argument above supplies that bound.

14 The sharp four-state amplitude bound under all triplet tests

The additional F inequalities sharpen the contrast-cone bound from one half to one quarter. The key restriction concerns the first three entries of the slow eigenfunction, rather than monotonicity of all four entries.

Lemma 4 (First-three arithmetic restriction)**.** *Let a positive reversible four-state generator have distinct ordered energies, a finite positive bath, a simple slow mode, and $F_{ijk} \le 0$, $G_{ijk} \le 0$ on every original-rate triplet. Orient its slow left eigenfunction by the nonnegative contrast coordinates of Theorem 4. Then*

$$w_1 \ge w_2, \qquad w_1 + w_2 \ge 2w_3, \qquad w_4 \le 0.$$

Proof. Write $q_i = \pi_i(\beta_b)$, $Q_j = \sum_{i \le j} q_i$, and use the unnormalized prefix contrasts $h^{(1)}, h^{(2)}, h^{(3)}$ from the contrast theorem. Their coordinate generator is

$$B = \operatorname{diag}(\eta_1, \eta_2, \eta_3)^{-1} H^\top D L H, \quad H = (h^{(1)}, h^{(2)}, h^{(3)}), \quad D = \operatorname{diag}(q).$$

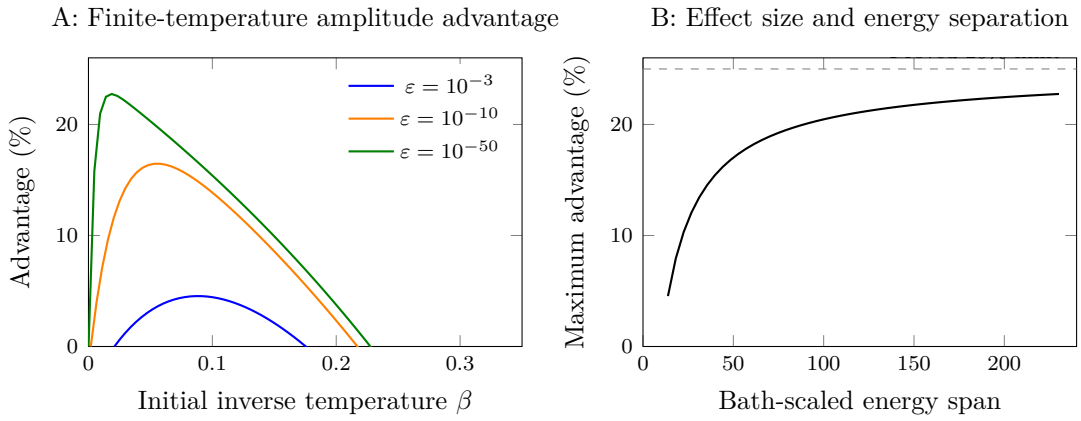


Figure 4: The unperturbed family of Theorem 13. Panel A plots $100[1 - |b_2(\beta_h)|/|b_2(\beta)|]$ with finite $\beta_h = [\log(1/\varepsilon)]^{-2}$. Panel B follows its thermal maximum as the energy span $2\log(1/\varepsilon)$ grows. The substantial limiting advantage is accompanied by increasing energy separation; the plot does not claim practical cooling gains at modest spans. Curves and maxima use 80-digit arithmetic and illustrate the analytic theorem. The separate certified 22.617% witness uses a different fixed hot preparation. The strict-triplet perturbation preserves the limiting result but is not plotted.

It is diagonally similar to the symmetric Metzler contrast matrix. Its Perron eigenvector x can be chosen nonnegative and nonzero, and $w = Hx$. Consequently $w_1 - w_2 = (Q_2/q_2)x_1 \geq 0$ and $w_4 = -(Q_3/q_4)x_3 \leq 0$. Put

$$s = \frac{(q_1 - q_2)q_3}{2q_2Q_3} > 0.$$

The remaining assertion is equivalent to $x_2 \geq sx_1$, since $w_1 + w_2 - 2w_3 = (2Q_3/q_3)(x_2 - sx_1)$.

We first make a reset subtraction, which is used only in this proof and does not alter the original triplet tests. The reduced conductances are $k_{ij} = a_{ij}/q_i$. Their minimum is in the first row, by all the G inequalities. Subtract $k_*(q\mathbf{1}^\top - Q_4 I)$, where $k_* = \min\{k_{12}, k_{13}, k_{14}\}$. All resulting rates are nonnegative; their generator is reversible and conservative, although it may be reducible. Every F and G is unchanged. On the nonstationary contrast space this operation adds $k_*Q_4 I$ to B, so its eigenvectors, simplicity and eigenvalue ordering are unchanged. The shifted largest contrast eigenvalue, denoted λ, is at most zero, because the shifted full matrix remains a Markov generator. At least one of the shifted downward rates a_{12}, a_{13}, a_{14} is zero. All quantities in the rest of this proof refer to this shifted generator.

If $x_1 = 0$, the assertion is immediate. Suppose $x_1 > 0$. Then $\delta = \lambda - B_{33} > 0$. To see strictness, symmetrize B: its largest eigenvalue is at least its third diagonal entry. Equality would force the third row and column of the positive semidefinite matrix $\lambda I - \operatorname{diag}(\sqrt{\eta})B\operatorname{diag}(1/\sqrt{\eta})$ to vanish. The third coordinate vector would then be a Perron eigenvector. Simplicity would force $x_1 = 0$, a contradiction. This also handles a zero shifted eigenvalue.

Eliminate x_3 from the eigenvector equations. The resulting two by two matrix is

$$\mathcal{B}_{ij} = B_{ij} + \frac{B_{i3}B_{3j}}{\delta}, \qquad i, j \leq 2.$$

It has nonnegative off-diagonal entries and is diagonally symmetrizable. The Schur complement of $\lambda I - B$ shows that λ is its simple largest eigenvalue, with eigenvector (x_1, x_2). Define

$$C = B_{21} + s(B_{22} - B_{11}) - s^2 B_{12}, \qquad D_* = (B_{23} - sB_{13})(B_{31} + sB_{32}).$$

Then

$$\delta\{\mathcal{B}_{21} + s(\mathcal{B}_{22} - \mathcal{B}_{11}) - s^2\mathcal{B}_{12}\} = \delta C + D_*.$$

A nonnegative value of this expression implies $x_2/x_1 \geq s$. Indeed, if the two off-diagonal entries of $\mathcal{B}$ are positive, its Perron ratio is the unique positive root of the concave quadratic in braces, viewed as a function of s. If those entries vanish, symmetrizability makes both vanish. Since $x_1 > 0$ and the largest eigenvalue is simple, one then has $x_2 = 0$ and $\mathcal{B}_{11} > \mathcal{B}_{22}$; the expression in braces is strictly negative. Thus this exceptional case cannot occur when the expression is nonnegative.

All G inequalities give $B_{31}, B_{32} \geq 0$, and direct simplification gives

$$B_{23} - sB_{13} = \frac{Q_4}{2q_1q_2Q_3}\{2a_{34}q_1q_2 - a_{14}q_2q_3 - a_{24}q_1q_3\} \geq 0.$$

Hence $D_* \geq 0$. If $C \geq 0$ the desired expression is already nonnegative. If $C < 0$, use $\lambda \leq 0$ to obtain

$$\delta C + D_* \geq -B_{33}C + D_*.$$

It remains to prove that this last, eigenvalue-free expression is nonnegative.

For brevity write the shifted downward rates as $(a, b, c, d, e, f) = (a_{12}, a_{13}, a_{14}, a_{23}, a_{24}, a_{34})$ and define

$$N = -2q_1q_2q_3F_{123} \geq 0, \qquad J = eq_1(2q_2 + q_3) - cq_2(2q_1 + q_3).$$

Exact multiplication yields

$$-B_{33}C + D_* = \frac{Q_2Q_4}{4q_1q_2^2Q_3^2}\,T, \quad T = (c + e + f)N + q_4\{2ceq_3(q_1 - q_2) + fJ\}.$$

The prefactor is positive. Since at least one of a, b, c is zero, nonnegativity follows from these three exhaustive cases:

1. If $c = 0$, then $J = eq_1(2q_2 + q_3) \geq 0$ and all terms of T are nonnegative.

2. If $a = 0$, $F_{124} \leq 0$ gives $cq_2(2q_1 + q_4) \leq eq_1(2q_2 + q_4)$. The ratio $(2q_2 + z)/(2q_1 + z)$ strictly increases with z and $q_3 > q_4$. It follows, without division by c or e, that $J \geq 0$. Thus $T \geq 0$.

3. If $b = 0$, then $N = 2aq_2q_3(q_1 - q_2) + dq_1q_3(2q_2 + q_3)$. If $c \leq e$, one has $J \geq 0$. If $c > e$, the inequality $F_{124} \leq 0$ gives instead

$$N + q_4J \geq 2q_1q_2(q_3 - q_4)(c - e) + dq_1q_3(2q_2 + q_3) \geq 0.$$

Now write $T = (c + e)N + f(N + q_4J) + 2ceq_3q_4(q_1 - q_2) \geq 0$.

This proves $x_2 \geq sx_1$ and completes the lemma, including equality faces and every reducible reset-subtracted case. □

Theorem 14 (Sharp quarter-amplitude bound)**.** *Under the hypotheses of Lemma 4, for every* $0 < \beta_h < \beta_c < \beta_b$,

$$\frac{|b_2(\beta_c)|}{|b_2(\beta_h)|} < \frac{4}{3}.$$

Consequently every ordinary cooling amplitude advantage is strictly below 25%. *The bound is sharp as a supremum, by Theorem 13. The corresponding supremum of the asymptotic relative-entropy advantage is* 7/16, *and the time advance is strictly below* $\log(4/3)/|\lambda_2|$.

Proof. Let $m(\beta)$ be the conditional Gibbs expectation of w on states $1,2,3$. It is nondecreasing for $\beta \geq 0$. To verify this directly, shift $E_1 = 0$, put $a = E_2$, $b = E_3$, $t = e^{-\beta a}$ and $y = e^{-\beta b}$, and use conditional probabilities $(p_1, p_2, p_3) = (1, t, y)/(1 + t + y)$. The first-three vector is a constant plus

$$u(1, -1, 0) + v(1, 1, -2), \qquad u = (w_1 - w_2)/2 \geq 0, \quad v = (w_1 + w_2 - 2w_3)/6 \geq 0.$$

Furthermore

$$(p_1 - p_2)' = \frac{2at + by - (b - a)ty}{(1 + t + y)^2} > 0, \qquad (p_1 + p_2 - 2p_3)' = -3p_3' > 0.$$

The first inequality uses $t \leq 1$ and $b > a > 0$; the second uses that E_3 is the largest of the three energies. Thus m is nondecreasing, strictly unless the first three mode components coincide.

Let $P(\beta) = Z_3(\beta)/Z_4(\beta)$ be the Gibbs mass on the first three states. It strictly increases with β, since the excluded state has strictly higher energy. For finite positive β, $3/4 < P(\beta) < 1$. Write $P_h = P(\beta_h)$, $P_c = P(\beta_c)$, and $r = P_c/P_h < 4/3$. The full overlap is $b_2 = Pm + (1 - P)w_4$. Therefore

$$b_2(\beta_c) - r b_2(\beta_h) = P_c\{m(\beta_c) - m(\beta_h)\} + (1 - r)w_4 \geq 0.$$

The contrast theorem ensures both overlaps are strictly negative on cooling, so $|b_2(\beta_c)| \leq r|b_2(\beta_h)| < (4/3)|b_2(\beta_h)|$. The relative-entropy and time-advance statements follow from their respective quadratic and logarithmic dependence on the amplitude ratio. The existing quarter-limit family proves sharpness; none of these suprema is attained at finite parameters in the stated domain. □

15 A finite-span necessary condition for a specified advantage

The sharp quarter bound has a more restrictive form when the energy span and the two preparation temperatures are specified.

Corollary 4 (Finite-span cooling bound)**.** *Under the full four-state hypotheses of Theorem 14, put*

$$\Delta = E_4 - E_1 > 0, \qquad d = (\beta_c - \beta_h)\Delta > 0, \qquad p_{4,h} = \pi_4(\beta_h),$$

for any finite $0 < \beta_h < \beta_c < \beta_b$. The asymptotic slow-amplitude advantage satisfies

$$\mathcal{A} := 1 - \frac{|b_2(\beta_h)|}{|b_2(\beta_c)|} \leq p_{4,h}(1 - e^{-d}) < \frac{1 - e^{-d}}{4} < \frac{1 - e^{-\beta_b \Delta}}{4}. \tag{11}$$

These bounds also hold when the ordering is ordinary and $\mathcal{A} \leq 0$. Consequently, achieving $\mathcal{A} \geq g$ for a specified $0 < g < 1/4$ requires

$$(\beta_c - \beta_h)\Delta > -\log(1 - 4g), \qquad p_{4,h} > g.$$

In particular $\beta_b \Delta > -\log(1 - 4g)$ is necessary if the preparation temperatures may be chosen freely on the cooling side. These are necessary conditions only; no sufficiency or sharpness is asserted for these finite-span bounds.

Proof. Write $P_3(\beta) = 1 - \pi_4(\beta)$. The proof of Theorem 14 established

$$\frac{|b_2(\beta_c)|}{|b_2(\beta_h)|} \leq \frac{P_3(\beta_c)}{P_3(\beta_h)}.$$

Both overlaps are nonzero by the contrast theorem, so inversion gives $\mathcal{A} \le 1 - P_3(\beta_h)/P_3(\beta_c)$. Shift the energy origin to $E_1 = 0$, leaving $E_4 = \Delta$. Then $Z_3(\beta_c) \le Z_3(\beta_h)$ and

$$\frac{\pi_4(\beta_c)}{P_3(\beta_c)} = \frac{e^{-\beta_c \Delta}}{Z_3(\beta_c)} \ge e^{-d} \frac{\pi_4(\beta_h)}{P_3(\beta_h)}.$$

Multiplying by $P_3(\beta_h)$ gives

$$\frac{P_3(\beta_h)}{P_3(\beta_c)} \ge P_3(\beta_h) + e^{-d} p_{4,h},$$

which proves the first bound. The highest-energy state has strictly less than one-quarter Gibbs mass at finite positive β_h, so the second inequality is strict. The last is strict because $\beta_c - \beta_h < \beta_b$. Solving $g < (1 - e^{-d})/4$ gives the required dimensionless separation; also $g \le \mathcal{A} \le p_{4,h}(1 - e^{-d}) < p_{4,h}$. □

For example, amplitude advantages of at least 10%, 20%, and 24% require, respectively,

$$d > \log(5/3) \approx 0.5108, \qquad d > \log 5 \approx 1.6094, \qquad d > \log 25 \approx 3.2189.$$

They also require $p_{4,h} > 0.10$, 0.20, and 0.24, respectively. These are restrictions on initial inverse-temperature separation and highest-state occupation, not guarantees that a generator satisfying them exhibits an effect. The quantity $\mathcal{A}$ concerns the asymptotic slow amplitude, not a finite-time distance or an absolute relaxation-time reduction.

15.1 An exact infinite-temperature endpoint criterion

The full triplet premise also removes the possible hidden interior cooling interval: any anomaly begins at the high-temperature end of the Gibbs curve. Only an endpoint derivative is needed to decide its existence.

Corollary 5 (Cooling is decided by one endpoint covariance)**.** *Under the hypotheses and orientation of Lemma 4, an ordinary asymptotic cooling effect exists if and only if*

$$b_2'(0) < 0, \quad \textit{equivalently} \quad \mathrm{Cov}_{\mathrm{unif}}(w, E) > 0.$$

Here the uniform law is on all four states; $\beta = 0$ is used only as an analytic endpoint, not as an allowed witness preparation. If this condition holds, there is exactly one finite positive stationary point of b_2. It lies in $(0, \beta_b)$, and $|b_2|$ has its unique cooling maximum there. The anomalous local ordering region is exactly $(0, \beta_{\mathrm{turn}})$. If $b_2'(0) \ge 0$, then $b_2'(\beta) > 0$ for every finite $\beta > 0$, including when the endpoint derivative is zero, and no ordinary cooling effect occurs. With a simple slow mode these conclusions apply to every fixed norm asymptotically.

Proof. Write $b = b_2$. The contrast theorem gives $b < 0$ on cooling and $b'(\beta_b) > 0$, as well as strict heating monotonicity. The derivative at the bath is strictly positive directly: every prefix-contrast numerator vanishes there with a strictly positive derivative, and the nonnegative Perron coordinate vector is nonzero.

Put $C = w_3 - w_4$. First suppose $C \ge 0$. By the arithmetic restriction,

$$m(0) = \frac{w_1 + w_2 + w_3}{3} \ge w_3 \ge w_4,$$

where m is the conditional first-three Gibbs expectation from the sharp bound. Since m is nondecreasing, $b = Pm + (1 - P)w_4$ gives

$$b' = Pm' + P'(m - w_4) > 0 \qquad (\beta \ge 0).$$

To justify strictness, if the first three components are not all equal, the elementary contrast decomposition in the preceding proof gives $m' > 0$. Otherwise m is constant, and nonconstancy of the full mode, its zero bath mean and $w_4 \le 0$ force $m > w_4$; then $P' > 0$ supplies strictness. Thus this case has no cooling anomaly.

Now suppose $C < 0$. Use the derivative-curve representation (5): $b' = g\Phi$, where $g = \pi_1' > 0$,

$$\Phi(u) = A + Bu + Cv(u), \qquad u(\beta) \searrow -1 \text{ as } \beta \to \infty.$$

The derivative-curve lemma makes Φ strictly concave in u. It extends continuously to the two endpoints of $[-1, u(0)]$, with

$$\Phi(-1) = w_1 - w_2 \ge 0, \qquad \operatorname{sign}\Phi(u(0)) = \operatorname{sign} b'(0).$$

If $\Phi(u(0)) \ge 0$, strict concavity puts its value strictly above the nonnegative chord at every interior point. Hence $b'(\beta) > 0$ for every finite $\beta > 0$, even in the endpoint equality case.

If $\Phi(u(0)) < 0$, its positive value at the bath and its nonnegative left endpoint imply that it has exactly one zero between $u(\beta_b)$ and $u(0)$. Indeed a concave function has an interval as its nonnegative superlevel set; strict concavity and positivity at the bath rule out an interior zero on the left-hand interval. On the right-hand interval the slope of chords is decreasing, so after reaching zero the function cannot return to nonnegative values. The zero is therefore unique, with negative values to its right and positive values to its left. Since u decreases with β, b' is negative for $0 \le \beta < \beta_{\text{turn}}$ and positive thereafter. The turning point lies strictly before the bath. On cooling, $|b| = -b$, proving the stated maximum and anomalous interval. Strict negativity persists at small finite positive inverse temperatures, so the criterion supplies admissible finite witnesses.

Finally $b'(0) = -\operatorname{Cov}_{\text{unif}}(w, E)$ by the fixed-mode covariance identity. No interpretation of a stationary point alone is used. □

The covariance is explicitly

$$\operatorname{Cov}_{\text{unif}}(w, E) = \frac{1}{4}\sum_{i=1}^{4} w_i E_i - \frac{1}{16}\Big(\sum_i w_i\Big)\Big(\sum_i E_i\Big).$$

Thus within the triplet-failure family there is no need to scan the initial cooling temperatures or solve an interior derivative equation merely to decide existence. Section 15.3 gives an equivalent evaluation using one cubic root and original-rate matrix products, avoiding construction of the slow eigenvector. This is an asymptotic criterion; it does not classify transient distance crossings.

15.2 Exact classification when only one mode is faster

The full triplet premise is especially restrictive when all but one of the nonstationary modes share the slow rate. The resulting classification holds in every dimension at least four and completely covers slow spectral degeneracy in four states.

Theorem 15 (A single faster mode must mix an adjacent pair)**.** *Let $n \ge 4$, and let M be an irreducible reversible n-state generator with distinct energy levels, positive finite bath temperature, and $F_{ijk}, G_{ijk} \le 0$ on every original-rate energy-ordered triplet. Write $q = \pi(\beta_b)$, $D = \operatorname{diag}(q)$ and $R = q\mathbf{1}^\top - I$. Suppose*

$$0 > \lambda_2 = \cdots = \lambda_{n-1} > \lambda_n.$$

Then for exactly one $k \in \{1, \dots, n-1\}$,

$$M = \rho R + \delta J^{(k)}, \qquad \rho > 0, \quad \delta > 0,$$

where $J^{(k)}$ is supported on the adjacent pair $(k, k+1)$,

$$J^{(k)}_{k,k+1} = \frac{q_k}{q_k + q_{k+1}}, \qquad J^{(k)}_{k+1,k} = \frac{q_{k+1}}{q_k + q_{k+1}},$$

and its diagonal is set by zero column sums. Conversely, every such matrix obeys all the triplet inequalities, with eigenvalues 0, $-\rho$ of multiplicity $n-2$, and $-\rho-\delta$. If all nonzero eigenvalues coincide, then $M = \rho R$.

Every generator in these classes preserves stochastic order. Consequently, for cooling or heating Gibbs preparations, the farther preparation has at least as large a convex divergence from the bath at every $t \geq 0$. This holds for every convex function finite on positive arguments; it is strict at finite times for strictly convex functions and distinct preparations. In particular there is no ordinary or strong asymptotic effect in KL or weighted Euclidean distance.

Proof. Use normalized weights $q_1 > \cdots > q_n > 0$. Set $\rho = -\lambda_2$ and $\delta = \lambda_2 - \lambda_n$. The reversible spectral resolution gives

$$M = \rho R - \frac{\delta}{N} D z z^\top, \qquad q^\top z = 0, \qquad N = z^\top D z > 0,$$

where z is a real fast left eigenvector. Substitution of $a_{ij} = \rho q_i - \delta q_i z_i z_j / N$ into the original tests gives

$$\begin{aligned} G_{ijk} &= -\frac{\delta q_j}{N} z_k (z_i - z_j), \\ F_{ijk} &= \frac{\delta}{N} \left\{ (q_i - q_j) z_i z_j - z_k \left[q_i z_i - q_j z_j + \frac{q_k}{2}(z_i - z_j) \right] \right\}. \end{aligned}$$

Thus $G \leq 0$ means $z_k(z_i - z_j) \geq 0$. When $z_i = z_j = a$, the second identity is $F_{ijk} = \delta(q_i - q_j) a(a - z_k)/N$; when $z_k = 0$ it is $F_{ijk} = \delta(q_i - q_j) z_i z_j / N$.

Let m be the largest index with $z_m \neq 0$ and orient $z_m > 0$. Zero bath mean excludes $m = 1$. If $m \geq 4$, the tests $G_{ijm} \leq 0$ order the entire prefix:

$$z_1 \geq z_2 \geq \cdots \geq z_{m-1}.$$

If some $i < j \leq m-2$ had $z_i > z_j$, the test $G_{ij,m-1} \leq 0$ would force $z_{m-1} \geq 0$. Every earlier entry would then be nonnegative, contradicting zero bath mean because $z_m > 0$ and all later entries vanish. Hence

$$z_1 = \cdots = z_{m-2} = a.$$

Since $m - 2 \geq 2$, $F_{12m} \leq 0$ gives $0 \leq a \leq z_m$. Zero bath mean forces $z_{m-1} < 0$, and $F_{12,m-1} \leq 0$ then forces $a = 0$. Therefore the support is exactly $(m-1, m)$.

If $m = 3$, the condition $n \geq 4$ supplies the zero component z_4. The test $G_{123} \leq 0$ gives $z_1 \geq z_2$, and zero bath mean forces $z_2 < 0$. Now $F_{124} \leq 0$ gives $z_1 \geq 0$, whereas $F_{134} \leq 0$ gives $z_1 \leq 0$. Thus $z_1 = 0$, and the support is exactly $(2, 3)$. If $m = 2$, zero bath mean immediately gives support exactly $(1, 2)$. This proves that only an adjacent pair can occur.

For support $(k, k+1)$ choose $z_k = -1/q_k$, $z_{k+1} = 1/q_{k+1}$. Then $-D z z^\top / N = J^{(k)}$ with the displayed rates. This proves necessity and uniqueness of the pair. It also proves strict positivity of all off-diagonal rates, even if only nonnegative rates were assumed.

For the converse put $d = \delta/(q_k + q_{k+1})$. The only nonzero triplet tests are

$$\begin{aligned} F_{k,k+1,l} &= -(q_k - q_{k+1}) d && (l > k+1), \\ F_{i,k,k+1} &= -(q_k + q_{k+1}/2) d && (i < k), \\ G_{i,k,k+1} &= -q_k d && (i < k). \end{aligned}$$

They are all negative; every omitted test is zero. The matrix $J^{(k)}$ has eigenvalue -1 on the right eigenvector Dz and vanishes on its $(n-1)$-dimensional kernel. The reset R is $-I$ on the zero-sum probability-perturbation subspace. This proves the spectrum. If all nonzero eigenvalues coincide, the reversible spectral resolution directly gives $M = \rho R$.

For stochastic monotonicity, take adjacent differences $d_i = f_{i+1} - f_i$ of functions. The backward reset generator induces $-I$ on these differences. The difference matrix of $J^{(k)}$ has only column k nonzero: its diagonal entry is -1, its entry in row $k-1$, when present, is $q_{k+1}/(q_k + q_{k+1})$, and its entry in row $k+1$, when present, is $q_k/(q_k + q_{k+1})$. Thus the difference generator is Metzler. The earlier reversible stochastic-order argument gives all-time convex-divergence ordering. Strict convexity makes its supporting-line inequality strict for unequal evolved distributions, which remain unequal at finite times because e^{tM} is invertible. This argument never selects a scalar vector from the repeated slow eigenspace. The strong-effect exclusion also follows directly from the earlier all-G projection theorem. □

Corollary 6 (The complete four-state degenerate boundary)**.** *Under the full four-state triplet premise, a double slow eigenvalue forces reset plus one adjacent-pair mixing term, and a triple slow eigenvalue forces pure reset. Both cases have all-time convex-divergence ordering. In particular every ordinary asymptotic cooling counterexample in KL or weighted Euclidean distance has a simple slow eigenvalue. Strict failure of all eight linear tests also forces spectral simplicity at the slow rate, since the classified degenerate generators have several triplet equalities.*

The inequalities $F, G \le 0$ are the primary hypotheses if zero rates are initially allowed; they agree with the original branchwise predicate for positive rates. This avoids defining κ by division by a possibly zero rate. The classification then implies positivity.

The reset-plus-pair family itself is prior work. Section II of Gelbwaser-Klimovsky, *Efficient protocol for the Markovian Mpemba effect in N-level systems*, arXiv:2609.09988v1, 9 September 2026 introduces the single-pair states-deviation protocol (SPSD). Our $\delta > 0$ adjacent-pair family is its positive-deviation case, with $\Delta = \delta/[\rho(q_k + q_{k+1})]$. That source already shows that positive pair deviation can activate the triplet test only when a level lies strictly between the pair. The necessity proved here is a converse classification: under the one-fast-mode spectrum and all triplet failures, every admissible generator must belong to that adjacent-pair family. The all-time divergence conclusion is an application of the established stochastic-order method, not a new construction or principle.

The restriction $n \ge 4$ is essential for the rank-one classification. For $n = 3$, take $q = (3/5, 3/10, 1/10)$, $z = (1, 2, -12)$, and $M = R - Dzz^\top/(z^\top Dz)$. All rates are positive, its spectrum is $0, -1, -2$, and $F_{123} = 0$, $G_{123} = -2/9$, but the fast mode uses all three states. There is no repeated slow eigenvalue in that dimension; the example pinpoints why an additional triplet is needed in the proof. For $n > 4$, the theorem concerns slow multiplicity exactly $n-2$ (or $n-1$ for reset). It does not classify smaller repeated slow eigenspaces with two or more distinct faster modes. No publication-priority claim is made for this classification.

Exact rank-one identities and general pair formulas are audited by `research/one_fast_mode_certificate.py`. The script additionally checks explicit symbolic matrices in dimensions four through eight, including their projector identities, spectra, triplet tables and Metzler difference matrices; the arbitrary-dimensional statement is proved by the support argument above, not inferred from those checks.

The restriction on multiplicity matters structurally. A five-state counterexample to extending the monotonicity conclusion to every repeated slow space has $q = (24, 12, 4, 2, 1)/43$ and spectrum $0, -1, -1, -2, -3$. It obeys all $F, G \le 0$, yet its adjacent-difference matrix has entry $K_{21} = -14/27$. The exact construction is given in `research/five_state_degenerate_nonmonotone.txt` and checked by its companion script. This rules out that stronger structural inference; it does not by itself exhibit a thermal divergence reversal.

15.3 A scalar rate test without computing a slow eigenvector

The endpoint criterion can be evaluated from the generator and one root of a cubic. This uses the classical polynomial formula for a spectral projector; the point is an explicit implementation of the preceding problem-specific criterion, not a new projector construction.

Let $q = \pi(\beta_b)$, $D = \operatorname{diag}(q)$, and put

$$a = \tfrac{1}{4}\mathbf{1} - q, \qquad d = -\tfrac{1}{4}(E - \bar{E}\mathbf{1}), \qquad \bar{E} = \tfrac{1}{4}\sum_i E_i, \qquad h = D^{-1}a.$$

Thus a is the infinite-temperature displacement and $d = \pi'(0)$. Define $\tau = \operatorname{tr} M$, $\sigma = [(\operatorname{tr} M)^2 - \operatorname{tr}(M^2)]/2$ and $A_j = h^\top M^j d$ for $j = 0, 1, 2$. The scalar polynomial

$$\mathcal{R}(x) = A_0 x^2 + (A_1 - \tau A_0)x + A_2 - \tau A_1 + \sigma A_0$$

has degree at most two. Let $p(x) = \det(xI - M)/x$ and let r be its largest root.

Corollary 7 (Rate evaluation of the triplet endpoint test)**.** *For a positive reversible four-state generator satisfying all original triplet failures, if r is simple, ordinary asymptotic cooling occurs if and only if $\mathcal{R}(r) > 0$. Equality means absence. If the largest nonzero root is repeated, cooling is absent by Theorem 15. Thus the full-premise decision requires a cubic root and a scalar quadratic evaluation, with no choice or numerical normalization of a slow eigenvector.*

Proof. Write the other two roots as s, t, permitting $s = t < r$. On the invariant zero-sum subspace the spectral projector onto r is

$$P_r = \frac{(M - sI)(M - tI)}{(r - s)(r - t)} = \frac{M^2 - (\tau - r)M + (\sigma - \tau r + r^2)I}{p'(r)}.$$

This identity is asserted on that subspace, not on the stationary direction. Reversibility ensures diagonalizability even if $s = t$. Since d has zero sum, it follows that $h^\top P_r d = \mathcal{R}(r)/p'(r)$. For any nonzero left slow eigenvector w, detailed balance gives

$$P_r = \frac{Dww^\top}{w^\top Dw}, \qquad \frac{\mathcal{R}(r)}{p'(r)} = \frac{(w^\top a)(w^\top d)}{w^\top Dw} = \frac{b(0)b'(0)}{w^\top Dw}.$$

Here $w^\top q = 0$, so $b(0) = w^\top a$. Orienting the mode by the contrast theorem makes $b(0) < 0$. Corollary 5 therefore makes the product positive exactly when cooling occurs. Finally $p'(r) = (r - s)(r - t) > 0$, proving the sign rule, including equality. □

All quantities except the single cubic root are explicit matrix products of the original rates, bath weights, and energies. The test still uses global spectral information through r; it is not a new local triplet test. For rational bath weights and commensurate energies, the common positive energy unit can be removed and the remaining sign can be decided by ordered algebraic arithmetic. For arbitrary real energies, floating evaluation near zero does not settle exact equality. This corollary concerns existence of an asymptotic effect, not every transient crossing for a specified distance.

15.4 Exact rational-input classification without commensurate energies

Within the full triplet-failure family, the endpoint criterion admits a terminating exact decision procedure even when the energy gaps are incommensurate. The additional ingredient is Baker's linear-independence theorem for logarithms, rather than a polynomial substitution for the whole Gibbs curve. This is a consequence of an established transcendence theorem; no new number-theoretic independence claim is made.

Theorem 16 (Rational-input endpoint decision)**.** *Consider a positive reversible four-state generator with rational entries, distinct ordered Gibbs bath weights and positive finite bath temperature. Suppose all original triplet predicates fail. Ordinary asymptotic cooling in relative entropy or weighted Euclidean distance can be decided by a terminating exact procedure using rational arithmetic, integer factorization, real algebraic root isolation and rational logarithm enclosures. Commensurate energy gaps are not required. A repeated slow eigenvalue is handled by the preceding degenerate-boundary classification, which excludes an effect in these distances. When the slow mode is simple, the same decision applies asymptotically to every fixed norm. The procedure concerns this full-triplet family; it is not a classifier for arbitrary unconstrained rational generators or for transient distance crossings.*

Proof. The unique normalized stationary law q is rational, since it solves a rational linear system. Equivalently one may supply rational ordered bath weights and rational downward rates, construct the reverse rates by detailed balance, and check the hypotheses exactly. In bath energy units take $E_i = -\log q_i$; an additive energy shift does not affect the criterion. All logarithms below are the real logarithms of positive numbers.

Let r be the largest root of the rational nonzero characteristic cubic $p(x) = \det(xI - M)/x$. Determine its multiplicity exactly. If it is repeated, the full-triplet degeneracy theorem returns absence of an effect. Otherwise write

$$\tau = \operatorname{tr} M, \qquad \sigma = \frac{\tau^2 - \operatorname{tr}(M^2)}{2}, \qquad K_r = M^2 - (\tau - r)M + (\sigma - \tau r + r^2)I.$$

Define the rational vectors and matrix

$$a = \tfrac{1}{4}\mathbf{1} - q, \qquad h = D^{-1}a, \qquad T = I - \tfrac{1}{4}\mathbf{1}\mathbf{1}^\top, \quad D = \operatorname{diag}(q),$$

and the row vector

$$c^\top = \tfrac{1}{4}h^\top K_r T \in \mathbb{Q}(r)^{1\times 4}.$$

The endpoint scalar from the spectral-projector test is

$$\mathcal{R}(r) = \sum_{i=1}^{4} c_i \log q_i. \tag{12}$$

Indeed $\pi'(0) = \frac{1}{4}T(\log q_i)_i$, and on the zero-sum probability subspace the slow spectral projector is $K_r/p'(r)$. Since $\pi'(0)$ has zero sum, this suffices even though the polynomial need not annihilate the stationary direction on the full space. For an oriented slow left mode w, the projector identity gives

$$\frac{\mathcal{R}(r)}{p'(r)} = \frac{(w^\top a)(w^\top \pi'(0))}{w^\top D w}.$$

Here $p'(r) > 0$ for a simple largest nonzero root, including when the two faster roots coincide, and $w^\top a = b_2(0) < 0$ by the contrast theorem. Thus the single-endpoint criterion says cooling occurs exactly when $\mathcal{R}(r) > 0$; equality means absence, not an undecided threshold.

Factor the numerators and denominators of the four positive rationals q_i. Let $\mathcal{P}$ be the resulting finite set of primes and write

$$q_i = \prod_{\ell\in\mathcal{P}} \ell^{v_{\ell i}}, \qquad v_{\ell i} \in \mathbb{Z}.$$

Then

$$\mathcal{R}(r) = \sum_{\ell\in\mathcal{P}} \gamma_\ell \log \ell, \qquad \gamma_\ell = \sum_i v_{\ell i} c_i \in \mathbb{Q}(r).$$

Each γ_ℓ has an exact zero test in the real algebraic field specified by the isolated root r.

The precise external theorem needed is the following form of Baker's result: if logarithms of nonzero algebraic numbers, together with $2\pi i$, are linearly independent over $\mathbb{Q}$, those logarithms are linearly independent over $\overline{\mathbb{Q}}$. This is stated in the opening paragraph of A. Baker, *Linear forms in the logarithms of algebraic numbers (II)*, Mathematika 14 (1967), 102–107, referring to the original 1966 paper, Mathematika 13, 204–216. For distinct positive primes, rational independence of their real logarithms follows from unique prime factorization: clear the denominators of a rational relation and exponentiate. Adjoining $2\pi i$ preserves that independence because all the prime logarithms are real. Baker's theorem therefore gives

$$\mathcal{R}(r) = 0 \quad \Longleftrightarrow \quad \gamma_\ell = 0 \text{ for every } \ell \in \mathcal{P}.$$

This is linear independence over algebraic coefficients, not algebraic independence of the logarithms. There is no branch ambiguity: only real logs of primes greater than one are used. A rational factor equal to one has empty factorization and contributes zero, so $\log 1$ is never included in the independent set.

If every coefficient vanishes, return equality and absence. Otherwise the sum is a nonzero real number. Refine the rational isolating interval for r, evaluate the algebraic coefficients by directed rational interval arithmetic, and enclose each logarithm by a convergent rational series. For example, for a prime ℓ put $z = (\ell - 1)/(\ell + 1) \in (0, 1)$. The partial sum and a valid tail bound are

$$L_N = 2\sum_{j=0}^{N} \frac{z^{2j+1}}{2j+1}, \qquad 0 < \log \ell - L_N \le \frac{2z^{2N+3}}{(2N+3)(1-z^2)}.$$

All quantities on the right are rational and the bound tends to zero. Refining all intervals produces an enclosure of $\mathcal{R}(r)$ whose width tends to zero. Its nonvanishing guarantees that eventually the enclosure excludes zero, deciding its sign. No unproved transcendental equality oracle or floating-point tolerance enters the decision. □

The theorem establishes termination, not a practical complexity bound. Integer factorization and very small nonzero logarithmic forms can be expensive; no uniform precision or runtime promise is made. It also does not remove the full-generator spectral dependence: isolating the one slow cubic root remains necessary, although a full slow eigenvector and a scan of initial temperatures are avoided. For arbitrary nonrational input data, the analytic scalar criterion remains valid but this rational-input termination argument does not apply.

The implementation is `research/exact_rational_triplet_classifier.py`. It checks the full premise exactly, factors the supplied rational weights, and uses the selected real algebraic embedding throughout. Range reduction to logarithms of numbers between one and two accelerates the rational series. An optional resource limit returns an unresolved result, never absence. With no such limit the theorem gives termination. The original counterexample, whose energy gaps are not all commensurate, is classified as cooling by an exact positive endpoint enclosure. An independent implementation audit includes commensurate comparisons with the Sturm classifier, scale invariance, degenerate cases and invalid-input rejection.

16 Counterexamples in every higher dimension

The failure is not restricted to exactly four states. A fast, high-energy state can be added without repairing any of the triplet tests.

Theorem 17 (Dimension extension)**.** *Suppose an n-state positive reversible generator, with distinct energies and a positive finite bath, has a simple slow mode and a strict ordinary cooling witness at two finite positive initial temperatures. Suppose every original-rate triplet predicate fails. There is an $(n+1)$-state positive reversible system with the same properties, retaining every original off-diagonal rate. If all original F, G inequalities are strict, every inequality in the extension is strict as well. Consequently, counterexamples form nonempty open sets for every state count $n \geq 4$.*

Proof. Use units $\beta_b = 1$ and unnormalized bath weights $q_i = e^{-E_i}$. Choose $A > 0$ so large that $-nA$ lies strictly below every original nonzero eigenvalue. Add a new highest state of weight $\epsilon = e^{-E_{n+1}}$, with downward rates $a_{i,n+1} = A$ for every $i \leq n$ and reverse rates $a_{n+1,i} = A\epsilon/q_i$. Recompute the diagonal to conserve probability. All rates are positive and detailed balance holds. Original triplet tests are unchanged. In a new triplet $i < j < n+1$, writing $t = e^{-(E_j - E_i)} < 1$, $u = e^{-(E_{n+1} - E_j)} > 0$, and $C = a_{ij} > 0$, one has

$$G = A(t-1) < 0, \qquad F = -(1-t)(C + Au/2) < 0.$$

Thus every added triplet fails strictly.

As $\epsilon \downarrow 0$, the full generator tends to

$$\begin{pmatrix} M & A\mathbf{1} \\ 0 & -nA \end{pmatrix}.$$

Its spectrum is the original spectrum together with $-nA$. The original slow eigenvalue remains simple and separated. Its left eigenvector extends from w to

$$\left(w, \frac{A \sum_{i=1}^{n} w_i}{\lambda_2 + nA} \right).$$

The denominator is nonzero by the choice of A. Ordinary finite-dimensional perturbation theory for a simple eigenvalue now supplies a continuously normalized slow eigenvector for sufficiently small positive ϵ, with a spectral gap bounded away from zero. This argument uses the generator directly; it does not symmetrize at the singular limiting bath weight.

For either fixed initial inverse temperature $\beta > 0$, the new Gibbs weight is $\epsilon^\beta \to 0$, so its preparation converges to the old Gibbs vector with a zero extra coordinate. The two slow overlaps therefore converge to the original overlaps. Their strict inequality persists for sufficiently small positive ϵ. Every energy is then finite and distinct. Induction extends the four-state counterexample to every larger dimension. Starting from the strict four-state example, all triplet margins, rate positivity, spectral simplicity and the overlap inequality are strict; continuity gives an open neighborhood at each finite extension in the energy/downward-rate parametrization of reversible systems. □

The construction adds rapidly emptying, initially weakly populated high-energy states. It establishes dimensional persistence, not an enhancement of the effect or a claim that the added states are experimentally useful. In particular, the finite positive witness temperatures must be fixed before taking the high-energy limit; the argument would not apply to a preparation at exactly infinite temperature.

The standard-library checker `five_state_extension_certificate.py` adds $E_5 = 9000 \log 10$ to the primary four-state witness, with $A = 1000$. It verifies all ten original Boolean triplet predicates, all twenty positive rates, detailed balance and column conservation. Exact rational inertia places the new fast eigenvalue in $(-4001, -3999)$ and preserves the original three spectral brackets. A weighted residual estimate controls the original four eigenvector coordinates; the new row equation controls the fifth without dividing by its extremely small bath weight. The same finite preparations $\beta_h = 1/1000$, $\beta_c = 1/100$ retain a rigorously certified overlap advantage greater than 10^{-4}. The fifth state's thermal weights are exactly 10^{-9} and 10^{-90} at those preparations.

17 Hidden cooling advantages approaching unity with dimension

The sharp four-state bound does not give a dimension-independent bound. Cloning the high-energy state produces an increasingly large normalization change along the initial Gibbs curve, while retaining a uniformly isolated slow relaxation mode.

Theorem 18 (A dimension-dependent achievable advantage)**.** *For every integer $n \geq 4$ there is a family of strictly positive, reversible n-state generators at bath inverse temperature one, with all energies distinct and all original $F_{ijk}, G_{ijk} < 0$, such that finite $0 < \beta_h < \beta_c < 1$ satisfy*

$$1 - \frac{|b_2(\beta_h)|}{|b_2(\beta_c)|} \longrightarrow 1 - \frac{3}{n}.$$

The slow eigenvalue tends to -3 and its separation from the next mode is greater than $1/4$ for all sufficiently small family parameters. The corresponding leading asymptotic KL advantage tends to $1 - 9/n^2$. Consequently no dimension-independent upper bound less than one can hold for these amplitude or leading KL advantages under the full triplet premise. Sharpness is asserted only for $n = 4$, by the preceding four-state theorem; no matching upper bound is claimed for $n > 4$.

Proof. Fix n, put $m = n - 3$, and choose strictly decreasing clone weights

$$r_j = \frac{2^{m-j}}{2^m - 1}, \qquad j = 1, \ldots, m, \qquad \sum_j r_j = 1.$$

Start with the quarter-limit family M_ε, its normalized bath weights q_i, and its left slow vector w from Theorem 13. Replace its fourth state by m states $c_1, \ldots, c_m$ with weights $q_4 r_j$. Equivalently the unnormalized bath weights are

$$\widehat{q}^{(n)} = (1, 1 - \varepsilon/3, 3/5, \varepsilon^2 r_1, \ldots, \varepsilon^2 r_m).$$

Their sum remains the four-state partition function. For small positive ε they give n distinct ordered energies $E_i = -\log \widehat{q}_i^{(n)}$.

Keep the original rates among the first three states. From each clone to low state $i \leq 3$ set $a_{i,c_j} = a_{i4}$, and in the reverse direction set $a_{c_j,i} = r_j a_{4i}$. Within the clone block use conditional reset at rate four: $a_{c_j,c_k} = 4r_j$ for $j \neq k$. Set all diagonals by conservation, giving a generator $\overline{M}$. All rates are positive and detailed balance is immediate. Lumping the clone probabilities exactly recovers M_ε. The additional $(m-1)$-dimensional subspace consists of probability perturbations supported on the clones whose sum is zero. On this subspace the generator is scalar, with eigenvalue

$$-4 - \sum_{i=1}^{3} a_{i4} < -4.$$

Thus $\overline{M}$ retains the simple slow eigenvalue -3, gap exceeding $1/3$, and left eigenvector

$$\overline{w} = (1, -4/5, w_3, -\varepsilon, \ldots, -\varepsilon).$$

It has zero bath mean and weighted norm at most one for small ε.

Here are all the original-rate triplet checks before a small repair. Triplets in the first three states are unchanged. For two low states $i < j \leq 3$ and a clone c_l,

$$G_{ijc_l} = G_{ij4}, \qquad F_{ijc_l} = F_{ij4} + \frac{(r_l - 1)q_4}{2q_j} G_{ij4}.$$

For one low state i and two clones $c_j < c_k$,

$$G_{ic_jc_k} = r_j(a_{4i} - 4), \qquad F_{ic_jc_k} = (r_j + r_k/2)(a_{4i} - 4).$$

All-clone triplets have $F = G = 0$. The original rates satisfy $0 < a_{i4} < 2$ throughout $0 < \varepsilon \le 1/30$. This bound is proved by three exact Bernstein checks on the rational generator. Their finite limits are

$$(a_{14}, a_{24}, a_{34}) \longrightarrow (220, 220, 132)/169.$$

Consequently $a_{4i} = a_{i4}q_4/q_i \le (10/3)\varepsilon^2 < 4$, so the one-low triplets are strictly negative. For the two-low/clone triplets, $G_{ij4} \le 0$ and $-G_{ij4} \le a_{j4} < 2$, while $q_4/q_j \le (5/3)\varepsilon^2$. Thus the possible positive defect in F is at most $(5/3)\varepsilon^2$. Every G is already nonpositive.

Define an auxiliary generator H at the new bath weights. On the first three states its reduced conductances are $k_{12} = 100, k_{13} = 10, k_{23} = 11$. Between low state i and any clone use $k_{i,c_j} = i$; among ordered clones use constant downward rate four, with reverse rates fixed by detailed balance. All its tests are strictly negative throughout $0 < \varepsilon \le 1/30$. The first-three tests are exactly those of the certified auxiliary generator in the quarter-limit construction. Write $Q = 13/5 - \varepsilon/3 + \varepsilon^2$ for the unchanged partition sum. For low pairs $12, 13, 23$ and clone c_l, the F values of H are

$$\frac{-1 - 98\varepsilon/3 - \varepsilon^2 r_l/2}{Q}, \qquad \frac{-24/5 - \varepsilon^2 r_l}{Q}, \qquad \frac{-21/5 + 3\varepsilon - \varepsilon^2 r_l/2}{Q}.$$

Each is at most $-1/3$ on the interval, and the G values are strictly negative. For one low state i and clones $j < k$ the tests are

$$G = iq_{c_j} - 4 < 0, \qquad F = G(1 + q_{c_k}/(2q_{c_j})) < 0.$$

For three clones they are $G = 4(t-1) < 0$ and $F = 4(t-1)(1 + u/2) < 0$, since the r_j are distinct and ordered.

Now put $\widetilde{M} = \overline{M} + 6\varepsilon^2 H$. Linearity makes every G strict. In two-low/clone triplets,

$$F(\widetilde{M}) \le (5/3)\varepsilon^2 - 6\varepsilon^2/3 = -\varepsilon^2/3 < 0.$$

All other F tests are already negative or acquire a strict negative contribution from H. Thus all tests are strict on the entire interval $0 < \varepsilon \le 1/30$.

The repair does not spoil the spectrum or limiting overlaps. Let $D_n = \mathrm{diag}(q^{(n)})$. The symmetric matrix $D_n^{-1/2} H D_n^{1/2}$ has operator norm at most

$$C_n = 2\{103 + 4(m-1)\}.$$

Indeed all rates leaving a low state sum to at most 103, and those leaving a clone sum to at most $3 + 4(m-1)$; each upward internal clone rate is at most its downward rate four. Gershgorin applied to the column generator and reversibility place its eigenvalues in $[-C_n, 0]$. This argument avoids any assumption of uniformly bounded reduced conductances inside the increasingly rare clone block. Weyl's inequality gives slow separation at least $1/3 - 12C_n\varepsilon^2 > 1/4$ for sufficiently small ε. The isolated eigenvector perturbation estimate in the quarter-limit proof gives, with consistent orientation and equal weighted norm,

$$\|\widetilde{w} - \overline{w}\|_{q^{(n)}} = O_n(\varepsilon^2).$$

Since $q_i^{(n)} \ge r_m\varepsilon^2/3$ for small ε,

$$\max_i |\widetilde{w}_i - \overline{w}_i| = O_n(\varepsilon).$$

Every thermal overlap changes by at most this quantity, uniformly in initial inverse temperature. The constants may depend on the fixed integer n; no joint limit with n and ε is used.

Finally put $L = \log(1/\varepsilon)$ and choose $\beta_h = L^{-2}$, $\beta_c = L^{-1/2}$. At β_h every unnormalized Gibbs weight tends to one. At β_c each clone weight $\exp(-2\sqrt{L})r_j^{1/\sqrt{L}}$ tends to zero, while the first three weights tend to one. Since the first three limiting slow components sum to $-2/15$ and all clone components tend to zero,

$$b_h \longrightarrow -\frac{2}{15n}, \qquad b_c \longrightarrow -\frac{2}{45}.$$

The repair error tends to zero, so the ratio tends to $3/n$. Both limiting overlaps are nonzero for each fixed n, and $3/n < 1$, which proves an admissible strict cooling witness for sufficiently small positive ε. The leading KL ratio is the square of the amplitude ratio, giving the stated KL advantage. Letting n grow after taking these fixed-dimensional limits proves the absence of a dimension-independent bound below one. □

The checker `research/high_dimension_amplitude_certificate.py` records the exact cloning identities and six Bernstein sign certificates. It also constructs the rational five-state member with $\varepsilon = 1/1000$ and clone weights $(2/3, 1/3)$, verifies every strict triplet test, positivity, conservation and detailed balance, and certifies a perturbed slow gap of at least $124037/375000 > 1/4$. This illustrative finite member verifies the generator construction; it is not presented as a separately interval-certified thermal witness. The all-dimensional thermal advantage is established by the limit proof.

There is also a separately certified finite five-state thermal witness. For $m = 2$, every outgoing rate sum of H is at most 100, so its symmetric operator norm is at most 200. Write $\eta = 1200\varepsilon^2$ for the perturbation bound. For $\varepsilon \le 1/200$, the gap stays above $1/3 - 2\eta \ge 41/150 > 1/4$, and the oriented eigenvector estimate gives

$$\|\widetilde{w} - \overline{w}\|_{q^{(5)}} \le \frac{\sqrt{2}\,\eta}{1/3 - \eta} < 6000\varepsilon^2.$$

The normalization is the same weighted norm as $\overline{w}$, which is at most one. Since the smallest normalized bath weight is at least $\varepsilon^2/9$, every component and thermal overlap changes by less than 18000ε. At $\varepsilon = 10^{-50}$, $\beta_h = 1/10000$ and $\beta_c = 1/50$, integer-power root bounds, combined with that error, enclose the repaired amplitude advantage between 37.6778168708% and 37.6778168731%. The exact rational endpoints and reproducible computation are in `research/five_state_large_advantage_certificate.py` and its JSON receipt. This witness exceeds the four-state supremum. Its very large energy span remains a mathematical construction, not evidence of a practical cooling protocol.

18 A precise reduction of the remaining dimensional bound

The construction above and the contrast bound leave a specific structural question. The next proposition isolates a sufficient additional inequality; it does not assert that all the original F tests force that inequality in arbitrary dimension.

Proposition 1 (Conditional three-state reduction)**.** *Let $n \ge 4$, let the energies be distinct and ordered, and let a positive reversible generator at $0 < \beta_b < \infty$ have a simple slow mode. Suppose every original $G_{ijk} \le 0$. Orient the slow left eigenfunction w by its nonnegative weighted-prefix contrast coordinates, as in Theorem 4. If additionally*

$$w_1 + w_2 \ge 2w_3, \tag{13}$$

then every finite cooling pair $0 < \beta_h < \beta_c < \beta_b$ satisfies

$$\frac{|b_2(\beta_c)|}{|b_2(\beta_h)|} < \frac{n}{3}.$$

In particular its amplitude advantage is strictly below $1 - 3/n$.

Proof. Write $q_i = \pi_i(\beta_b)$, $Q_k = \sum_{i \le k} q_i$, and

$$h_i^{(k)} = \begin{cases} 1, & i \le k, \\ -Q_k/q_{k+1}, & i = k+1, \\ 0, & i > k+1. \end{cases}$$

The contrast theorem supplies $w = \sum_{k=1}^{n-1} x_k h^{(k)}$ with $x_k \ge 0$, not all zero. Put $f = x_1 h^{(1)} + x_2 h^{(2)}$; it is supported on the first three states. Every contrast with $k \ge 3$ is constant on these states, so

$$f_1 - f_2 = w_1 - w_2 = \frac{Q_2}{q_2} x_1 \ge 0, \qquad f_1 + f_2 - 2f_3 = w_1 + w_2 - 2w_3 \ge 0.$$

Thus the first three entries of f are a constant plus $u(1,-1,0) + v(1,1,-2)$ with $u, v \ge 0$.

Let $m_f(\beta)$ be the conditional Gibbs mean of f on states $1, 2, 3$. This mean is increasing when $f \ne 0$. Indeed, shifting $E_1 = 0$ and writing $a = E_2$, $b = E_3$, $t = e^{-a\beta}$, $y = e^{-b\beta}$, the conditional probabilities are $(p_1, p_2, p_3) = (1, t, y)/(1 + t + y)$ and

$$(p_1 - p_2)' = \frac{2at + by - (b-a)ty}{(1+t+y)^2} > 0, \qquad (p_1 + p_2 - 2p_3)' = -3p_3' > 0.$$

Here $0 < a < b$ and $0 < t \le 1$ for $\beta \ge 0$. If $f \ne 0$, its zero bath mean prevents its first three entries from being constant, so at least one of u, v is positive. Consequently $m_f(\beta_b) = 0$ and $m_f(\beta) < 0$ for $0 \le \beta < \beta_b$.

For $j \le n$, write $P_j(\beta) = Z_j(\beta)/Z_n(\beta)$ for the Gibbs mass on the first j states. If $j < n$, their lower energies give $j/n < P_j(\beta) < 1$ for finite $\beta > 0$. The overlap of f is $P_3 m_f$, and hence, when $f \ne 0$,

$$0 < \frac{|\langle f \rangle_{\beta_c}|}{|\langle f \rangle_{\beta_h}|} = \frac{P_3(\beta_c)}{P_3(\beta_h)} \frac{|m_f(\beta_c)|}{|m_f(\beta_h)|} < \frac{n}{3}.$$

For each $k \ge 3$, the conditional mean of $h^{(k)}$ on its first $k+1$ states equals

$$m_k(\beta) = 1 - \left(1 + \frac{Q_k}{q_{k+1}}\right) \frac{e^{-\beta E_{k+1}}}{Z_{k+1}(\beta)}.$$

It is strictly increasing, since E_{k+1} is the largest energy in that conditional system, and it vanishes at β_b. Its cooling values are therefore negative. The same factorization gives

$$\frac{|\langle h^{(k)} \rangle_{\beta_c}|}{|\langle h^{(k)} \rangle_{\beta_h}|} < \frac{n}{k+1} \le \frac{n}{4}.$$

For $k + 1 = n$, the mass factor is identically one and strictness instead comes directly from the strict increase of the negative conditional mean; this handles the full-support contrast without asserting $P_n < 1$.

The nonzero terms in $b_2(\beta) = \langle f \rangle_\beta + \sum_{k=3}^{n-1} x_k \langle h^{(k)} \rangle_\beta$ are all negative on cooling. Its ratio of absolute overlaps is thus a weighted average of the corresponding component ratios, with positive weights proportional to their hot absolute overlaps. Each ratio is strictly below $n/3$, proving the assertion. □

To state the remaining gap precisely, let S_n be the supremum of

$$1 - \frac{|b_2(\beta_h)|}{|b_2(\beta_c)|}$$

over positive reversible n-state generators with distinct ordered energies, a positive finite bath, a simple slow mode, all original $F_{ijk}, G_{ijk} \le 0$, and finite cooling pairs. The construction in Theorem 18 and the general contrast bound give

$$1 - \frac{3}{n} \;\le\; S_n \;\le\; 1 - \frac{2}{n}.$$

Every individual finite pair has advantage strictly below the displayed upper bound; that strictness does not replace the non-strict inequality for the supremum. For $n = 4$, Theorem 14 settles $S_4 = 1/4$. For $n > 4$, proving that the full original triplet premise forces (13) would, by the proposition and the existing construction, settle $S_n = 1 - 3/n$. This all-dimensional arithmetic implication is not proved here. A counterexample to that intermediate implication alone would not establish that the amplitude supremum exceeds $1 - 3/n$; a thermal-overlap violation would still be needed.

Proposition 2 (A constant rare-tail subfamily)**.** *Let a positive reversible n-state generator, $n \ge 4$, have distinct ordered energies, a positive finite bath, and all original $F, G \le 0$. Suppose its oriented simple slow eigenfunction is constant on states $4, \dots, n$, and $Q_T := \sum_{k=4}^{n} q_k < q_3$. Then $w_1 + w_2 \ge 2w_3$ and the amplitude bound of Proposition 1 holds. The full dynamics need not be lumpable.*

Proof. Orthogonally compress the reversible backward generator onto functions constant on the tail, using its q-weighted inner product. The compressed four-state generator has weights (q_1, q_2, q_3, Q_T), unchanged low-low rates, and

$$a_{iT} = \sum_{k \ge 4} \frac{q_k}{Q_T} a_{ik}, \qquad a_{Ti} = \sum_{k \ge 4} a_{ki} = \frac{Q_T}{q_i} a_{iT}.$$

Its rates are positive, its diagonal conserves probability, and it is reversible. The original slow eigenfunction belongs to this subspace, so it remains an eigenfunction. The Rayleigh variational principle bounds every compressed nonstationary eigenvalue above by the original slow eigenvalue. Equality is simple because the original slow eigenspace is one-dimensional. Hence this is also the compressed simple slow mode.

For $i < j \le 3$, direct substitution gives

$$G_{ijT} = \sum_{k \ge 4} \frac{q_k}{Q_T} G_{ijk}, \qquad F_{ijT} = \sum_{k \ge 4} \frac{q_k}{Q_T} F_{ijk} + \sum_{k \ge 4} \frac{q_k (Q_T - q_k)}{2 Q_T q_j} G_{ijk}.$$

All coefficients are nonnegative, and the test on 123 is unchanged. The compressed weights are strictly ordered because $Q_T < q_3$. The four-state arithmetic lemma therefore applies. Its orientation agrees with the original one: the first two contrast coordinates are unchanged, and the third is nonnegative because the common tail value is $w_n \le 0$. The preceding conditional bound completes the proof. □

This compression is different from eliminating states by a Schur complement. Its test preservation is the displayed identity, not an assumption about effective rates. The restrictions of a constant tail mode and $Q_T < q_3$ are substantive; aggregation can reorder energies when the latter fails.

18.1 An exact obstruction to a higher-dimensional trace proof

The successful four-state arithmetic proof does not extend merely by replacing its one tail coordinate with a zero-energy Schur complement. The following exact five-state example rules out that intermediate closure claim; it does not disprove the slow-mode arithmetic conjecture.

Take normalized bath weights $q = (100, 4, 3, 2, 1)/110$. In the pair order

$$12, 13, 14, 15, 23, 24, 25, 34, 35, 45,$$

start with reduced conductances

$$k^{(0)} = (1, 999/1000, 999/1000, 0, 1, 999/1000, 0, 2, 0, 2).$$

For $i < j$ set

$$\eta = 10^{-6}, \qquad k^*_{ij} = k^{(0)}_{ij} + \eta/q_i - \eta/q_1, \qquad k^*_{ji} = k^*_{ij}.$$

The initial F tests are negative and the G tests are nonpositive. Adding the term η/q_i amounts to adding the strictly interior constant-downward-rate generator. Subtracting η/q_1 is a reset translation, which leaves every F, G unchanged. Thus k^* has all triplet tests strict, is nonnegative, and has minimum zero on edge15. Now use the strictly positive physical rates

$$a_{ij} = q_i(1 + k^*_{ij}) \quad (i \neq j)$$

with a conserving diagonal. This is reversible, has every original triplet predicate false with strict $F, G < 0$, and its minimum reduced conductance is exactly one.

Subtract this minimum reset and trace out states5 and4. The resulting three-state generator is the genuine conserving Schur complement; in reduced-conductance coordinates, elimination of a highest state l updates the remaining off-diagonal entries by

$$k_{ij} \longmapsto k_{ij} + \frac{q_l k_{il} k_{jl}}{\sum_{a<l} q_a k_{al}}.$$

Both denominators are positive in this example. Exact rational arithmetic gives, for the resulting effective rates,

$$F^{\mathrm{trace}}_{123} = \frac{452114029270345354793}{30223906046743763750000} > 0.$$

The complete exact checks are in `research/reset_trace_obstruction.py` and its JSON certificate.

This disproves the implication that full original triplet failure, after subtracting the minimum reset, forces $F_{123} \leq 0$ on the three-state trace. The original-rate inequalities and the effective-rate test above concern different generators. In particular this calculation provides no anomalous thermal pair and makes no claim that the original five-state slow vector violates $w_1 + w_2 \geq 2w_3$. The higher-dimensional proof, if that inequality is true, must retain more of the slow eigenvalue dependence or use a different argument.

18.2 An exact linear-feasibility reduction for a proposed slow mode

The remaining arithmetic conjecture can be attacked without repeatedly computing eigenvectors or imposing a semidefinite constraint. This is an exact reduction of the unresolved question, not a proof of the inequality.

Proposition 3 (Graph-coupling feasibility for the arithmetic conjecture)**.** *Fix an integer $n \geq 4$, positive ordered bath weights $q_1 > \cdots > q_n$, normalized to sum one, and a strictly positive vector*

$x \in \mathbb{R}^{n-1}$. *Let H contain the unnormalized weighted Helmert columns and let $\eta = H^\top DH = \operatorname{diag}(\eta_1, \dots, \eta_{n-1})$. For each pair of contrast coordinates introduce $\ell_{ij} = \ell_{ji}$, and define*

$$B_{ij} = \frac{\ell_{ij}}{\eta_i} \quad (i \neq j), \qquad B_{ii} = -\sum_{j \neq i} \frac{\ell_{ij} x_j}{\eta_i x_i}, \qquad M^{(0)} = \left(HB\eta^{-1}H^\top D\right)^\top.$$

For fixed q, x, every original F, G evaluated on $M^{(0)}$ is linear in ℓ. Consider the finite linear program

$$\text{maximize } t \quad \text{subject to } \sum_{i<j} \ell_{ij} = 1, \qquad \ell_{ij} \geq t,$$
$$F_{abc}(M^{(0)}) \leq -t, \quad G_{abc}(M^{(0)}) \leq -t \qquad (a < b < c).$$

Its optimum is positive if and only if Hx is a simple slow left mode of a strictly positive reversible physical generator having every F, G strict, with that generator allowed an arbitrary reset and time scale.

Consequently a counterexample to $w_1 + w_2 \geq 2w_3$ for a simple slow mode under the full weak triplet premise exists in a fixed dimension if and only if some positive ordered q and positive x, with

$$x_2 < \frac{(q_1 - q_2)q_3}{2q_2(q_1 + q_2 + q_3)} x_1,$$

give positive optimum. If such a counterexample exists, q, x, ℓ may all be chosen rational. Equivalently, for each fixed dimension its existence is an explicit existential formula over the real numbers using polynomial inequalities after positive denominators are cleared.

Proof. By construction $Bx = 0$ and ηB is symmetric. More explicitly, for every real vector y,

$$-y^\top \eta B y = \sum_{i<j} \ell_{ij} x_i x_j \left(\frac{y_i}{x_i} - \frac{y_j}{x_j}\right)^2.$$

Thus B is negative semidefinite in its weighted inner product, with kernel exactly $\operatorname{span}(x)$ when the graph is connected. With $X = \operatorname{diag}(x)$, the matrix $X^{-1}BX$ is a Markov backward generator with off-diagonal entries $\ell_{ij}x_j/(\eta_i x_i)$; it is reversible with weights $\eta_i x_i^2$. If every $\ell_{ij} > 0$, its zero eigenvalue is simple and all others are strictly negative. Since $H^\top D\mathbf{1} = 0$, $M^{(0)}$ conserves probability, is reversible at q, and induces exactly B on the contrast coordinates. It need not have nonnegative rates. For sufficiently large $\rho > 0$, however,

$$M = M^{(0)} + \rho(q\mathbf{1}^\top - I)$$

has every rate strictly positive. Reset addition annihilates all the linear F, G tests. Its nonstationary eigenvalues are those of B minus ρ, so Hx is a simple slow left eigenvector with rate $-\rho$. This proves sufficiency of a positive LP margin.

Conversely take a physical generator with strict tests and slow coordinates $x > 0$. The ordered-conductance theorem makes its contrast matrix Metzler; strict G makes every off-diagonal contrast entry positive. Subtract its slow eigenvalue times the identity in contrast space. The resulting matrix annihilates x, is symmetrizable by η, and therefore has exactly the displayed graph-coupling form with every $\ell_{ij} > 0$. This shift corresponds to a reset translation, so does not change F, G. Rescale time to make $\sum \ell_{ij} = 1$. There are finitely many strict inequalities, hence a common positive margin exists and the LP optimum is positive.

For the global counterexample equivalence, start from a simple slow mode with a strict violation of the arithmetic inequality, allowing weak triplet failures. Add a sufficiently small generator whose every downward rate is one, with detailed-balance reverse rates. Its tests are $G = t - 1 < 0$ and $F =$

$(t-1)(1+u/2) < 0$. The perturbed generator therefore has every test strict. Spectral simplicity and the strict arithmetic violation persist by continuity. The contrast matrix now has strictly positive off-diagonal entries, so its Perron coordinates are positive. The preceding construction applies. Conversely a positive LP margin and the displayed strict coordinate inequality give a physical counterexample by reset addition.

All strict conditions are continuous in the positive variables q, x, ℓ. Rational points are dense in their domains (including $\sum q_i = 1$ and $\sum \ell_{ij} = 1$), so a successful strict example can be replaced by rational data while keeping every strict inequality. The new matrix is then constructed from those rational q, x, ℓ; its slow mode is exactly Hx. This does not assume that the original generator already had a rational eigenvector. The Helmert formulas and all tests are rational functions with positive denominators on this domain. Clearing those denominators gives the claimed finite existential polynomial system. This establishes an exact decision reduction, without claiming that general quantifier elimination is computationally practical. □

For a fixed prescribed mode on the weak boundary, connectivity rather than positivity of every ℓ is the exact simplicity condition. It can still be encoded linearly: require $\ell \geq 0$, weak $F, G \leq 0$, and a positive common lower bound on the sum of ℓ crossing each nontrivial cut of the contrast-coordinate graph. Thus the strict LP above is not incorrectly being used to reject all weakly realizable modes at a fixed x. Its strict formulation is sufficient and necessary for the global strict-violation search by the perturbation argument.

19 Approaches explored

The initial route comparison, including explicit continue/change criteria, is recorded in `research/approaches.md`. The decisive work covered both theorem construction and adversarial falsification.

1. **Reversible spectral geometry.** Symmetrization and the Dirichlet form gave reliable eigenvalue ordering and an equilibrium-orthogonal search space. The obstacle was that rate inequalities do not make the slow-mode orientation convex. Spectral certificates remain essential; this route did not yield the full absence theorem.

2. **Order and covariance.** The first calculation was $K = BLH_0$. Its Metzler subfamily yielded the theorem above. An exact auxiliary example in `research/theorem_notes.md` disproves the intermediate claim that all triplet failures force a monotone slow eigenvector. This identified the missing implication rather than assuming it.

3. **Exponential-polynomial zero counting.** Multiplying by $Z(\beta)^2$ gives

$$Z^2 b_2' = -\sum_{i<j} (w_j - w_i)(E_j - E_i) e^{-\beta(E_i+E_j)}.$$

   A direct pairwise expansion has at most six terms in four states, but the Gibbs Wronskian representation is sharper: it gives at most $n-2$ stationary points and makes the four-state derivative curve strictly convex. This is the decisive route to the finite classification in Section 9. The Chebyshev-system machinery is classical; the explicit identity and its application are proved here.

4. **Feasible-rate geometry.** The identity $F = hG + dH$ converted apparent branch enumeration into eight linear inequalities. A search enumerated extremal rays after removing the reset direction, then sampled positive mixtures. It found the first root-search candidate after 67,000 systems. Extremal rays were used for discovery only: we did not assume eigenvector properties are inherited by mixtures.

5. **Backward eigenvector design.** Starting from the discovered thermal mode, we prescribed rational q and rational w with $q^\top w = 0$, then solved the linear eigenvector equations for rates. This produced the rational-mode companion and an exact eigenvalue. The remaining spectrum was checked independently; realization of an arbitrary faster eigenmode would not suffice.

6. **Lumping and perturbation.** The near-neutral highest state explains the observed mode. No adiabatically eliminated rate was substituted into the original predicate. Strict-margin continuity supplies a rigorous open family; a separate singular-perturbation theorem was unnecessary once a finite positive example was certified.

7. **Difficult regimes and targeted sampling.** Broad positive-rate sampling missed the effect. Signed-cone directions plus reset restoration found an independent witness. This concentrated the search on unequal energy gaps and narrow high-temperature windows, where a coarse temperature grid is weak.

8. **Algebraic proof.** Rational bath weights made the selected Gibbs probabilities algebraic even though the energy gaps are not commensurate. Exact root brackets and rational interval arithmetic proved the two-temperature inequality. Commensurate energies were therefore not needed.

9. **Additional idea: reset lineality.** The reset generator lies in the lineality space of all the triplet inequalities and acts as a scalar on nonstationary modes. This allowed signed directions to be made physical without changing their mode geometry, and gives the explicit counterexample family above. The September paper also studies the fully degenerate reset point. Section 4.6 traces the general algebra to established equal-input generators; only its problem-specific use is needed here.

10. **Additional idea: thermal dual cones.** The cone spanned by Gibbs tangent vectors would have a dual consisting of modes with controlled derivative sign. The first calculation was the covariance expansion. The counterexample shows that the original rate cone's slow modes cross the required thermal dual. A complete dual-cone characterization remains an extension, not a premise used in the proof.

20 Verification and reproducibility

20.1 Exact certificate construction

Run from the workspace root:

```
python3 research/adversarial_certificate.py
python3 research/adversarial_rational_mode_certificate.py
```

The primary script uses SymPy rational/integer arithmetic to check positivity, column sums, detailed balance and the original Boolean predicates. It derives the characteristic polynomial and cofactor vector and verifies its eigenvector residual and bath expectation modulo $C(x)$. The actual slow root is enclosed between

$$-\frac{42820190741029}{2155193997917} \quad \text{and} \quad -\frac{135549509489200}{6822377112669}.$$

Opposite exact polynomial signs certify this interval. Two disjoint negative intervals similarly contain the other roots; degree three proves completeness. The interval evaluation of P_1 excludes zero, so division by it is justified.

For each $q_i^{1/n}$, $n = 1000, 100$, the script proposes adjacent grid points L_i, U_i of mesh 10^{-24}, then checks exactly

$$0 \le L_i, \qquad L_i^n \le q_i \le U_i^n.$$

Strict monotonicity of the positive nth-power map proves the enclosure. Approximate arithmetic only proposes the endpoints; it never accepts an inequality. Polynomial Horner evaluation, products and division use exact interval endpoints. All denominators are verified to exclude zero. The final interval comparison is also exact. Full rational endpoints, rather than merely rounded decimals, are saved in the JSON certificates.

The tighter certified margin in the primary normalization is greater than

$$0.000103407628000118473315 27.$$

The deliberately widened report bounds are conservative. Scaling w consistently changes the displayed numbers but never the conclusion; search logs may use a unit weighted norm, whereas the final report fixes $w_1 = 1$.

20.2 A separate proof using inertia and residual bounds

The primary example also admits a verification that does not use its cubic coefficients, cofactor polynomials, previously computed root brackets or certificate JSON files. The standalone script `research/standalone_primary_check.py` starts only from the six printed downward rates and four bath weights. It uses Python's standard-library integers and fractions, with explicit rejection checks that remain active under optimization. It rebuilds the generator and evaluates the original two-branch predicates directly, without invoking Lemma 1. The following argument explains its complete spectral and thermal certificate.

Exact identification of the slow eigenspace. Put $L = M^\top$, $D_q = \mathrm{diag}(q)$ and $H = D_q L$. Detailed balance makes H symmetric. For rational ξ, exact symmetric elimination of $H - \xi D_q$ into a unit triangular factor, a diagonal factor and the transpose triangular factor gives the following pivot signs, eliminating coordinates in the order $1, 2, 3, 4$:

ξ	Pivot signs	Number of eigenvalues of L above ξ
-337	$+,+,+,+$	4
-336	$+,+,-,+$	3
-60	$+,-,+,+$	3
-59	$+,-,+,-$	2
-20	$+,-,+,-$	2
-19	$+,-,-,-$	1

All pivots are nonzero; the script records their exact rational values. Each elimination step replaces the trailing block by $A_{ij} - A_{ik}A_{kj}/A_{kk}$ and is a congruence. Congruence preserves the counts of positive and negative eigenvalues. Also $H - \xi D_q$ is congruent to $D_q^{1/2} L D_q^{-1/2} - \xi I$. Thus the counts drop by one across each of $(-337, -336)$, $(-60, -59)$ and $(-20, -19)$. These account for three simple negative eigenvalues; $L\mathbf{1} = 0$ accounts for the fourth. In particular the slow nonzero eigenspace is one-dimensional and lies in $(-20, -19)$. No approximate eigensolver is used in this identification.

Residual control replaces an exact eigenvector formula. Define the following exact rational trial values (each decimal terminates):

$$\widehat{r} = -19.86836952145135, \qquad \widehat{w}_1 = 1, \quad \widehat{w}_2 = -0.75893729814493, \quad \widehat{w}_3 = -0.61211930644096,$$

$$\widehat{w}_4 = -\frac{q_1\widehat{w}_1 + q_2\widehat{w}_2 + q_3\widehat{w}_3}{q_4} = -\frac{193315388213}{50000000000000}.$$

Use the weighted norm $\|f\|_q^2 = \sum_i q_i f_i^2$. Direct rational multiplication verifies

$$q^\top \widehat{w} = 0, \qquad \|L\widehat{w} - \widehat{r}\widehat{w}\|_q^2 < 10^{-16}.$$

These trial values are not accepted as eigenvalues or eigenvectors: only their explicitly checked residual bound is used.

Let z be the orthogonal projection of $\widehat{w}$ onto the true slow eigenspace in this weighted inner product. The operator L is self-adjoint there. Its stationary component in $\widehat{w}$ vanishes because $q^\top \widehat{w} = 0$. The two faster eigenvalues are below -59, whereas $\widehat{r} > -20$, so their distances from $\widehat{r}$ exceed 39. Expanding in a weighted orthonormal eigenbasis gives

$$\|L\widehat{w} - \widehat{r}\widehat{w}\|_q^2 \geq 39^2\|\widehat{w} - z\|_q^2, \qquad \|\widehat{w} - z\|_q < \frac{10^{-8}}{39}.$$

Since $\|\widehat{w}\|_q \geq 1$, the projection is nonzero. The minimum bath weight is $1/400$, hence each coordinate error is less than $20\,10^{-8}/39$. For every probability vector p this implies

$$|z^\top p - \widehat{w}^\top p| < \delta, \qquad \delta = \frac{1}{195000000}. \tag{14}$$

Independent thermal enclosures. For $n = 1000$ and 100, integer binary search finds m_i satisfying

$$\left(\frac{m_i}{10^{12}}\right)^n \leq q_i < \left(\frac{m_i + 1}{10^{12}}\right)^n.$$

These comparisons use integer powers and positive denominators. Signed sums of the endpoint bounds enclose $\sum_i \widehat{w}_i q_i^{1/n}$; positive sums enclose its partition function. Taking all numerator/denominator endpoint quotients and adding the error in Eq. (14) proves the deliberately widened rational intervals

$$z^\top \pi(1/1000) \in (-0.093745434, -0.093745422),$$
$$z^\top \pi(1/100) \in (-0.093848841, -0.093848830).$$

Both overlaps are negative, and their absolute-value difference exceeds $0.000103396 > 10^{-4}$. This proves the cooling reversal for the actual simple slow mode, rather than for the trial vector. The paired right mode is $D_q z/(z^\top D_q z)$, giving unit left-right overlap. Here the left mode is z, whose first coordinate is in $(1-\delta, 1+\delta)$, rather than exactly one. Dividing by this positive coordinate recovers the main report's normalization; the checker verifies that the normalized advantage still exceeds 10^{-4}.

Independence and its limits. The original certificate uses symbolic characteristic polynomials and cofactor intervals; this checker uses rational congruence and a spectral residual bound. Its standalone implementation requires no research data files or third-party packages. Ten adversarial tests include an isolated execution with only the checker present, identical results under normal and optimized Python, rejection of a transposed generator, a false triplet premise, a wrong mode, a poor eigenvector and reversed temperatures, plus analytically known arithmetic examples. These tests exercise failure paths; they do not themselves prove the checker correct.

This is a separate implementation and a different proof within the same model-assisted project, not independent human authorship, formal proof-assistant verification or external peer review. It still trusts Python's integer/fraction arithmetic and the mathematical argument above. It independently verifies the primary asymptotic counterexample; the finite-time crossing brackets remain supported by their separate uniformization certificate. All methods still require review of the correspondence to the published claim in Section 2.3. A reviewer can run the standalone checker in an empty directory and audit the argument above without trusting the discovery search or any stored numerical output.

20.3 Lean verification: coverage and reproduction

The companion library in `formal/` verifies three results in Lean: the rational-mode counterexample in Appendix B, its initial-to-eventual relaxation reversal in a specified quadratic distance, and the sharp four-state 25% amplitude supremum. These are kernel-checked proofs over exact rational and real numbers. The library does not formalize the entire report. In particular, its explicit witness is the Appendix B generator, whose two modified rates distinguish it from the primary generator in Theorem 1 and the plotted finite-time crossings.

Counterexample and actual evolution. The theorem `Mpemba.appendixB_counterexample` collects positivity of all twelve off-diagonal rates, column conservation, detailed balance, strictly ordered energies, failure of every original triplet predicate, the rational left eigenpair, algebraic simplicity and slow-mode ordering, and the strict thermal-overlap comparison at $\beta_h = 1/1000$ and $\beta_c = 1/100$. States $1, \dots, 4$ are indexed by $0, \dots, 3$ in Lean; `M i j` retains the destination-first convention. The predicate includes both inclusive κ branches and both strict sign tests. The equivalence between its failure and $F \le 0$, $G \le 0$ is also proved.

The weights q_i^{β} are proved equal to $e^{-\beta E_i}$ with $E_i = -\log q_i$. Rational integer-power comparisons establish enclosures for the real Gibbs probabilities; no externally computed interval is accepted as a premise. The proved overlap intervals are

$$-0.09376 < b_2(1/1000) < -0.09375, \qquad -0.09386 < b_2(1/100) < -0.09385.$$

They suffice for strict reversal, but are wider than the Appendix B intervals supplied by the separate Python certificate.

The formal distance is

$$D_q(p) = \sum_{i=1}^{4} \frac{(p_i - q_i/Q)^2}{q_i}, \qquad Q = \sum_i q_i. \tag{15}$$

Thus $QD_q(p)$ is Pearson chi-square divergence from $\pi_b = q/Q$; the square root of D_q gives the same ordering in the associated weighted Euclidean norm. For the Appendix B generator, the library constructs the full spectral evolution, proves its initial condition and master equation, and proves uniqueness among global differentiable solutions. The theorem `Mpemba.overlap_implies_eventual_distance_order` proves that strict absolute slow-overlap ordering for normalized initial vectors implies strict D_q ordering at all sufficiently late times. This uses a proved spectral limit, rather than assuming the asymptotic expansion. Separately, the initial Gibbs distances obey

$$D_q(\pi(1/100)) < 24 < D_q(\pi(1/1000)).$$

The theorem `Mpemba.master_equation_distance_reversal` transfers the initial-to-eventual reversal to any global solutions with these initial conditions; `Mpemba.reversal_at_a_finite_positive_time` gives existence at a finite positive time. No numerical crossing time or KL-distance reversal is certified by this Lean bridge. The KL argument in Section 2.4 and the primary generator's certified KL and TV crossing brackets retain their separate analytic and computational support.

Sharp amplitude supremum. The general formal model has bath weights $(1, t, u, v)$ with $0 < v < u < t < 1$, positive downward rates, upward rates fixed by detailed balance, and conservative diagonals. The bath inverse temperature is one. Its `SimpleSlowMode` hypothesis specifies a nonzero bath-orthogonal left eigenfunction, its eigen-equation, a one-dimensional eigenspace, and the largest Rayleigh quotient on the bath-orthogonal subspace. These are explicit spectral hypotheses; the required mode shape and nonzero cooling overlaps are derived from them and the four original triplet failures.

For $0 < \beta_h \le \beta_c < 1$, the theorem `Mpemba.Quarter.full_generator_amplitude_advantage_below_quarter` proves

$$A = 1 - \frac{|b_2(\beta_h)|}{|b_2(\beta_c)|} < \frac{1}{4}.$$

The definition `Mpemba.Quarter.attainableAdvantages` restricts to strictly ordered finite cooling preparations with strict amplitude reversal and all the generator and spectral conditions just stated. Theorems `Mpemba.Quarter.sharp_quarter_isLUB` and `Mpemba.Quarter.sharp_quarter_not_attained` prove that $1/4$ is this set's least upper bound and is not attained. The theorem `Mpemba.Quarter.exists_advantage_arbitrarily_close` explicitly gives an admissible A with $1/4 - \delta < A < 1/4$ for every $\delta > 0$.

The formal sharpness construction is a simpler rational family than the uniformly gapped family used elsewhere in this report. It retains

$$q = (1, 1 - \varepsilon/3, 3/5, \varepsilon^2), \quad w = (1, -4/5, -1/3 - 4\varepsilon/9 + 5\varepsilon^3/3, -\varepsilon), \quad \lambda_2 = -3,$$

fixes $a_{12} = 1$, and defines the remaining five downward rates as rational functions in `formal/Mpemba/SharpFamily.lean`. Lean proves positivity, every original triplet failure and slow-mode simplicity for all sufficiently small positive ε. Here $F_{124} = G_{234} = 0$, which is allowed by the original predicate's strict tests. With $\varepsilon = e^{-L^2}$, $\beta_h = L^{-3}$ and $\beta_c = L^{-1}$, all sufficiently large finite L give admissible preparations, and Lean proves

$$b_2(\beta_h) \longrightarrow -1/30, \qquad b_2(\beta_c) \longrightarrow -2/45, \qquad A \longrightarrow 1/4.$$

This verifies sharpness for actual admissible generators. The stronger uniform spectral-gap estimate and the perturbation making all eight triplet inequalities strict are not included in this formalization. The 25% quantity is a slow-amplitude advantage; no percentage reduction in cooling time follows from this formal statement.

Reproduction and trust boundary. All 27 proof modules are supplied in `formal/`, with a README, a library entry point, an axiom audit and the verification script. The `lean-toolchain` file pins Lean to version 4.19.0. The `lakefile.toml` configuration pins mathlib to commit

`c44e0c8ee63ca166450922a373c7409c5d26b00b`

Keep these files and `lake-manifest.json` with the sources in the `formal` directory: the manifest fixes the transitive dependencies. On another machine, install elan (the Lean toolchain manager), Git, Python 3 and a POSIX shell, then run from the package root:

```
cd formal
lake exe cache get
./check.sh
```

Elan selects the pinned Lean version; Lake obtains the dependencies from the committed manifest. The cache command downloads precompiled mathlib dependencies. Initial setup needs network access. Do not run `lake update` to reproduce the committed dependency set. If the compiled cache downloader fails to link, its interpreter entry point can be used after Lake has fetched the package sources:

```
lake env lean --run .lake/packages/mathlib/Cache/Main.lean get
./check.sh
```

The README also gives a smaller cache download restricted to the imported mathlib modules. In this workspace, `check.sh` automatically uses the local elan installation when present; otherwise it uses `lake` on the executable search path.

The verification run on 12 October 2026 built the complete library and passed 43 named-theorem axiom audits. Successful execution ends with

```
PASS: 43 theorem axiom audits; standard logical axioms only.
```

The script checks the number and names of the reports, requires the main results to be included, and rejects dependencies beyond `propext`, `Classical.choice`, and `Quot.sound`. The proof sources have no `sorry`, `admit`, custom axioms or `native_decide` calls. Python parses the axiom reports; it does not supply mathematical premises to Lean. The usual trust in Lean's kernel, the toolchain and the computing environment remains.

The formal coverage is limited to the results specified above. It does not extend to this report's general KL bridge, classifiers, higher-dimensional bounds, degenerate-mode classifications, or time-advance corollaries. Formal verification also does not establish publication priority, external peer review, or fidelity of the conjecture's transcription. Reviewers must still compare the formal definitions and hypotheses with the manuscript and the original source claim.

20.4 Figure and finite-time verification

Run `research/physics_plots.py` for the figures and data, and `research/finite_time_certificate.py` for exact finite-time bounds. Plot values use 80-digit mpmath arithmetic. Direct matrix exponentiation and reversible symmetric spectral propagation agree within $6\,10^{-81}$ at ten checks spanning $t = 0$ to 0.4. The plot CSV files and numerical metrics are saved under `output/figures/`. Those numerical comparisons are distinct from the exact certificates.

The finite-time certificate uses uniformization with $\nu = 253$ and the exactly stochastic matrix $P = I + M/\nu$:

$$e^{tM}p(0) = e^{-\nu t}\sum_{k=0}^{\infty}\frac{(\nu t)^k}{k!}P^k p(0).$$

It retains terms through $k = 250$. Each omitted probability component is at most one; the scalar positive-series tail is bounded by its first omitted term divided by $1 - \nu t/252$. The same positive Taylor bounds give a rational enclosure of $e^{-\nu t}$. Initial Gibbs probabilities use the already-certified algebraic brackets. For logarithms it uses

$$\log r = 2\sum_{k=0}^{59}\frac{z^{2k+1}}{2k+1} + R, \quad z = \frac{r-1}{r+1}, \quad |R| \le \frac{2|z|^{121}}{121(1-z^2)}.$$

All interval operations round outward onto a rational grid of mesh 10^{-50}. Exact rational endpoint signs prove the displayed KL and TV crossing brackets. The script uses floating point only to print readable values. The root estimates shown on the figures are numerical; their enclosing crossing intervals are exact.

20.5 Executed discovery calculations

The discovery environment used Python with NumPy 1.26.4, SciPy 1.13.1 and SymPy 1.14.0. Seeds and actual parameters are retained in the scripts and output files. Floating-point discovery outputs are not mathematical certificates.

`research/cone_search.py` used seed 10411, energy gaps $\exp U[-4,2]$, 1,000 ray mixtures per energy set, and 150 cooling inverse temperatures from 0.001 to 0.999. It stopped after 67 energy sets, 67,000 systems, on a strict candidate. Coefficients of the ray mixtures were $\exp U[-9,3]$; a reset direction restored positivity. SVD acceptance tolerances for rays were 10^{-8} and 10^{-9}, and deduplication tolerance 10^{-6}. These are discovery tolerances only.

The independent positive-rate search used seed 934256 and 20,000,000 draws. Gaps were $10^{U[-3,1]}$, rates $10^{U[-5,5]}$ rescaled by their maximum; 2,020,118 draws passed the predicates. It found no accepted witness on its 18-point cooling/heating grid, under the margins and spectral filters in `research/adversarial_search.py`. The signed search used seed 742908, gaps $10^{U[-3,0.7]}$, and normal signed variables scaled by $10^{U[-3,3]}$, alternating an additional destination-weight factor. It stopped at 9,950,000 draws (573,427 feasible) with a witness. Full grids, filters and stopping records are retained in the scripts and `research/adversarial_notes.md`. Neither a failed search nor its sample count was used as proof. A separate model-assisted verification checked the primary candidate at 70 digits, then reviewed the exact certificate's matrix orientation, branch logic, root exhaustion and interval operations. The rational-mode companion was checked separately by symbolic equations and root counts.

No experiments, empirical measurements, messages to authors, or external publication were performed. All generator tests use original rates. All promised executed certificate checks passed.

20.6 Verification of the expanded results

The general contrast identity is checked symbolically for dimensions two through seven in `ordered_conductance_certificate.py`; the displayed all-dimension proof does not depend on those finite checks. A separate numerical audit compares 5,000 heating preparations across 1,000 generators in dimensions 3, 4, 5, 7 and 10, with no violation of the proved weighted-Euclidean ordering.

The classifier audit records 20,000 side classifications, 600 comparisons against exact polynomial Sturm tests, and 40 high-precision Wronskian checks, with no remaining discrepancy. The tests use independent covariance evaluations with exponential scaling to avoid underflow. They validate an implementation; the general classification rests on the analytic proof. The energy-threshold and degenerate-mode scripts give exact rational sharpness examples. Floating evaluation alone does not certify equality cases.

The exact commensurate classifier passes sixteen boundary, multiplicity and algebraic embedding checks. A cross-audit of 100 systems compares 200 side decisions with a separate geometric classifier: 70-digit symmetric diagonalization supplies modes, which are then converted to floating-point inputs for that independent implementation. All comparisons agree, with no unresolved boundary decision. This cross-audit exposed and led to repair of a scaled-algebraic-root representation bug; its first failure log is retained. The tests support the implementation and do not replace its exact Sturm argument. The pure-transient pair and five-state extension have separate exact certificates, including two crossing roots and ten strict triplet failures, respectively.

The exact inverse constructor is audited with ten checks, including affine invariance of the proposed mode, rejection of excluded strong-mode directions and invalid inputs, and independent rational LDL inertia of the resulting generators. The inverse constructor's zero-slack setting is complete for the original inequalities; positive slack strengthens them and cannot certify original infeasibility. The quarter-limit family is certified over an entire parameter interval by 30 exact Bernstein sign certificates. Its finite thermal witness uses integer-power root enclosures, with an analytic perturbation bound for the strict-triplet variant. The amplitude upper bound and its sharpness construction have direct analytic proofs. Commands, dependencies and individual scope limitations are listed in `RUNME.txt`. The one-fast-mode boundary classification has 1,694 exact symbolic checks covering its generic identities and converse pairs in dimensions four through eight; its all-dimension support proof is separate from

those finite checks. The scalar rate test is a direct spectral-projector identity, not a new general projector theorem.

20.7 Boundary audit

The disproof does not require zero rates, $\beta_b = 0$, a zero-temperature bath, equality of energies or spectral degeneracy. The source numerically discusses equal and zero rates but does not supply a general finite-temperature boundary proof. If zero rates are allowed, irreducibility must still be checked; otherwise bath equilibrium need not be unique. The printed κ quotient is undefined if $C = 0$. A closure defined through F, G requires a stated extension, not silent division by zero.

For a degenerate slow eigenspace the scalar b_2 must be replaced by the full slow spectral projection. Its norm or entropy contribution can depend on several coefficients. Reversibility prevents Jordan blocks but not this ambiguity. Continuity of a selected scalar mode cannot be assumed across an eigenvalue crossing. Our strict spectral separation avoids these issues. The spectral theorem assumes a simple slowest nonzero eigenvalue; no unstated limiting argument is used. The all-time divergence corollary separately covers degenerate spectra under its reversible monotonicity assumptions.

21 Remaining gap

The original ordinary-cooling implication is false. Its strong-effect and asymptotic-heating exclusions survive in every dimension by Theorem 4, under weaker hypotheses. Theorem 9 classifies all simple four-state modes once the full slow eigenvector is known, with a separate finite prescription for degenerate modes and specified distances.

The inverse problem has a separate exact solution at fixed rational bath weights and rational slow-mode direction: Theorem 12 reduces realizability under every triplet failure to eight half-planes intersecting an open disk. A continuously certified family approaches the sharp 25% amplitude-advantage supremum, with a uniform spectral gap and a perturbation making all triplet inequalities strict. The new upper bound follows from a restriction on the first three components of the slow eigenfunction. Within this full triplet-failure family, ordinary cooling occurs exactly when the uniform energy–mode covariance is positive; any cooling maximum is unique. The broader G-only contrast family retains its different sharp bound $1 - 2/n$.

The scalar rate test in Section 15.3 avoids computing the full slow eigenvector, but retains the largest nonzero cubic root. A comparably sharp criterion using local original-rate inequalities alone is not established. Other remaining tasks are an efficient certified implementation for general nonrational input and its transcendental equality cases, and a classification of transient reversals for distances such as relative entropy and total variation. Sections 10.1 and 11 settle the exact commensurate implementation and the specified-pair finite-time weighted-Euclidean test, respectively. Section 15.4 removes the commensurability restriction for exact rational inputs under the full triplet premise, using Baker's logarithmic independence theorem. The unrestricted contrast-cone finite-time heating exclusion is for weighted Euclidean distance. Finite-time KL ordering is additionally proved on a uniform neighborhood of the bath, on both sides, while tridiagonal contrast generators preserve heating order for every convex divergence at all finite temperatures. The explicitly classified degenerate full-premise systems separately exclude reversals for every convex divergence at every time. Neither the scalar classification nor the sixteen-term degenerate prescription classifies all transient crossings. Supporting classical tools and the rank-one realization are not claimed as new; publication priority for the problem-specific consequences requires further expert review.

22 Priority research goal: robust finite-time cooling

Research priority recorded 12 October 2026; verified rate-model outcome below. The objective is a physically motivated thermalization rate model with a substantial, robust *finite-time* Mpemba advantage under moderate parameter ranges. This takes priority over extending the asymptotic amplitude extremum. The sharp 25% bound concerns a slow-mode amplitude; it does not establish a 25% reduction in cooling time. The existing extreme-parameter sharpness families are therefore mathematical benchmarks, not solutions of this goal.

22.1 Physical model and scope

Start with thermally activated transitions between four metastable states, using the energy-landscape framework of Lu and Raz (2017). For a transition from j to i, use

$$a_{ij} = \nu_{ij} \exp[-\beta_b(B_{ij} - E_j)], \qquad B_{ij} = B_{ji} > \max(E_i, E_j), \qquad \nu_{ij} = \nu_{ji} > 0.$$

Then $a_{ij}e^{-\beta_b E_j} = a_{ji}e^{-\beta_b E_i}$, so detailed balance is automatic. Both preparations evolve with this same bath generator. Begin with a common attempt frequency and a low-dimensional, physically specified barrier geometry. Choose that geometry before optimizing the effect. Merely expressing an arbitrary reversible matrix as six independently fitted barriers does not establish a physical mechanism. A convincing candidate must explain its state interpretation, barrier constraints, metastable coarse-graining and separation from intrawell relaxation; a continuum landscape or other independent microscopic model should test the rate approximation when feasible.

The preferred extension of the present paper also retains positive rates, a simple slow mode and failure of every original-rate triplet test. Check and report those predicates separately. A physically useful example that fails this additional premise may meet the finite-time research objective, but is not a stronger counterexample to the triplet conjecture. Similarly, sparse topologies and additional state counts must be labeled as changes of scope, not silently substituted for the four-state construction.

22.2 Provisional quantitative benchmarks

These are initial research targets, not established experimental standards. Set $k_B T_b = 1$ and fix the reference attempt frequency ν_0. Use ordered energies with adjacent gaps at least 0.1 and total span at most 5; take barrier excesses $B_{ij} - \max(E_i, E_j)$ between 2 and 6. Start with $\nu_{ij} = \nu_0$, and require the ratio of the largest to smallest nonzero directed rate to be at most 10^3. Restrict preparations to $1.25 \le T_c/T_b < T_h/T_b \le 10$. These constraints jointly define the initial search box; they do not guarantee that any target is feasible. Report the actual rate range and relaxation time, not just the energy range. Any wider box, variable prefactors, or changed threshold must be recorded as a separate study.

For $x \in \{h, c\}$, let $p_x(t) = e^{Mt}\pi(T_x)$ and use relative entropy in nats as the primary distance,

$$D_x(t) = \sum_i p_{x,i}(t) \log \frac{p_{x,i}(t)}{\pi_{b,i}}, \qquad A_D(t) = 1 - \frac{D_h(t)}{D_c(t)}, \qquad \tau = \frac{1}{|\lambda_2|}.$$

Require normal initial ordering $D_h(0) > D_c(0)$. Target both:

1. **A resolved distance advantage.** Find a contiguous interval $[t_1, t_2] \subset (0, 5\tau]$ of length at least 0.25τ on which $A_D(t) \ge 0.10$ and $D_c(t) \ge 10^{-3}$. The distance floor excludes a large percentage arising only in an arbitrarily small asymptotic tail.

2. **Time saved to a common threshold.** Define $t_x(\delta) = \inf\{t \geq 0 : D_x(t) \leq \delta\}$ and $A_t(\delta) = 1 - t_h(\delta)/t_c(\delta)$. Target $A_t(\delta) \geq 0.05$ throughout the predeclared band $10^{-3} \leq \delta \leq 10^{-2}$, with both initial distances above 10^{-2}. Report absolute time savings in units of ν_0^{-1} as well as percentages and τ units. These are post-quench relaxation times; preparation time and energy cost require a separate accounting for any operational claim.

Also report total-variation and Pearson chi-square trajectories to expose distance dependence. A KL success is not automatically a success for every distance. Common rate rescaling cannot improve either percentage and must not be presented as a cooling mechanism.

22.3 Robustness, verification and decision record

The target is a neighborhood, not a single tuned optimum. Test simultaneous ±5% variations in the independent positive energy gaps, barrier excesses and preparation-temperature excesses $T_x/T_b - 1$, rebuilding detailed balance each time. For models with independently variable prefactors, perturb those too. Specify parameter correlations imposed by the physical geometry. Seek a neighborhood inside the declared parameter box in which both performance benchmarks survive; quantify a smaller verified neighborhood if the target tolerance fails. If the result is advertised as an extension of the counterexample, the triplet predicates must also survive throughout that neighborhood.

Use sampled perturbations and optimization for discovery, followed by independent high-precision matrix-exponential and first-passage checks. Certify interval-wide distance and threshold claims with interval bounds or analytic estimates; a time grid or a random perturbation cloud alone does not certify robustness. Record seeds, search boxes, tolerances, rejected candidates and all benchmark revisions. The deliverable is a compact physical model, explicit parameters, a finite-time performance figure, reproducible calculations and a stated robustness domain, with physical approximation errors distinguished from numerical errors.

The first milestone was to specify and justify the constrained barrier geometry, then screen the declared box and analyze the best candidates. Sections 22.4–22.6 now establish both benchmarks on the full requested neighborhood for an idealized thermal-collision rate model. Material and continuum validation remain outside that verified conclusion. At the time this priority was recorded, no new candidate meeting these benchmarks had been established. If a model or box fails, report the bounded search evidence or a proved obstruction and the next justified model change. Failure to find a candidate is not a general impossibility theorem. New evidence and any resulting conclusions belong in this comprehensive research report before being promoted into the focused paper.

22.4 Physical-model study: finite-time gains and certified robustness domains

Status on 12 October 2026. A constrained four-state Arrhenius network now meets both predeclared finite-time benchmarks throughout the simultaneous ±5% parameter box. Continuous interval certificates establish at least 10% KL-distance advantage on $0.4 \leq t/\tau \leq 0.8$ and at least 5% time savings for every KL threshold in $[10^{-3}, 10^{-2}]$. The whole box also lies inside the declared physical parameter ranges; see Section 22.5. This is a separate physical-model study: two triplets satisfy the original test, so it does not strengthen the paper's all-triplet-failure counterexample.

Model chosen before screening. Let the energies be $E = (0, g_1, g_1 + g_2, g_1 + g_2 + g_3)$, with basins $A = \{1, 4\}$ and $B = \{2, 3\}$. Their energy levels are interleaved. Use a common attempt frequency ν_0 and

$$B_{ij} = \max(E_i, E_j) + h_{ij}, \qquad h_{14} = h_A, \quad h_{23} = h_B, \quad h_{12} = h_{13} = h_{24} = h_{34} = h_X.$$

The three shared activation excesses represent two types of intrabasin rearrangement and a more costly change of basin. They were not fitted independently on six edges or reconstructed from a

chosen eigenvector. The dated model specification is `research/physical_model_design.txt`. This is a phenomenological network of metastable configurations; it is not yet derived from a specified material or continuum potential. Section 22.6 supplies an exact stochastic energy-conserving collision realization of these same rates, with its bath and contact assumptions stated explicitly.

The refined parameters, rounded here for readability, are

$$\begin{aligned}(g_1, g_2, g_3) &= (0.830894205,\ 0.116352749,\ 2.729136959),\\ (h_A, h_B, h_X) &= (2.114063920,\ 2.822657579,\ 5.640080424),\\ (T_c/T_b, T_h/T_b) &= (3.869407740,\ 9.472085162).\end{aligned}$$

The certified model uses the full decimal inputs in `research/physical_rate_refined.json`, not the rounded display. Its energy span is $3.67638\,k_BT_b$ and its directed rate ratio is approximately 584.94. The smallest and largest directed rates are approximately $0.000206426\,\nu_0$ and $0.120746\,\nu_0$; $\tau = 147.232\,\nu_0^{-1}$. All rates are strictly positive and detailed balance follows from the symmetric saddles.

Mechanism and physical interpretation. The nominal decay rates in attempt-frequency units are approximately 0.0067920, 0.116138 and 0.130813. Thus the two faster modes decay about 17 and 19 times as quickly as the slow mode. Normalize the slow left eigenfunction to have first component one. Numerically it is $(1, -1.24304, -1.24253, 0.92447)$, nearly constant within each basin. The cold and hot preparations put respectively 0.46591 and 0.47963 of their probability in basin A, compared with 0.55459 at the bath. The hotter preparation therefore has less of the slow basin-population mismatch, although it starts with a larger total KL distance. Its additional disturbance within the basins relaxes quickly. These numbers support the interpretation; exact lumpability is not assumed. The relevance of metastable populations is established in prior work, including Biswas, Rajesh and Pal (2023), Section IV and Fig. 2. Their piecewise-linear Langevin model relates initial well populations to relaxation. Their Section VI also exhibits the effect without a metastable minimum: metastability is not a general necessary condition. The citation supplies a mechanism precedent, not a derivation of our four-state rates, an exact lumping theorem, or our robustness certificate. Neither that general mechanism nor Arrhenius kinetics is claimed new here.

The barriers are only 2.11–$5.64\,k_BT_b$ above the higher endpoint. Consequently a Kramers-type asymptotic rate approximation could have material errors even with a clear separation of the network's eigenvalues. The 17-fold spectral separation is between network modes, not a measurement of intrawell relaxation. A microscopic derivation must establish that separate scale and quantify the coarse-graining error. No experiment, absolute clock calibration, preparation cost or continuum validation is claimed by this calculation. The separate collision realization defines an exact thermalizing model without a large-barrier approximation; it does not validate a material's continuum escape-rate law.

Finite-time evidence. Direct 80-digit matrix exponentials give the following times, in ν_0^{-1} units:

KL threshold	Cold time	Hot time	Saved time	Saving
10^{-2}	40.2978	27.6018	12.6960	31.5055%
$10^{-2.5}$	125.0952	105.4404	19.6548	15.7118%
10^{-3}	209.9825	190.3277	19.6548	9.3602%

These spot checks are supplemented by outward interval arithmetic. The certificate proves at least 8.12% time savings for *every* threshold in $[10^{-3}, 10^{-2}]$, with both initial distances above the band and the hot preparation initially farther away. On the fixed time interval $[58.8927, 117.7854]\,\nu_0^{-1}$ (endpoints rounded here), it proves $D_h \le 0.9 D_c$ and $D_c > 0.00349$. An interval LDL inertia calculation brackets

the slow eigenvalue and checks that this interval is longer than 0.25τ and ends before 5τ. The more precise endpoints are in the certificate receipt. The nominal gain is therefore resolved at finite distances and finite times, rather than inferred solely from a slow-mode amplitude.

What the interval certificate establishes. The checker uses uniformization with $\rho = 1/2$:

$$e^{Mt}p = e^{-\rho t}\sum_{k=0}^{\infty}\frac{(\rho t)^k}{k!}(I + M/\rho)^k p.$$

The transition matrix is nonnegative and stochastic for the exact input. Each component of its iterates lies in $[0, 1]$, so a geometric upper bound on the remaining Poisson mass encloses the truncated tail. All exponential and logarithmic evaluations use 60-digit outward interval arithmetic. Relative entropy decreases along either relaxation. On a time cell $[t_i, t_{i+1}]$, it therefore suffices to prove $D_h(t_i) < 0.9D_c(t_{i+1})$. On a threshold cell $[\delta_i, \delta_{i+1}]$, it suffices to bound $1 - t_h(\delta_i)/t_c(\delta_{i+1})$ from below. Eighty time cells and sixty threshold cells pass; interval sign checks bracket each threshold root. This is an executable numerical certificate with stated mathematical dependencies, not a new Lean formalization.

Robustness: discovery evidence and continuous verification. The refined candidate was checked at all $2^8 = 256$ corners and 3,000 seeded interior points of the independent $\pm 5\%$ box in the three energy gaps, three barrier excesses and two temperature excesses. At 31 thresholds per point, the smallest observed time saving was 5.1203%. The shortest sampled distance-advantage window exceeded 1.08τ. These samples found no failure of either performance target. Those samples alone do not exclude an interior parameter or threshold failure. The subsequent continuous certificates are distinguished from the samples in Section 22.5. Three further seeded adversarial searches varied all eight uncertain parameters and the threshold continuously. None found a saving below 5.1203%. These local/global numerical optimization attempts strengthen the screening evidence but do not prove the global minimum. The physical range constraints are interval-certified for the whole box: the energy span is below 3.86021, every gap exceeds 0.11053, barrier excesses lie in $[2.00836, 5.92209]$, the rate ratio is below 993.753, and both preparations remain in the declared range.

An initial nominal optimization gave about 16.76% minimum sampled time savings, but 441 of 1,256 subsequent perturbation cases failed at least one target or range constraint. A second search guarding the physical box and individual parameter changes still failed in 50 cases. Three refinements against adverse combinations produced the present candidate. Retaining these failures is important: checking one parameter at a time would have overstated robustness. The first two searches used 19,728 and 14,973 objective evaluations; the three refinements used 9,977, 10,094 and 9,743. These bounded searches do not establish global optimality or a no-go result for other models.

The numerical original-triplet signs are $(F, G) = (-,-), (+,+), (+,+), (-,-)$ for $123, 124, 134, 234$, respectively; the positive margins are large compared with numerical error. Thus this example is outside the all-triplet-failure premise. Both finite-time benchmarks have now been certified on the full $\pm 5\%$ box, independently of that additional triplet premise. The physical approximation check appears in Section 22.7; it does not establish a continuum or material realization. Scripts and exact input provenance are listed in `RUNME.txt`.

The nominal total-variation and Pearson chi-square trajectories are retained alongside KL in `output/figures/physical_finite_time.csv`. On the sampled grid in $0.4 \le t/\tau \le 0.8$, their distance advantages are approximately 12.50% and 23.43%, respectively, compared with 23.39% for KL. These diagnostics support the qualitative reversal for the other distances at the nominal point; the continuous robustness certificates below concern KL only.

22.5 Continuous verification of the full ±5% parameter box

Verified scope on 12 October 2026. Both predeclared finite-time benchmarks hold throughout the simultaneous ±5% uncertainty box. For every parameter vector in that box,

$$D_h(t) \le 0.9 D_c(t), \quad D_c(t) \ge 10^{-3}, \qquad 0.4 \le t/\tau \le 0.8,$$

and

$$t_h(\delta) \le 0.95 t_c(\delta) \qquad \text{for every } 10^{-3} \le \delta \le 10^{-2}.$$

The distance interval has length 0.4τ and ends before 5τ. These are continuous parameter, time and threshold claims, not inferences from a perturbation cloud. They concern KL distance only.

The eight independently perturbed coordinates are the three gaps, three shared barrier excesses and two temperature excesses $T_x/T_b - 1$. The generator and Gibbs preparations are rebuilt throughout the box; no independently chosen matrix entries or eigenvectors are perturbed. This preserves the physical geometry and detailed balance exactly. The uncertainty domain remains inside all predeclared moderate ranges, as recorded in Section 22.4.

Threshold inequalities and an exact temperature reduction. For fixed physical parameters, the simple slow eigenvalue $\lambda < 0$ and $\tau = -1/\lambda$ do not depend on the preparation temperatures. Set $u = t/\tau$ and $\mathcal{D}_x(T, u) = D_x(T, u\tau)$. Separate interval checks prove that both preparations have distance above 10^{-2} at $u = 0.1$ and below 10^{-3} at $u = 2$ throughout the box. KL is strictly decreasing along every nonstationary trajectory of this irreducible reversible generator. Each threshold root in the specified band is therefore unique and belongs to $(0.1, 2)$.

Two additional continuous certificates prove

$$\partial^2_{T_c} \mathcal{D}_c \le 0 \quad (0.1 \le u \le 2), \qquad \partial_{T_h} \mathcal{D}_h \le 0 \quad \text{whenever } 10^{-3} \le \mathcal{D}_h \le 10^{-2}, \quad 0.1 \le u \le 2.$$

Both cover all six physical parameters and the full relevant temperature interval. The other preparation temperature is irrelevant to each single-preparation distribution and may be fixed in that check. Let T_c^-, T_c^+ and T_h^- denote the box's cold endpoints and lowest hot temperature. For any fixed threshold, write $u_x = t_x/\tau$ and $m = \min\{u_c(T_c^-, \delta), u_c(T_c^+, \delta)\}$. At time m, both endpoint cold distances are at least δ. Concavity in T_c puts every interior cold distance above the chord, so $u_c(T_c, \delta) \ge m$. For the hot preparation, implicit differentiation at its threshold root gives

$$\partial_{T_h} u_h = -\frac{\partial_{T_h} \mathcal{D}_h}{\partial_u \mathcal{D}_h} \le 0.$$

Thus the slowest hot arrival is at T_h^-. It remains to prove just two boundary-temperature inequalities, still continuous in all six physical parameters and in u:

$$10^{-3} \le \mathcal{D}_c(T_c^{\pm}, u) \le 10^{-2} \quad \Longrightarrow \quad \mathcal{D}_h(T_h^-, 0.95u) \le \mathcal{D}_c(T_c^{\pm}, u).$$

The two completed certificates imply

$$u_h(T_h, \delta) \le u_h(T_h^-, \delta) \le 0.95m \le 0.95 u_c(T_c, \delta).$$

This proves the whole temperature domain; checking its endpoints without the derivative certificates would not suffice.

Temperature derivatives. For the Gibbs distribution $q(T)$, let $U = \sum_i q_i E_i$ and $V = \mathrm{Var}_q(E)$. The checker propagates

$$q_i' = q_i \frac{E_i - U}{T^2}, \qquad q_i'' = q_i \left[\frac{(E_i - U)^2 - V}{T^4} - \frac{2(E_i - U)}{T^3} \right]$$

through the same semigroup as q. Since their masses vanish,

$$D_T = \sum_i p_i' \log(p_i/\pi_i), \qquad D_{TT} = \sum_i \left[\frac{(p_i')^2}{p_i} + p_i'' \log(p_i/\pi_i) \right].$$

Normal initial ordering also follows analytically: for a Gibbs preparation above the bath, $\partial_T D(\pi(T)\|\pi_b) = (\beta_b - \beta)\,\mathrm{Var}_{\pi(T)}(E)/T^2 > 0$. The early endpoint bounds and contractivity place both initial distances above the threshold band.

Semigroup enclosures. Conservation of mass reduces probability deviations to three coordinates. A fixed basis close to the nominal eigenbasis transforms the reduced generator to R. Each constant edge matrix is transformed before being multiplied by its uncertain rate, retaining cancellation of intrabasin flux. An interval characteristic-polynomial bracket isolates λ; interval Newton and a Schur-complement enclosure tighten it, while implicit differentiation encloses its parameter derivatives. Reversibility supplies real, diagonalizable eigenvalues. Explicit spectral bounds verify that both remaining eigenvalues lie below λ.

Let P be the slow spectral projector and $Q = I - P$. The implementation uses two complementary fast-block enclosures. Writing R in a scalar plus 2×2 block form with first row (r_{00}, b) and bottom-right block F, normalize its slow right vector to $(1, v)^\top$. Set $H = F - vb$, $W = (-v, I_2)$ and $V_f = Q[:, 2:3]$. Then

$$e^{Rt} = e^{\lambda t} P + V_f e^{Ht} W.$$

For $Ht = D + C$ with D diagonal and C off diagonal, the first two Dyson terms have diagonal $e^{D_{ii}}$ and off-diagonal $C_{ij} e^{(D_{11}+D_{22})/2}\,\mathrm{sinhc}((D_{11} - D_{22})/2)$. With $m = \max_i D_{ii}$ and $c = \|C\|_\infty$, the omitted terms have norm at most $e^m(e^c - 1 - c)$. For each parameter derivative the remainder norm is at most

$$e^m \left[\|D'\|_\infty (e^c - 1 - c) + \|C'\|_\infty (e^c - 1) \right].$$

Outward bounds on these expressions are added to the value and derivative intervals. This is an enclosure of the truncation error, not an assertion that the approximation error is roundoff-sized.

For the distance, endpoints and two boundary-temperature time checks, the checker intersects this enclosure with the exact identity

$$e^{Rt} = e^{\lambda t} P + e^{at} \left[\cosh(dt) Q + \frac{\sinh(dt)}{d} (R - aI) Q \right],$$

where the fast eigenvalues have sum $2a$ and squared half-gap $d^2 \geq 0$. Entire-function evaluations of $\cosh\sqrt{z}$ and $\sinh\sqrt{z}/\sqrt{z}$ handle coincident fast modes without dividing by an uncertain gap. Intersecting two valid enclosures retains validity and avoids a fixed Dyson truncation floor on small boxes.

Sign verification and coverage. KL is evaluated using its chain rule into a binary basin distance plus two conditional binary distances. The binary terms use $\phi(z) = (1 + z)\log(1 + z) - z$, with its correct monotone branches. Interval automatic differentiation and centered mean-value bounds reduce dependency overestimation. For the time margin $F = \mathcal{D}_c(u) - \mathcal{D}_h(0.95u)$, either of

$$F + \alpha(\mathcal{D}_c - 10^{-2}) \geq 0, \qquad F + \alpha(10^{-3} - \mathcal{D}_c) \geq 0, \qquad \alpha \geq 0,$$

is a sufficient certificate on the threshold band. Heuristics choose multipliers, but acceptance always requires a whole-box outward sign bound. Sign-definite partial derivatives may move the minimum of this same expression to a face. A face's being outside the threshold band never excludes the original box.

Basic binary64 operations are widened outward by one neighboring value; MPFR evaluates decimal inputs and transcendental functions with directed rounding. Compilation excludes fast-math and floating-point contraction. An unresolved sign causes subdivision, never acceptance. Closed child boxes share their boundary. A coverage run succeeds only when the queue is empty; a node-limit exit is incomplete, not a certificate or a physical failure. The input rounding encloses the exact requested ±5% box.

Proof obligation	Tree nodes	Accepted leaves
Distance window	23,351	11,676
Both early endpoints	105	53
Cold late endpoint	1	1
Cold temperature concavity	135,721	67,861
Hot temperature monotonicity on band	52,785	26,393
Hot late endpoint	1	1
Time saving: T_c^-, T_h^-	54,425	27,213
Time saving: T_c^+, T_h^-	65,045	32,523

Every listed run ended with zero pending cells and no enclosure errors. Earlier ±1% certificates and incomplete direct eight-parameter coverage attempts remain in the record. The temperature reduction supersedes those incomplete attempts without shrinking the requested domain.

Reproducibility and limits. Run `research/run_robust_finite_time_certificate.py` to compile `research/physical_temperature_reduction.cpp` and `research/physical_temperature_reduced_combined.cpp`, then replay all eight obligations. The runner checks embedded inputs against `research/physical_rate_refined.json` and records source, binary and log hashes in `research/robust_finite_time_certificate/certificate.json`. The independent checker `research/check_temperature_reduction.py` uses the original four-state generator and 70-digit direct matrix exponentials: 6,744 value checks for KL and its first two temperature derivatives, plus 360 parameter/time derivative checks, pass at the nominal point, all 256 corners and 24 seeded interior points as specified in its receipt. Finite differences are implementation cross-checks, not the justification of the analytic interval derivative bounds.

This is an executable interval proof with reviewable mathematical and software dependencies, not a Lean formalization or external peer review. The exact stochastic realization in Section 22.6 supplies a bath interpretation of the same rate model. Numerical certification does not establish a material's rate law, an experimentally calibrated cooling clock, preparation costs or experimental feasibility.

22.6 An energy-conserving thermal-collision realization of the same rates

The Arrhenius network also has an exact stochastic bath realization that does not invoke a large-barrier escape approximation. This clarifies what can be physically modeled without claiming a particular material or a continuum potential. The system retains the same four energies, the same three shared saddle excesses, and the same eight uncertain parameters.

Give each unordered pair ij an independent Poisson contact clock of rate ν_0. At a contact, draw a fresh reservoir energy $Y \geq 0$ with density $\beta_b e^{-\beta_b Y}$. This is a canonical reservoir with constant density

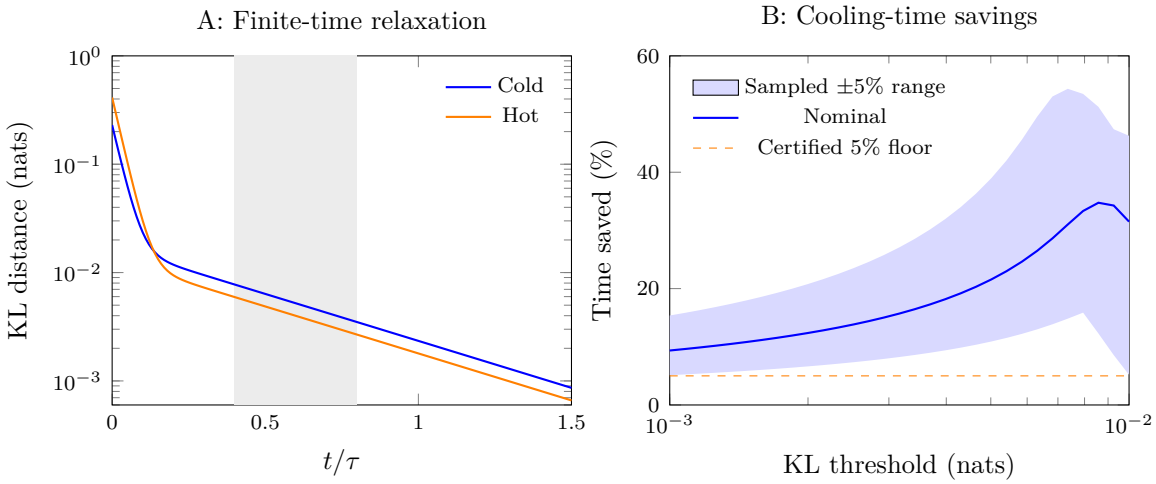


Figure 5: The constrained Arrhenius candidate of Section 22.4. The shaded time interval in A has an interval certificate of at least 10% KL-distance advantage with cold distance above 10^{-3} nats. The distance claim now holds throughout the simultaneous ±5% box. In B, the nominal model has a certified lower bound of 8.12% time savings throughout the threshold band. The shaded parameter envelope uses 256 corners and 3,000 sampled interior points at 31 thresholds; it is not a certificate for the whole parameter box. The dashed 5% floor is separately certified throughout the full simultaneous ±5% box for every threshold in the band (Section 22.5). The approximate nominal relaxation time is $\tau = 147.232\,\nu_0^{-1}$.

of energy states. If the system is in endpoint j, the contact changes it to i precisely when

$$E_j + Y \geq B_{ij}.$$

After an accepted contact, the outgoing reservoir energy is $Y' = Y + E_j - E_i$. Otherwise the state and reservoir energy are unchanged. Contacts on pairs not containing the current state do nothing. Because $B_{ij} > \max(E_i, E_j)$, accepted moves always have $Y' \geq 0$. They conserve the combined energy exactly.

Rate reduction and detailed balance. Integrating out the independent reservoir mark yields

$$a_{ij} = \nu_0 \Pr(Y \geq B_{ij} - E_j) = \nu_0 e^{-\beta_b(B_{ij}-E_j)}.$$

This is the original generator, with its original physical contact clock; no residence time is deleted and no rate is rescaled to improve the advantage. For an accepted move the reverse contact is allowed because $E_i + Y' = E_j + Y \geq B_{ij}$. Applying the map twice returns the original state and energy, and the translation $Y \mapsto Y'$ preserves its integration measure. The joint canonical density $e^{-\beta_b(E_j+Y)}$ is also unchanged. Thus each contact kernel, and their Poisson superposition, obey detailed balance. These identities hold throughout the uncertainty box, not just at the nominal point.

Equivalently, with $h_{ij} = B_{ij} - \max(E_i, E_j)$, the rate is $\nu_0 e^{-\beta_b h_{ij}} \min\{1, e^{-\beta_b(E_i-E_j)}\}$. The acceptance factor is the familiar Metropolis form (1953). The original method accepts downhill proposals and accepts uphill proposals with the Boltzmann factor; rejected proposals still contribute to its configuration averages (Section II, pp. 5–6 of the Los Alamos report). Auxiliary-energy simulation methods also have a long history. Creutz (1983), pp. 1411–1412, introduces a retained energy-carrying demon with conserved combined energy. He also explicitly describes redrawing its energy from the Boltzmann distribution before each visit, recovering Metropolis sampling. Thus a refreshed thermal

auxiliary energy is itself an established device. Our specified pair clocks and symmetric barrier gates provide the rate and time interpretation used here; neither cited sampling method alone establishes that physical clock or this model's finite-time performance. No new thermostat or sampling algorithm is claimed here. The contribution of this construction is an explicit bath interpretation of the particular constrained rate family and its finite-time clock.

A two-coordinate interpretation of the states. Label the ordered states by $(x, y) = (0,0), (1,0), (1,1), (0,1)$ and write

$$H(x,y) = \epsilon_x x + \epsilon_y y + Jxy, \quad (\epsilon_x, \epsilon_y, J) = (E_2, E_4, E_3 - E_2 - E_4).$$

Nominally these coefficients are approximately $(0.83089, 3.67638, -3.56003)\, k_B T_b$. The two basins fix x; their internal degree of freedom is y. Within basin A, its excited configuration is expensive, whereas the two configurations in basin B are nearly degenerate. The exact equilibrium basin free energies are

$$F_A(T) = -T \log(1 + e^{-E_4/T}), \qquad F_B(T) = E_2 - T \log(1 + e^{-(E_3 - E_2)/T}),$$

with $k_B = 1$ and $E_1 = 0$. Their different temperature dependence explains the changing basin populations used in the slow-mode mechanism. This interpretation does not assume exact lumpability into two basins. Changes of these Hamiltonian coefficients are correlated by the independently perturbed gaps; no additional independent uncertainty claim is made. Contacts may change one or both coordinates. Thus this is not a claim about standard single-spin-flip dynamics or an identified spin material.

Independent event-clock check and physical limits. An executable algebra check verifies the canonical tail integral, energy conservation and the contact involution. An independent simulation uses 500,000 trajectories per preparation and Poisson contact counts, with no fixed time step. It samples the thermal reservoir energies explicitly. For efficiency it thins only the state-independent gate $Y \geq h_{ij}$: the retained pair clock has rate $\nu_0 e^{-\beta_b h_{ij}}$, and the conditional reservoir energy is h_{ij} plus a fresh exponential energy. This exact thinning retains the waiting-time cost of rejected contacts. At five checked physical times, simulated probabilities agree with direct matrix exponentials within 0.00118 in every component, passing the declared six-standard-error diagnostic envelopes. This Monte Carlo check is supplementary; the collision-to-generator equality is analytic and the performance certificates do not depend on sampling. Code and receipts are `research/thermal_collision_model.py` and `research/thermal_collision_audit.json`.

The result is an idealized stochastic collision model. Its assumptions include fresh independent thermal contacts, a constant reservoir density of states, Poisson contact clocks and instantaneous energy-conserving updates. It is not a derivation from an autonomous Hamiltonian, an identified material or a measured bath. The costs of maintaining the reservoir/contact process and preparing the two initial temperatures are not included in the post-quench relaxation comparison. For the activated landscape interpretation, intrawell equilibration and the accuracy of a continuum escape-rate approximation remain separate physical questions. The collision realization avoids that approximation within its stated stochastic assumptions; it does not validate it for a material.

22.7 An explicit activated-intermediate check of the physical clock

The four-well study can be extended to retain time spent in transition regions. This gives a mesoscopic check of one coarse-graining assumption, not a derivation from a continuum potential or an experimental system. For each of the six well pairs ij, add an intermediate s_{ij} with energy B_{ij} and

phase-space weight $\eta > 0$. Well weights are one. At the bath temperature, in the same dimensionless energy convention, set

$$Q_{s_{ij},j} = 2\nu_0 e^{-(B_{ij}-E_j)}, \qquad Q_{j,s_{ij}} = 2\nu_0/\eta,$$

and use the corresponding pair of rates at endpoint i. All other off-diagonal rates vanish. The diagonal entries conserve probability. The stationary weights are $g_a e^{-E_a}$, where $g_j = 1$ and $g_{s_{ij}} = \eta$. Each edge obeys detailed balance because its equilibrium flux in either direction is proportional to $2\nu_0 e^{-B_{ij}}$.

An intermediate has total escape rate $4\nu_0/\eta$ and mean residence time $\eta/(4\nu_0)$. Partitioning Q into well and intermediate blocks gives the exact identity

$$Q_{ww} - Q_{ws}Q_{ss}^{-1}Q_{sw} = M.$$

Indeed, each entrance rate is twice the original well-to-well rate, and the next exit chooses either endpoint with probability 1/2. Thus the original generator is the exact trace generator when time in intermediates is deleted. The following calculation instead evolves the full ten-state generator in physical time, including those residence times. Both preparations are full weighted Gibbs distributions $p_a(T) \propto g_a e^{-E_a/T}$, including the intermediate populations. Distances are KL divergences from the full ten-state bath distribution.

Numerical results and independent checks. For each displayed η, the nominal parameters are unchanged from Section 22.4. The threshold grid contains 41 logarithmically spaced values in $[10^{-3}, 10^{-2}]$. The perturbation sample contains all 256 corners of the eight-parameter ±5% box and 500 seeded interior points for each fixed η.

η	Nominal minimum time saving	Sampled worst time saving
0.025	9.5759%	5.2719%
0.050	9.7883%	5.4207%
0.100	10.2036%	5.7115%

All 2,268 parameter samples passed the sampled finite-time benchmarks. The shortest sampled window with at least a 10% distance advantage was approximately 1.0889τ, with the cold distance at least 10^{-3} and times below 5τ; here τ is the full model's slow time. These are finite grids in parameters, thresholds and time, not certificates over their continuous ranges or over an interval of η values.

At $\eta = 0.05$, the cold and hot preparations have respectively about 1.90% and 4.07% probability in intermediates. Independently constructed 70-digit direct matrix exponentials agree with the spectral calculation to within 6.55×10^{-13} in KL distance at five checked times. Independent numerical root solves give cold/hot arrival times 210.67249/190.05131 at threshold 10^{-3} and 40.83628/27.51015 at 10^{-2}, in ν_0^{-1} units. These correspond to 9.78826% and 32.63307% time savings. The finite-time benefit therefore survives this explicit extension at the tested points; it is not an artifact of deleting intermediate residence times in this model.

Scope and remaining work. The original moderate bounds still describe the four well energies and their effective rates. They do *not* describe every state and rate of the expanded model: its highest transition-state energy is about $9.32\,k_B T_b$, and its fast intermediate exits make the full directed rate ratio exceed the original 10^3 cap. This extension is a separate test of the physical approximation, not a replacement candidate claimed to satisfy the original all-state bounds. The phase-space weight is a transition-region volume parameter, not an integer degeneracy. No continuum-potential derivation, material identification or absolute time calibration has been established.

Both finite-time benchmarks throughout the original four-state ±5% uncertainty box are now certified in Section 22.5. Earlier interval methods that could not close that box remain historical

incomplete attempts; their enclosure overestimation was not evidence of negative probabilities or a failed physical benchmark. The completed spectral and temperature-reduction proof concerns the four-state model. It does not turn the ten-state numerical checks into a continuous robustness theorem. The exact stochastic bath realization of Section 22.6 supplies an independent interpretation of the original rates. A specified material or continuum system would still require its own rate and preparation-weight validation.

The model and nominal scan are in `research/activated_intermediate_model.py`; the independent audit is `research/check_activated_intermediates.py`. Machine-readable results are `research/activated_intermediate_results.json` and `research/activated_intermediate_audit.json`.

22.8 Other open mathematical questions

A separate open mathematical question is finite-time relative-entropy heating order for general contrast-Metzler generators outside the quantified bath neighborhood. Theorems 5 and 6 now settle, respectively, all heating temperatures for tridiagonal contrast generators and both bath sides within $|\beta - \beta_b|(E_n - E_1) \le 2/\sqrt{5}$. The general case is still open here. Section 6.1 proves that the monotone-ratio step of the tridiagonal proof cannot simply be assumed for general admissible dynamics. A completed 1200-second search evaluated 6,740,275,200 sampled derivatives in dimensions 4, 5, 6 and 8 and found none negative; this is search evidence, not a proof. A subsequent 3602-second targeted four-state optimizer made 66,396,528 objective calls and found no negative score. Its best candidate remained positive under an independent 100-digit direct matrix exponential; neither search proves absence. The full four-state amplitude supremum and the location of its possible anomalous interval are now settled. For larger state counts, Theorem 18 supplies the achievable limit $1 - 3/n$ under all original tests, whereas the general contrast upper bound remains $1 - 2/n$. A sharp full-premise bound for $n > 4$ remains open between these bounds. Proposition 1 reduces a matching upper bound to the slow-mode inequality $w_1 + w_2 \ge 2w_3$. Section 18.1 gives an exact obstruction to one natural proof route: even after subtraction of its minimum reset, a generator satisfying every original triplet inequality can have a three-state trace violating $F_{123} \le 0$. The arithmetic implication itself remains unresolved in higher dimension. These questions exploit the new structure rather than searching again for strong effects that the theorem now excludes.

For external review, the package includes explicit generators, two distinct primary counterexample checks, finite-time certificates, the all-dimension contrast proof and symbolic checks, the finite classifier with polynomial-reference audits, and exact sharpness constructions. The package now also includes the Lean verification described in Section 20.3. The remaining analytic proofs and executable checks are not thereby formalized. All work belongs to the same project; no external human peer review is claimed.

Appendix A: Explicit slow-mode cofactor polynomials

The primary counterexample's exact left eigenvector is $w_i = P_i(r)/P_1(r)$, where r is the simple slow root specified in Theorem 1 and

$$\begin{aligned}
P_1(x) &= 3184000000x^3 + 1287155436000x^2 \\
&\quad + 74562002576870x + 532366138003157, \\
P_2(x) &= 3184000000x^2 + 9060415184000x + 532366138003157, \\
P_3(x) &= 89152000000x^2 + 14211422840400x + 532366138003157, \\
P_4(x) &= 79281600000x^2 + 28279185628400x + 532366138003157.
\end{aligned}$$

Appendix B: A rational slow-mode companion

The data and strict overlap reversal below are verified in Lean, together with an initial-to-eventual Pearson chi-square relaxation reversal. Section 20.3 states the exact coverage and distinguishes the Lean enclosures from the tighter Python-certified intervals printed here.

Keep the same q, a_{12}, a_{13}, a_{14} and a_{24}, but set

$$a_{23} = \frac{330154675693}{1470375000}, \qquad a_{34} = \frac{1527327066}{151990625}.$$

Use detailed balance and conservative diagonals as before. Then

$$w = \left(1, -\frac{759}{1000}, -\frac{48957}{80000}, -\frac{1}{250}\right), \qquad \lambda_2 = -\frac{4966671}{250000}$$

obey $M^\top w = \lambda_2 w$ and $q^\top w = 0$ by rational arithmetic. Here $N = 27567998009/16000000000$ and $v = \operatorname{diag}(q)w/N$.

The characteristic polynomial factors as

$$\frac{x(250000x + 4966671)R(x)}{63250716205000000000000},$$

where

$$\begin{aligned} R(x) = {} & 2530028648200000000x^2 \\ & + 1002356769715761942000x + 50981800562345565497767. \end{aligned}$$

One root of R lies in $(-337, -336)$ and the other in $(-60, -59)$, as exact endpoint signs show. The rational eigenvalue is therefore the simple slow mode. The companion certificate checks all original predicates strictly and proves

$$\begin{aligned} b_2(1/1000) &\in (-0.093755177882, -0.093755177881), \\ b_2(1/100) &\in (-0.093857287853, -0.093857287852). \end{aligned}$$